



APPROVED
FOR COMPLIANCE WITH AS MENTIONED(SEE
LETTER) REQUIREMENTS

ON BEHALF OF GOVT. OF INDIA

**INDIAN REGISTER OF SHIPPING
MUMBAI**

DATE: 29-Dec-2022

AS AMENDED

SEE PAGE NO: 2, 3, 9, 47-49,102-105.

SEE LETTER E-172074-228715

**FINAL
TRIM AND STABILITY BOOKLET
OF
KB 40
(OFFICIAL NO. MAR-2401-D)**

Rev.	Description	Date	Prepared	Review	
<u>Issued for</u>		<u>Approval</u>			
Client: K.B. SHIPPING & CO. (JAMNAGAR)		Project: PONTOON (YARD NO.CHISTI/025)			
Builder: CHISTI MARINE ENGINEERING WORKS					
Doc #		Document Title:		Prepared	
CM22-VI-112-0211		FINAL TRIM AND STABILITY BOOKLET		OM	
Scale	Size			Sheet	Date
NTS	A4			1 of 115	22-11-2022
				Checked	
				MG	
				Review 1	
				Review 2	
 Cybermarine www.cybermarine.net 10 Bukit Batok Crescent #12-01 The Spire Singapore 658 079 501 Technocity IT Premises Plot No X-5/3 Mahape Navi Mumbai 400 710 India					

CONTENTS

General

Page No

1.General Particulars	03
2.Stability Criteria	04
3.Notes to the Master	05
4.Draft & Trim Calculation	06
5.Worked Out Example	09
6.Definitions, Symbols and Sign Conventions	17
7.Metric Conversion Table	18

Ship's Data

1. Angle of Down flooding	19
2. Capacity Plan and Table	20
3. Table of Hydrostatic Particulars	22
4. Definition of KN	33
5. Volume used For KN Computation	34
6. Displacement Vs KN tables	35
7. Deadweight Table	46
8. Table of Maximum Permissible KG _{SHIP}	47

Summary of Conditions

51

Loading Conditions

1. Cond#0- Lightship	53
2. Cond#1- Lightship + Crane + Per. Ballast	58
3. Cond#2- Lightship + Load at Aft + Ballast Crane Lifting 36.6T @ 18m Radius Aft	63
4. Cond#3- Lightship + Load at Stbd + Ballast Crane Lifting 36.6T @ 18m Radius Stbd	76
5. Cond#4- Lightship + Load at Fwd + Ballast Crane Lifting 36.6T @ 18m Radius Fwd	89
6. Cond#5- Lightship + Crane + Ballast (Towing Condition) + Deck cargo	102

Appendix

I. Lightship Particulars	106
II. Wind Area at various drafts	107
III. Draft Survey report Approval Letter	111

GENERAL PARTICULARS

Name of the Vessel	KB 40
Client	K.B. SHIPPING & CO. (JAMNAGAR)
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI/025
Type of Vessel	PONTOON
Official No.	MAR-2401-D
Date of Keel Laid	02-07-2022
Port of Registry	MORMUGAO
Class Notation	HSU, PONTOON, DECK STRENGTHENED FOR 10T/m ² FOR CRANE OPERATION

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	15.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	2.200 M
Draft (Ext.)	2.212 M
Displacement (Ext.)	1569.15 T
Lightship	373.37 T
Deadweight	1195.78 T
Gross Tonnage	564
Net Tonnage	169

NOTE :

During unmanned tow minimum forward draft of 2.0 m to be maintained at all times for the compliance with slamming requirements(Refer "Structural plan" approved on 05 Jul 2022 for details)

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

- 1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.
- 2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.
- 3) The minimum range of stability should be:
For $L \leq 100\text{m}$: 20°
For $L \geq 150\text{m}$: 15°
For intermediate length by interpolation

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.
2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship's stability in various loading conditions.
3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.
4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.
5. In this booklet hydrostatic particulars and cross curves are given at following trims.

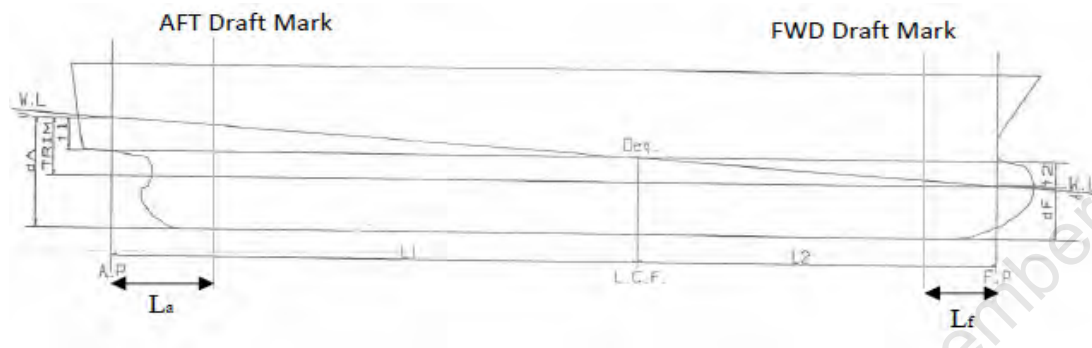
-0.50m, -0.25m, 0.00m, 0.25m, 0.50m

These trims are the expected operational trims.

Notes on Use of Cross Curves of Stability

The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

- a) Longitudinal Centre of gravity of the ship calculated as follows:

$$\text{LCG} = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

- b) Longitudinal centre of buoyancy, LCB
- c) Displacement, Δ
- d) Moment to change trim, MCT1cm
- e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$\text{Trim} = \frac{\Delta * (\text{LCB} - \text{LCG})}{\text{MCT1cm} * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{\text{LCF} * \text{Trim}}{\text{LBP.}} + T (\text{extreme})$$

$$T_{FP} = T_{AP} - \text{Trim}$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{\text{Trim} * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{\text{Trim} * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

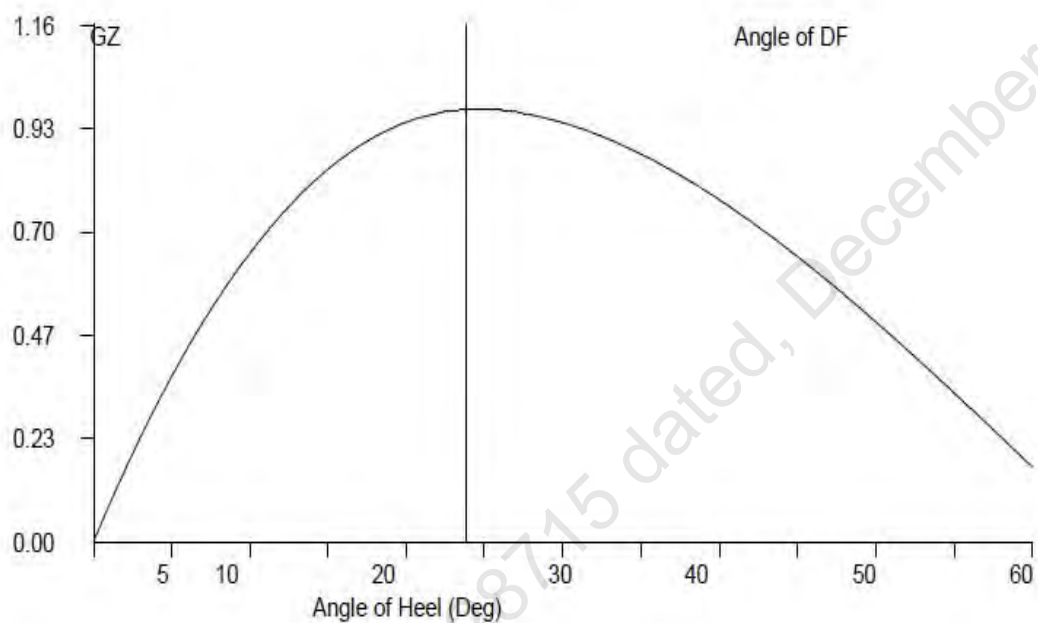
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

- 1) $GZ = KN - KGo \sin (\theta)$
- 2) Area under 0-10 = $10 * 0.01745 * [GZ0 + GZ10] / 2$
- 3) Area under 10-20 = $10 * 0.01745 * [GZ10 + GZ20] / 2$
- 4) Area under 20-30 = $10 * 0.01745 * [GZ20 + GZ30] / 2$
- 5) Area under 30-40 = $10 * 0.01745 * [GZ30 + GZ40] / 2$
- 6) Area under 40-50 = $10 * 0.01745 * [GZ40 + GZ50] / 2$
- 7) Area under 50-60 = $10 * 0.01745 * [GZ50 + GZ60] / 2$
- 8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
- 9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Examples

1 Loading the Vessel

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 5: Lightship + Crane + Ballast +Deck cargo

Displacement	1569.15	Tonnes	
LCG	24.993	M Fwd of AP	
VCG (KG)	3.461	M above Base Line	
TCG	0.000	M Off centerline	

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

$$\text{Free Surface moment for a Tank} = \frac{\text{Moment of Inertia (M}^4\text{) for a Tank} \times \text{Density of Liquid in Tank}}{12}$$

$$\text{KGo (Fluid)} = \text{KG (Solid)} + \text{Free Surface Correction.}$$

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

TANK NAME	VOL (Cu.M.)	SpGr	Weight (MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
03_PFWBTK.C	132.30	1.000	132.30	16.000	2116.80	0.000	0.00	1.500	198.45	91.87
05_PFWBTK.C	132.30	1.000	132.30	34.000	4498.20	0.000	0.00	1.500	198.45	91.87
TOTAL TANK			264.60	25.000	6615.00	0.000	0.00	1.500	396.90	183.74

FIXED WEIGHT			Weight (MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			373.37	24.970	9,323.05	0.000	0.00	3.000	1,120.11
CRANE			250.00	25.000	6,250.00	0.000	0.00	6.120	1,530.00
DECK CARGO			681.18	25.000	17,029.50	0.000	0.00	3.500	2,384.13
TOTAL FIXED WEIGHT			1,304.55	24.991	32,602.55	0.000	0.00	3.859	5,034.24

ITEM			Weight (MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			264.600	25.000	6615.000	0.000	0.000	1.500	396.900
TOTAL FIXED WEIGHT			1304.550	24.991	32602.549	0.000	0.000	3.859	5034.240
TOTAL WEIGHT			1569.150	24.993	39217.549	0.000	0.000	3.461	5431.140

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Free Surface Correction for VCG = FSM/Displacement

	0.117	m
VCGc	3.578	m

From the MaxVCG table,

Displacement	1569.15	T
Allowable Max VCG	5.093	m

VCGc < Maxvcg

PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact

Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	1569.150	T
----------------	----------	---

Draft ,Tm @ 25.0f =	2.212	m
LCB =	25.000	m
LCF =	25.000	m
MCT =	32.380	T-m/cm
KMT =	10.344	m

The vessels trim is to be determined as follow,

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{Displacement} / (\text{MCT} \times 100)$$

0.003 m, (+) is aft trim, (-) is fwd trim

Determining the Forward perpendicular, Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.0 (MS), Tmid	2.212	m
-------------------------	-------	---

Draft @ 50.0f (FP), Tfwd	=Tm - Trim/2
	2.210 m

Draft @ 0 (AP), Taft	=Tm + Trim/2
	2.214 m

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

From HYDROSTATICS

Hydrostatic Particulars			Drafts & Trims	
MCT	32.380	T-m	Draft (Mean)	2.212 m
LCB	25.000	m	Draft AP (USK)	2.214 m
			Draft FP (USK)	2.210 m
Metacentric Height				
KMt	10.344	m		
KG	3.461	m		
KG0	3.578	m		
GMt	6.766	m	Trim	0.003 m

The condition shown above is first worked using Even Keel hydrostatic particulars Program then interpolates the hydrostatic particulars from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			Drafts & Trims	
MCT	31.300	T-m	Draft (Mean)	2.212 m
LCB	24.993	m	Draft AP (USK)	2.214 m
			Draft FP (USK)	2.210 m
Metacentric Height				
KMt	10.344	m		
KG	3.461	m		
KG0	3.578	m		
GMt	6.766	m	Trim	0.004 m

Trim is calculated as follows:

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where Trim* is trim at which hydrostatics were calculated.

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Intact Stability (Primary Criterion)

Minimum Required

Displacement	1569.15	T
KGO	3.578	M
GMt	6.766	M

0.150 M

Residual Arm Curve

Angle (Deg)	0	10	20	30	40	50
KN (M)	-0.006	1.611	2.215	2.459	2.528	2.487
GZ (M)	-0.006	0.984	0.985	0.665	0.223	-0.258

	Computed values	Minimum Required
Residual Area from abs 0 deg to MAXRA	0.1781 m-rad	0.080 m-rad
Angle from Equilibrium to RAZero or flood	44.69 m-rad	20.00 Deg.
Angle from Equilibrium to 50% DK Imm Angle	3.03 m-rad	0.00 Deg.

Plot GZ values against the angles of heel as above.

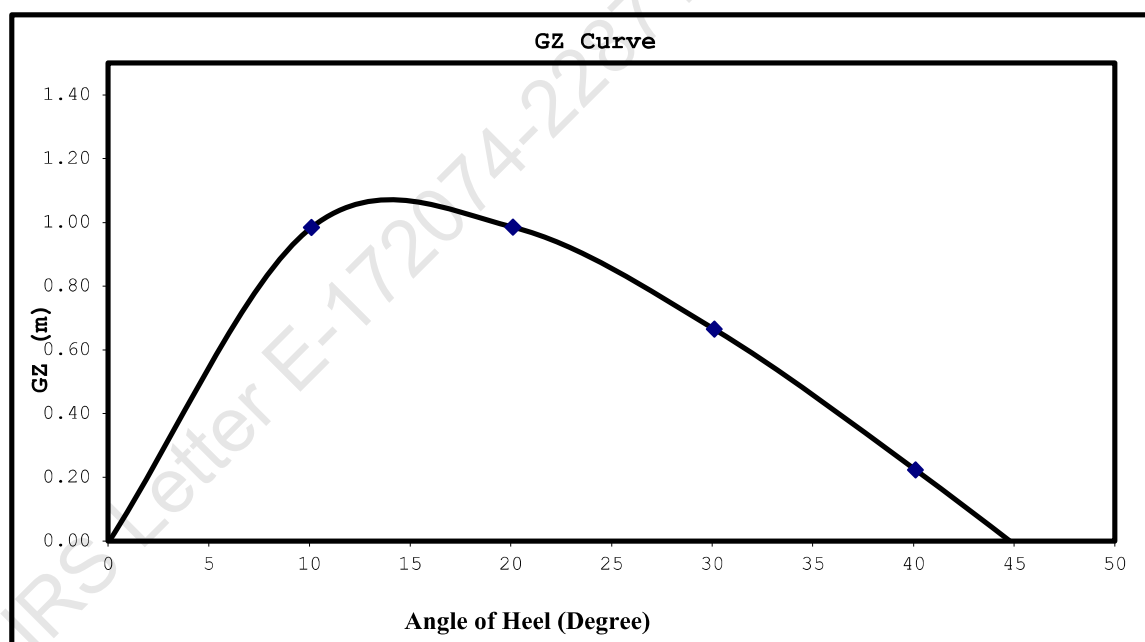
At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin.

This is the Tangent to the GZ Curve at the Origin. Find the Maxima of GZ

curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding

If maximum GZ occurs at an angle less than 30 degrees

then area upto this angle must be calculated.



Worked Out Example Contd.

Mean Draft	2.212	M
LCF from AP	25.000	M
LCB from AP	24.993	M
MCT	31.300	T-M/Cm
Trim	0.004	M
Draft (AP)	2.214	M
Draft (FP)	2.210	M
KG	3.461	M
KGo	3.578	M
GoMt	6.766	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (Ext) for the corresponding season, as per the assigned loadline.

BLANK CALCULATION SHEET

[illegible][illegible][illegible]

Free Surface Correction for VCG = FSM/Displacement

VCGc m
m

From the MaxVCG table,

Displacement T
Allowable Max VCG m

$$\boxed{\text{VCGc} < \text{Maxvcg}}$$

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement = T

Draft ,Tm @ 25.0f = m
LCB = m
LCF = m
MCT = T-m/cm
KMT = m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement / (MCTX100)
m, (+) is aft trim, (-) is fwd trim

Determining the Forward perpendicular, Midship and aft Perpendicular Draft

LBP = 50m
Draft @ 25.0 (MS), Tmid m

Draft @ 50.0f (FP), Tfwd =Tm - Trim/2
m

Draft@0 (AP), Taft =Tm + Trim/2
m

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floataction

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

<u>MULTIPLY BY</u>	<u>TO CONVERT FROM</u>	<u>TO OBTAIN</u>	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram's	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	<u>TO OBTAIN</u>	<u>TO CONVERT FROM</u>	<u>MULTIPLY</u>

Relation between Weight and volume

1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Angle of Downflooding

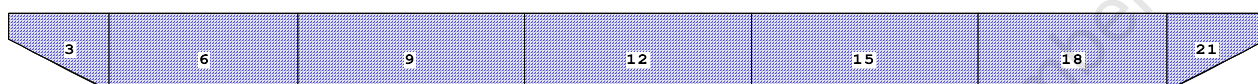
1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill of this opening is a 'potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

07-04-22 14:33:35 Cybermarine Technologies PTE. Ltd.
GHS 18.00 PONTON

CAPACITY PLAN AND TABLE

Profile View



Plan View

1	4	7	10	13	16	19
2	5	8	11	14	17	20
3	6	9	12	15	18	21

Tanks

1 01_VOIDTK.P.100% FRESH WATER	8 03_PFWBTK.C.100% FRESH WATER	16 06_VOIDTK.P.100% FRESH WATER
2 01_VOIDTK.C.100% FRESH WATER	9 03_VOIDTK.S.100% FRESH WATER	17 06_VOIDTK.C.100% FRESH WATER
3 01_VOIDTK.S.100% FRESH WATER	10 04_VOIDTK.P.100% FRESH WATER	18 06_VOIDTK.S.100% FRESH WATER
4 02_VOIDTK.P.100% FRESH WATER	11 04_VOIDTK.C.100% FRESH WATER	19 07_VOIDTK.P.100% FRESH WATER
5 02_VOIDTK.C.100% FRESH WATER	12 04_VOIDTK.S.100% FRESH WATER	20 07_VOIDTK.C.100% FRESH WATER
6 02_VOIDTK.S.100% FRESH WATER	13 05_VOIDTK.P.100% FRESH WATER	21 07_VOIDTK.S.100% FRESH WATER
7 03_VOIDTK.P.100% FRESH WATER	14 05_PFWBTK.C.100% FRESH WATER	
	15 05_VOIDTK.S.100% FRESH WATER	

07-04-22 14:33:35 Cybermarine Technologies PTE. Ltd.
GHS 18.00 PONTON

TANK STATUS

Trim: zero, Heel: zero

Part-----	Cu.M.----	SpGr----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
01_VOIDTK.P	39.1	1.000	39.13	2.339f	4.971p	1.914	40.83*
01_VOIDTK.C	39.6	1.000	39.61	2.335f	0.000	1.904	40.83*
01_VOIDTK.S	39.1	1.000	39.13	2.339f	4.971s	1.914	40.83*
02_VOIDTK.P	109.3	1.000	109.34	7.750f	4.981p	1.511	76.56*
02_VOIDTK.C	110.2	1.000	110.25	7.750f	0.000	1.500	76.56*
02_VOIDTK.S	109.3	1.000	109.34	7.750f	4.981s	1.511	76.56*
03_VOIDTK.P	131.2	1.000	131.21	16.000f	4.981p	1.511	91.88*
03_PFWBTK.C	132.3	1.000	132.30	16.000f	0.000	1.500	91.87*
03_VOIDTK.S	131.2	1.000	131.21	16.000f	4.981s	1.511	91.88*
04_VOIDTK.P	131.2	1.000	131.21	25.000f	4.981p	1.511	91.88*
04_VOIDTK.C	132.3	1.000	132.30	25.000f	0.000	1.500	91.87*
04_VOIDTK.S	131.2	1.000	131.21	25.000f	4.981s	1.511	91.88*
05_VOIDTK.P	131.2	1.000	131.21	34.000f	4.981p	1.511	91.88*
05_PFWBTK.C	132.3	1.000	132.30	34.000f	0.000	1.500	91.87*
05_VOIDTK.S	131.2	1.000	131.21	34.000f	4.981s	1.511	91.88*
06_VOIDTK.P	109.3	1.000	109.34	42.250f	4.981p	1.511	76.56*
06_VOIDTK.C	110.2	1.000	110.25	42.250f	0.000	1.500	76.56*
06_VOIDTK.S	109.3	1.000	109.34	42.250f	4.981s	1.511	76.56*
07_VOIDTK.P	39.1	1.000	39.13	47.661f	4.971p	1.914	40.83*
07_VOIDTK.C	39.6	1.000	39.61	47.665f	0.000	1.904	40.83*
07_VOIDTK.S	39.1	1.000	39.13	47.661f	4.971s	1.914	40.83*
Total Tanks----->			2,077.75	25.000f	0.000	1.553	1531.25
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

HYDROSTATIC PARTICULARS

15-02-19 10:56:08 Cybermarine Technologies PTE. Ltd.
GHS 16.50

HYDROSTATIC PROPERTIES

Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

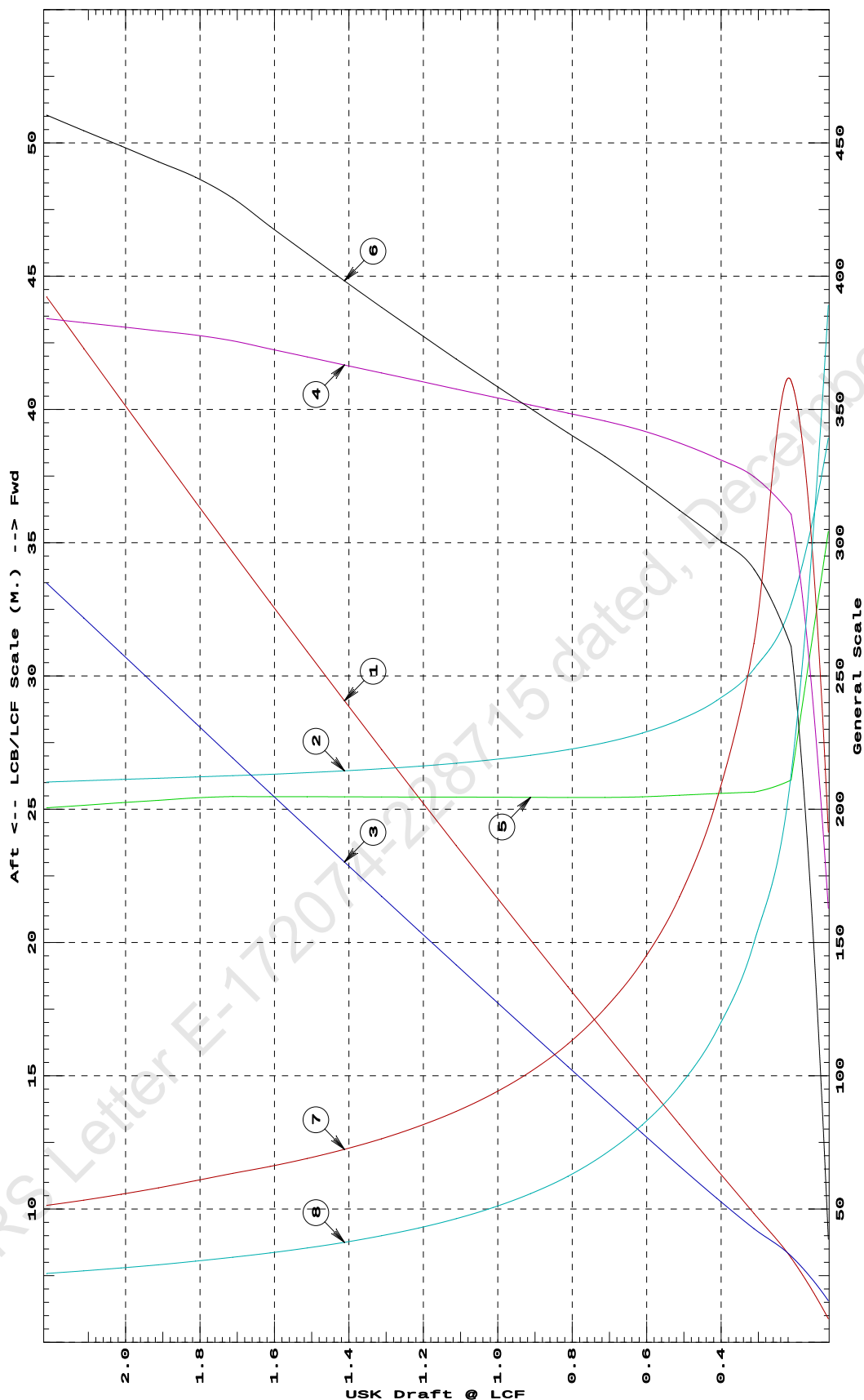
LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	35.31	38.933f	0.062	3.26	35.422f	2.70	382.72	155.640	
0.212	126.67	32.680f	0.129	6.21	26.091f	18.28	721.37	85.218	
0.312	193.60	30.258f	0.171	6.49	25.644f	20.31	524.43	59.944	
0.412	259.63	29.079f	0.216	6.63	25.587f	21.12	406.76	46.782	
0.512	326.98	28.354f	0.264	6.75	25.525f	21.85	334.10	38.458	
0.612	395.34	27.859f	0.314	6.84	25.465f	22.58	285.53	32.636	
0.712	464.24	27.501f	0.364	6.91	25.439f	23.27	250.56	28.275	
0.812	533.64	27.233f	0.414	6.97	25.443f	23.89	223.80	24.906	
0.912	603.65	27.026f	0.465	7.03	25.446f	24.52	203.11	22.305	
1.012	674.25	26.861f	0.515	7.09	25.451f	25.17	186.64	20.238	
1.112	745.46	26.726f	0.566	7.15	25.454f	25.83	173.21	18.554	
1.212	817.28	26.614f	0.618	7.21	25.457f	26.50	162.12	17.166	
1.312	889.68	26.520f	0.669	7.27	25.462f	27.18	152.75	15.996	
1.412	962.69	26.440f	0.721	7.33	25.465f	27.88	144.78	15.003	
1.512	1,036.38	26.371f	0.773	7.39	25.470f	28.59	137.93	14.149	
1.612	1,110.58	26.311f	0.825	7.45	25.473f	29.31	131.94	13.408	
1.712	1,185.40	26.258f	0.877	7.51	25.476f	30.05	126.74	12.763	
1.812	1,261.19	26.210f	0.929	7.56	25.427f	30.60	121.30	12.168	
1.912	1,337.62	26.163f	0.982	7.59	25.335f	31.00	115.89	11.627	
2.012	1,414.37	26.115f	1.034	7.62	25.243f	31.42	111.05	11.147	
2.112	1,491.44	26.068f	1.087	7.65	25.148f	31.82	106.66	10.716	
2.212	1,568.83	26.020f	1.139	7.68	25.053f	32.23	102.71	10.332	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

HYDROSTATIC PROPERTIES at 0.5 M. FWD TRIM



- 1 Displacement 1=4 MT
 2 LCB (use top scale)
 3 VCB (KB) 1=.004 M.
 4 Immersion 1=.02 MT/cm
 5 WPA 1=1.95 Sq.M.
 6 LCF (use top scale)
 7 Moment/Trim 1=.07 M.-MT/cm
 8 KML 1=2 M.
 KMT 1=.4 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
 Trim is per 50 M. "K" = BPL

15-02-19 10:56:08 Cybermarine Technologies PTE. Ltd.
GHS 16.50

HYDROSTATIC PROPERTIES

Trim: Fwd 0.250/50.000, No Heel, Fixed VCG = 0.000

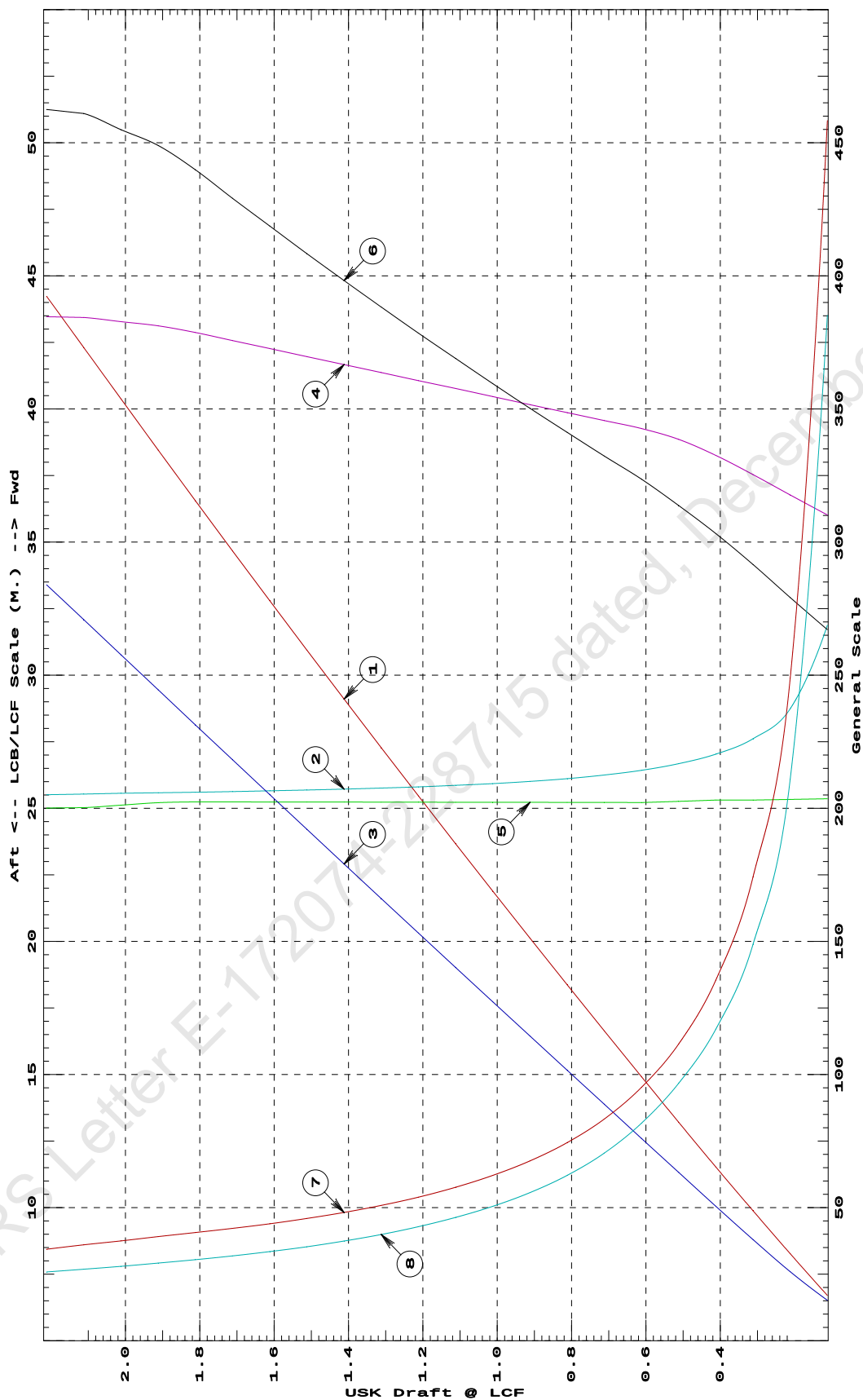
LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT-----	
0.112	67.99	31.878f	0.061	6.20	25.355f	18.70	1375.02	154.077	
0.212	130.85	28.739f	0.104	6.35	25.333f	19.54	746.69	84.364	
0.312	195.25	27.612f	0.152	6.51	25.309f	20.41	522.70	59.562	
0.412	261.01	27.032f	0.202	6.65	25.309f	21.22	406.53	46.823	
0.512	328.33	26.675f	0.253	6.77	25.258f	21.96	334.41	38.687	
0.612	396.62	26.426f	0.304	6.85	25.218f	22.65	285.59	32.693	
0.712	465.44	26.248f	0.355	6.91	25.219f	23.26	249.88	28.193	
0.812	534.86	26.114f	0.406	6.97	25.221f	23.89	223.30	24.847	
0.912	604.89	26.011f	0.458	7.03	25.223f	24.52	202.66	22.253	
1.012	675.51	25.929f	0.509	7.09	25.225f	25.16	186.26	20.193	
1.112	746.74	25.862f	0.561	7.15	25.227f	25.83	172.93	18.521	
1.212	818.57	25.806f	0.613	7.21	25.229f	26.49	161.82	17.132	
1.312	891.00	25.759f	0.664	7.27	25.230f	27.18	152.52	15.970	
1.412	964.03	25.719f	0.716	7.33	25.233f	27.88	144.57	14.979	
1.512	1,037.66	25.685f	0.768	7.39	25.234f	28.58	137.73	14.127	
1.612	1,111.90	25.655f	0.821	7.45	25.237f	29.31	131.80	13.391	
1.712	1,186.73	25.629f	0.873	7.51	25.238f	30.04	126.56	12.744	
1.812	1,262.18	25.605f	0.925	7.57	25.240f	30.79	121.97	12.177	
1.912	1,338.33	25.584f	0.978	7.62	25.209f	31.43	117.41	11.662	
2.012	1,415.08	25.561f	1.031	7.66	25.117f	31.84	112.52	11.180	
2.112	1,492.21	25.536f	1.083	7.69	25.024f	32.27	108.13	10.753	
2.212	1,569.14	25.510f	1.136	7.69	25.009f	32.37	103.16	10.342	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

HYDROSTATIC PROPERTIES at 0.25 M. FWD TRIM



- ① Displacement 1=4 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.004 M.
- ④ Immersion 1=.02 MT/cm
- ④ WPA 1=1.95 Sq.M.
- ⑤ LCF (use top scale)
- ⑥ Moment/Trim 1=.07 M.-MT/cm
- ⑦ KML 1=3 M.
- ⑧ KMT 1=.4 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
 Trim is per 50 M. "K" = BPL

15-02-19 10:56:08 Cybermarine Technologies PTE. Ltd.
GHS 16.50

HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

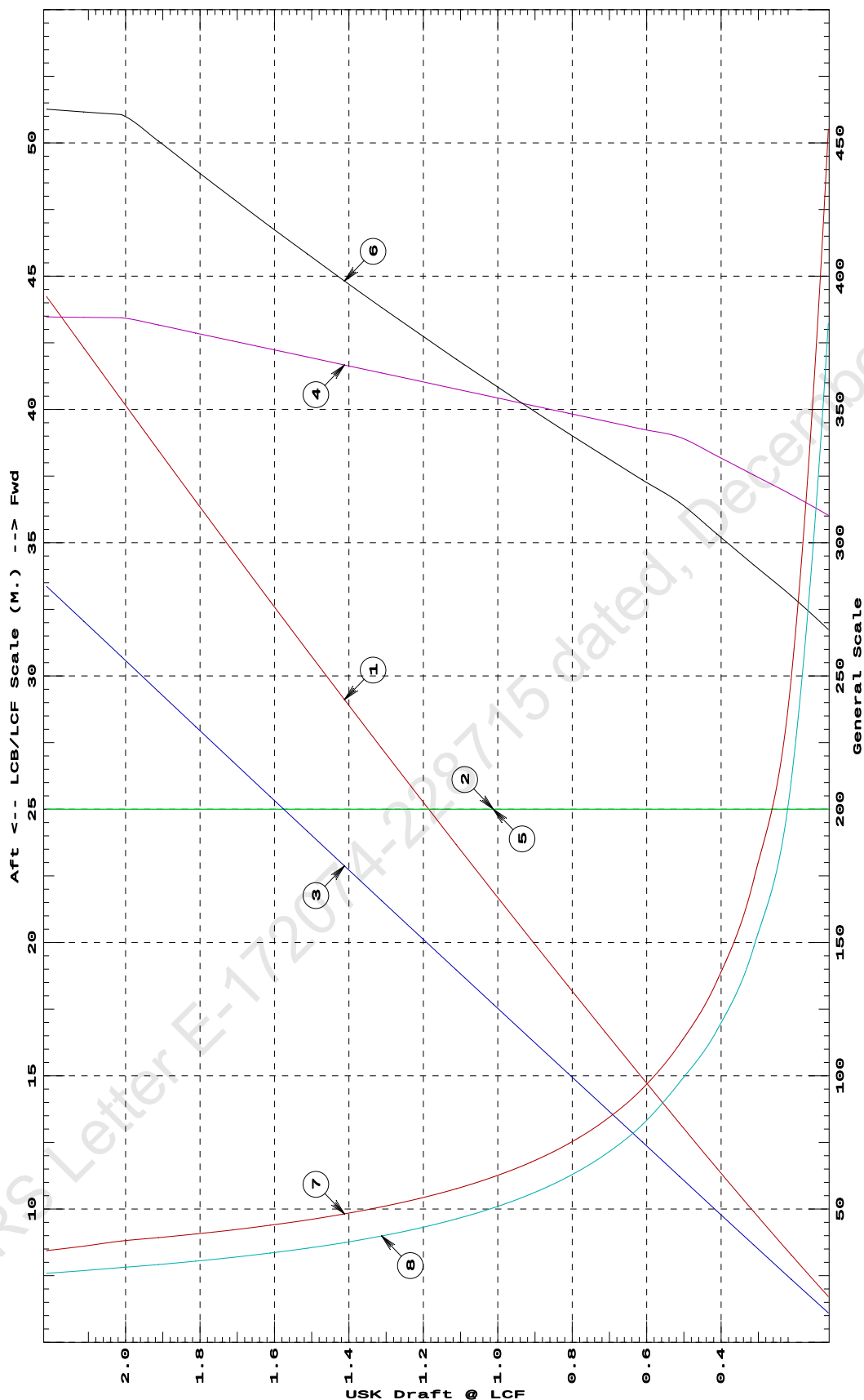
LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	68.53	25.000f	0.044	6.21	25.000f	18.72	1365.62	152.943	
0.212	131.38	25.000f	0.095	6.36	25.000f	19.61	746.26	84.110	
0.312	195.72	25.000f	0.146	6.51	25.000f	20.41	521.46	59.400	
0.412	261.52	25.000f	0.197	6.65	25.000f	21.24	406.00	46.750	
0.512	328.77	25.000f	0.249	6.79	25.000f	22.05	335.40	38.975	
0.612	396.99	25.000f	0.301	6.85	25.000f	22.65	285.27	32.654	
0.712	465.83	25.000f	0.352	6.91	25.000f	23.27	249.73	28.175	
0.812	535.26	25.000f	0.404	6.97	25.000f	23.88	223.09	24.822	
0.912	605.29	25.000f	0.456	7.03	25.000f	24.52	202.53	22.237	
1.012	675.92	25.000f	0.507	7.09	25.000f	25.16	186.15	20.181	
1.112	747.15	25.000f	0.559	7.15	25.000f	25.82	172.79	18.506	
1.212	818.99	25.000f	0.611	7.21	25.000f	26.50	161.77	17.125	
1.312	891.43	25.000f	0.663	7.27	25.000f	27.18	152.43	15.959	
1.412	964.46	25.000f	0.715	7.33	25.000f	27.87	144.50	14.970	
1.512	1,038.11	25.000f	0.767	7.39	25.000f	28.59	137.68	14.122	
1.612	1,112.35	25.000f	0.819	7.45	25.000f	29.30	131.72	13.382	
1.712	1,187.19	25.000f	0.872	7.51	25.000f	30.04	126.53	12.740	
1.812	1,262.64	25.000f	0.924	7.57	25.000f	30.79	121.92	12.171	
1.912	1,338.68	25.000f	0.977	7.63	25.000f	31.55	117.84	11.669	
2.012	1,415.33	25.000f	1.029	7.69	25.000f	32.24	113.91	11.217	
2.112	1,492.22	25.000f	1.082	7.69	25.000f	32.31	108.27	10.755	
2.212	1,569.15	25.000f	1.134	7.69	25.000f	32.38	103.19	10.344	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

HYDROSTATIC PROPERTIES at LEVEL TRIM



- ① Displacement 1=4 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.004 M.
- ④ Immersion 1=.02 MT/cm
- ④ WPA 1=1.95 Sq.M.
- ⑤ LCF (use top scale)
- ⑥ Moment/Trim 1=.07 M.-MT/cm
- ⑦ KML 1=3 M.
- ⑧ KMT 1=.4 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
 Trim is per 50 M. "K" = BPL

15-02-19 10:56:08 Cybermarine Technologies PTE. Ltd.
GHS 16.50

HYDROSTATIC PROPERTIES

Trim: Aft 0.250/50.000, No Heel, Fixed VCG = 0.000

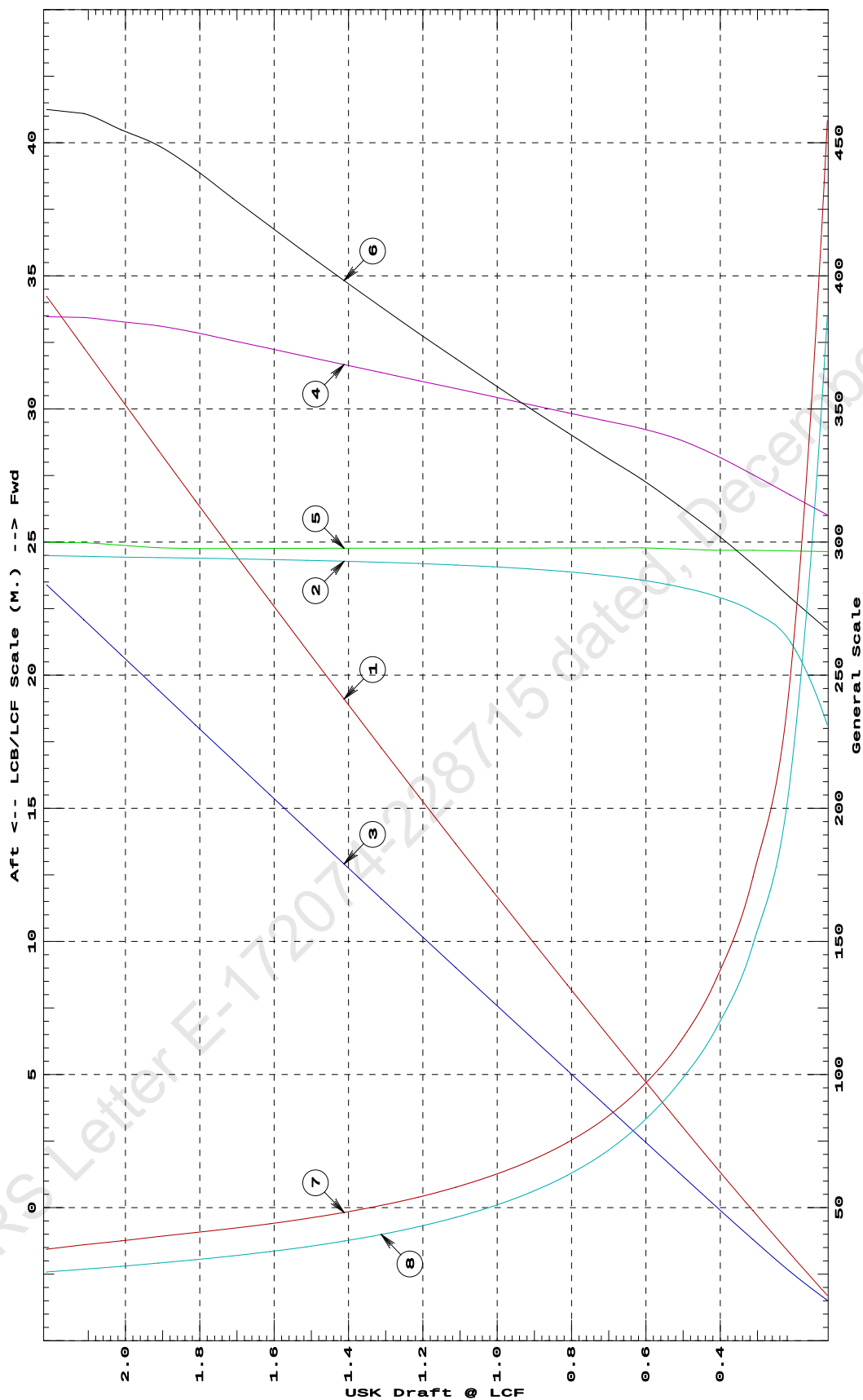
LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT-----	
0.112	67.99	18.121f	0.061	6.20	24.645f	18.70	1375.05	154.081	
0.212	130.84	21.260f	0.104	6.35	24.667f	19.54	746.72	84.368	
0.312	195.22	22.387f	0.152	6.51	24.691f	20.41	522.75	59.569	
0.412	260.99	22.967f	0.202	6.65	24.691f	21.22	406.56	46.826	
0.512	328.32	23.325f	0.253	6.77	24.742f	21.96	334.41	38.687	
0.612	396.63	23.574f	0.304	6.85	24.782f	22.65	285.59	32.692	
0.712	465.44	23.752f	0.355	6.91	24.781f	23.26	249.88	28.193	
0.812	534.88	23.886f	0.406	6.97	24.779f	23.89	223.29	24.846	
0.912	604.91	23.989f	0.458	7.03	24.777f	24.52	202.66	22.253	
1.012	675.53	24.071f	0.509	7.09	24.775f	25.16	186.25	20.192	
1.112	746.76	24.138f	0.561	7.15	24.773f	25.83	172.92	18.521	
1.212	818.59	24.194f	0.613	7.21	24.771f	26.49	161.82	17.132	
1.312	891.02	24.241f	0.664	7.27	24.770f	27.18	152.52	15.969	
1.412	964.05	24.281f	0.716	7.33	24.767f	27.88	144.57	14.978	
1.512	1,037.68	24.315f	0.768	7.39	24.766f	28.58	137.72	14.127	
1.612	1,111.92	24.345f	0.821	7.45	24.764f	29.31	131.80	13.391	
1.712	1,186.75	24.371f	0.873	7.51	24.762f	30.04	126.56	12.744	
1.812	1,262.19	24.395f	0.925	7.57	24.760f	30.79	121.97	12.177	
1.912	1,338.33	24.416f	0.978	7.62	24.791f	31.43	117.42	11.662	
2.012	1,415.08	24.439f	1.031	7.66	24.883f	31.84	112.52	11.180	
2.112	1,492.15	24.464f	1.083	7.69	24.976f	32.27	108.13	10.753	
2.212	1,569.13	24.490f	1.136	7.69	24.991f	32.37	103.16	10.342	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

HYDROSTATIC PROPERTIES at 0.25 M. AFT TRIM



- ① Displacement 1=4 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.004 M.
- ④ Immersion 1=.02 MT/cm
- ④ WPA 1=1.95 Sq.M.
- ⑤ LCF (use top scale)
- ⑥ Moment/Trim 1=.07 M.-MT/cm
- ⑦ KML 1=3 M.
- ⑧ KMT 1=.4 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
 Trim is per 50 M. "K" = BPL

15-02-19 10:56:08 Cybermarine Technologies PTE. Ltd.
GHS 16.50

HYDROSTATIC PROPERTIES

Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

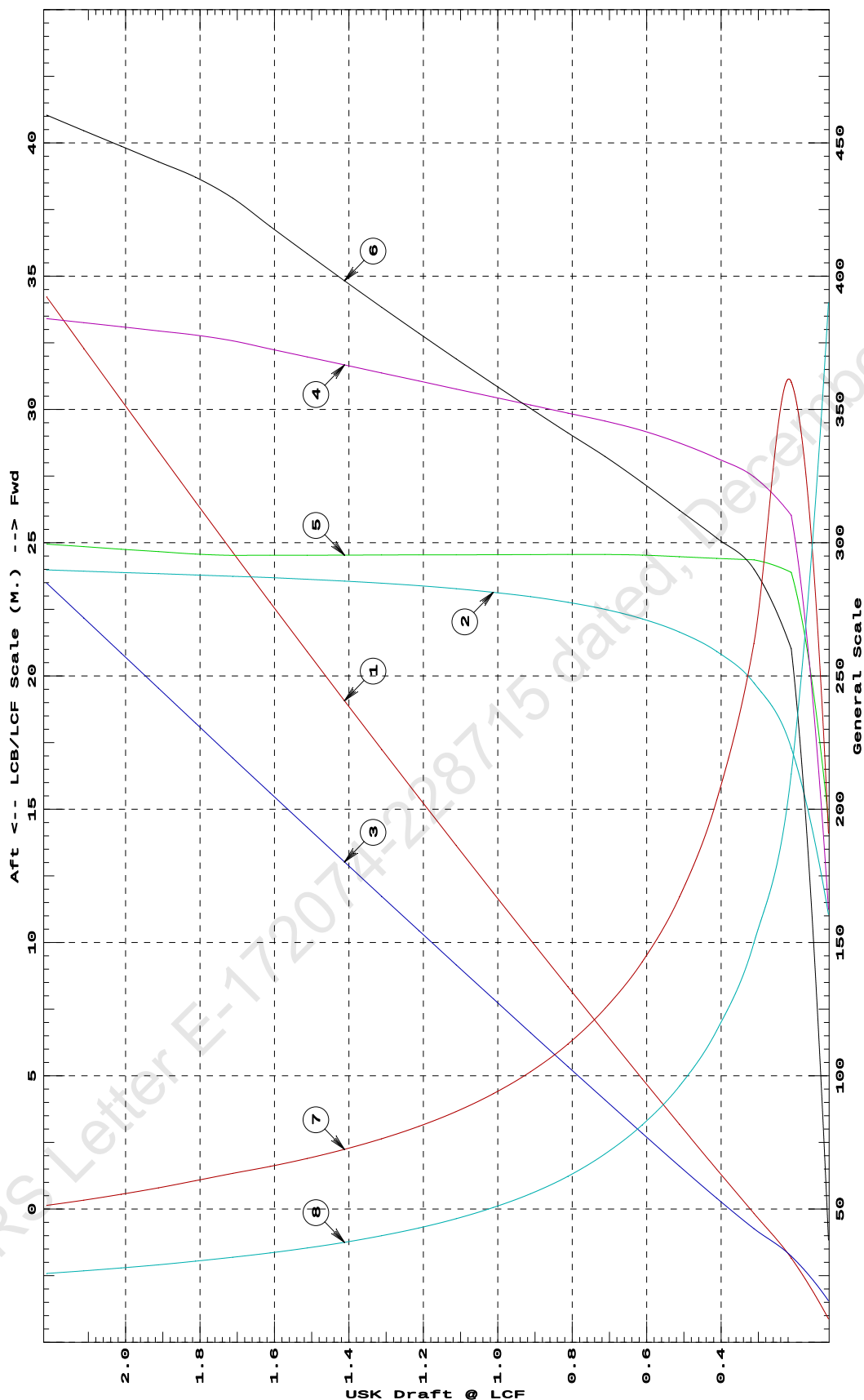
LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT-----	
0.112	35.14	11.050f	0.062	3.25	14.553f	2.68	381.82	155.997	
0.212	126.38	17.305f	0.129	6.21	23.886f	18.21	720.54	85.309	
0.312	193.59	19.742f	0.171	6.49	24.356f	20.31	524.44	59.946	
0.412	259.63	20.921f	0.216	6.63	24.413f	21.12	406.77	46.784	
0.512	326.97	21.646f	0.264	6.75	24.475f	21.85	334.11	38.459	
0.612	395.34	22.141f	0.314	6.84	24.535f	22.58	285.54	32.637	
0.712	464.31	22.499f	0.364	6.91	24.561f	23.27	250.53	28.272	
0.812	533.71	22.767f	0.414	6.97	24.557f	23.89	223.77	24.904	
0.912	603.73	22.974f	0.465	7.03	24.554f	24.52	203.09	22.303	
1.012	674.39	23.140f	0.515	7.09	24.549f	25.17	186.61	20.234	
1.112	745.60	23.274f	0.567	7.15	24.546f	25.83	173.19	18.551	
1.212	817.41	23.386f	0.618	7.21	24.543f	26.50	162.10	17.163	
1.312	889.82	23.480f	0.669	7.27	24.538f	27.18	152.73	15.994	
1.412	962.83	23.560f	0.721	7.33	24.535f	27.88	144.77	15.001	
1.512	1,036.44	23.629f	0.773	7.39	24.530f	28.59	137.92	14.149	
1.612	1,110.65	23.689f	0.825	7.45	24.527f	29.31	131.94	13.407	
1.712	1,185.47	23.742f	0.877	7.51	24.524f	30.05	126.73	12.762	
1.812	1,261.17	23.790f	0.929	7.56	24.573f	30.60	121.30	12.168	
1.912	1,337.60	23.837f	0.982	7.59	24.665f	31.00	115.89	11.627	
2.012	1,414.34	23.885f	1.034	7.62	24.757f	31.42	111.06	11.148	
2.112	1,491.42	23.932f	1.087	7.65	24.852f	31.82	106.67	10.716	
2.212	1,568.81	23.980f	1.139	7.68	24.947f	32.23	102.71	10.332	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

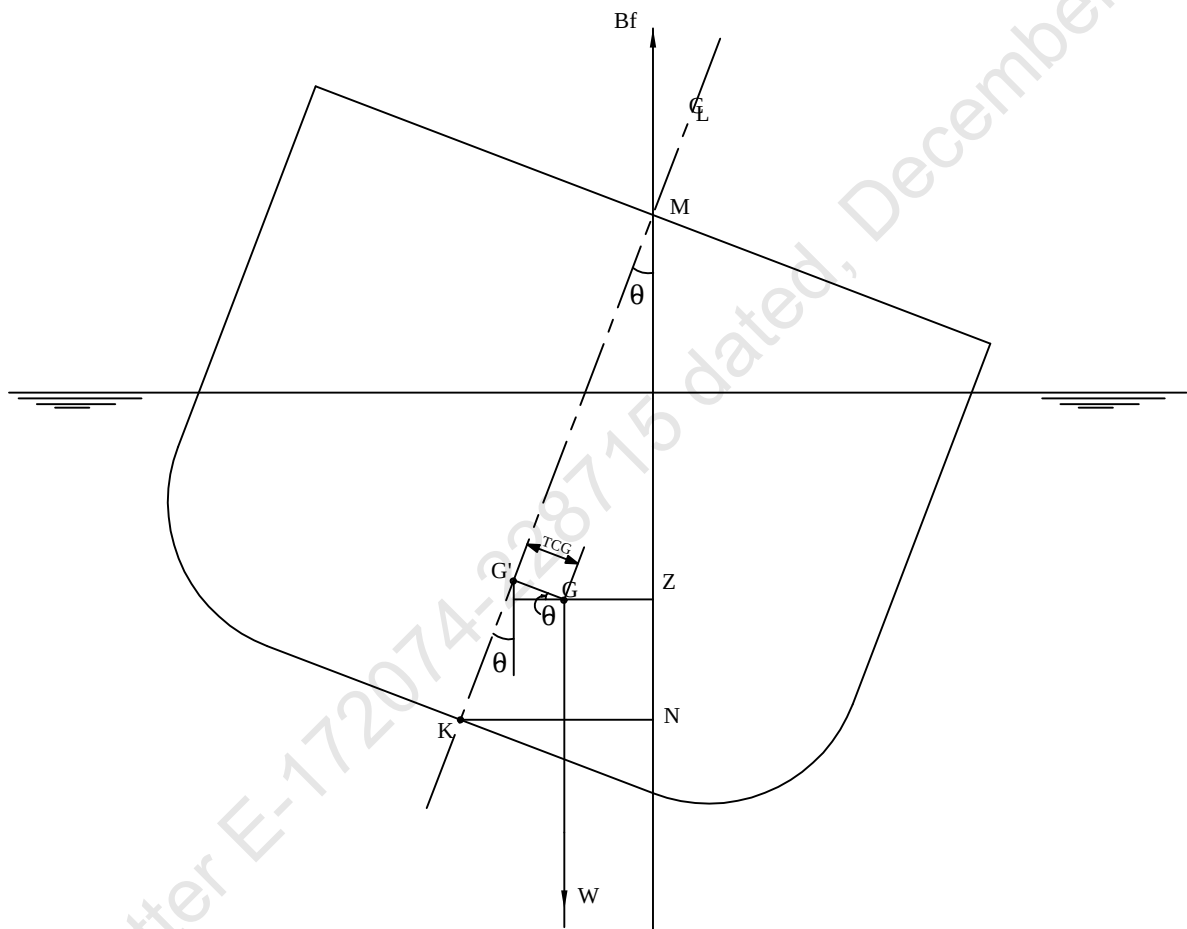
HYDROSTATIC PROPERTIES at 0.5 M. AFT TRIM



- ① Displacement 1=4 MT
 ② LCB (use top scale)
 ③ VCB (KB) 1=.004 M.
 ④ Immersion 1=.02 MT/cm
 ⑤ WPA 1=1.95 Sq.M.
 ⑥ LCF (use top scale)
 ⑦ Moment/Trim 1=.07 M.-MT/cm
 ⑧ KML 1=2 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
 Trim is per 50 M. "K" = BPL

KN CONCEPT



$$\underline{GZ = KN - KG' \sin \theta - TCG \cos \theta}$$

Profile View

VOLUME USED FOR KN COMPUTATION



CROSS CURVE TABLES

15-02-19 10:56:39 Cybermarine Technologies PTE. Ltd.

GHS 16.50

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

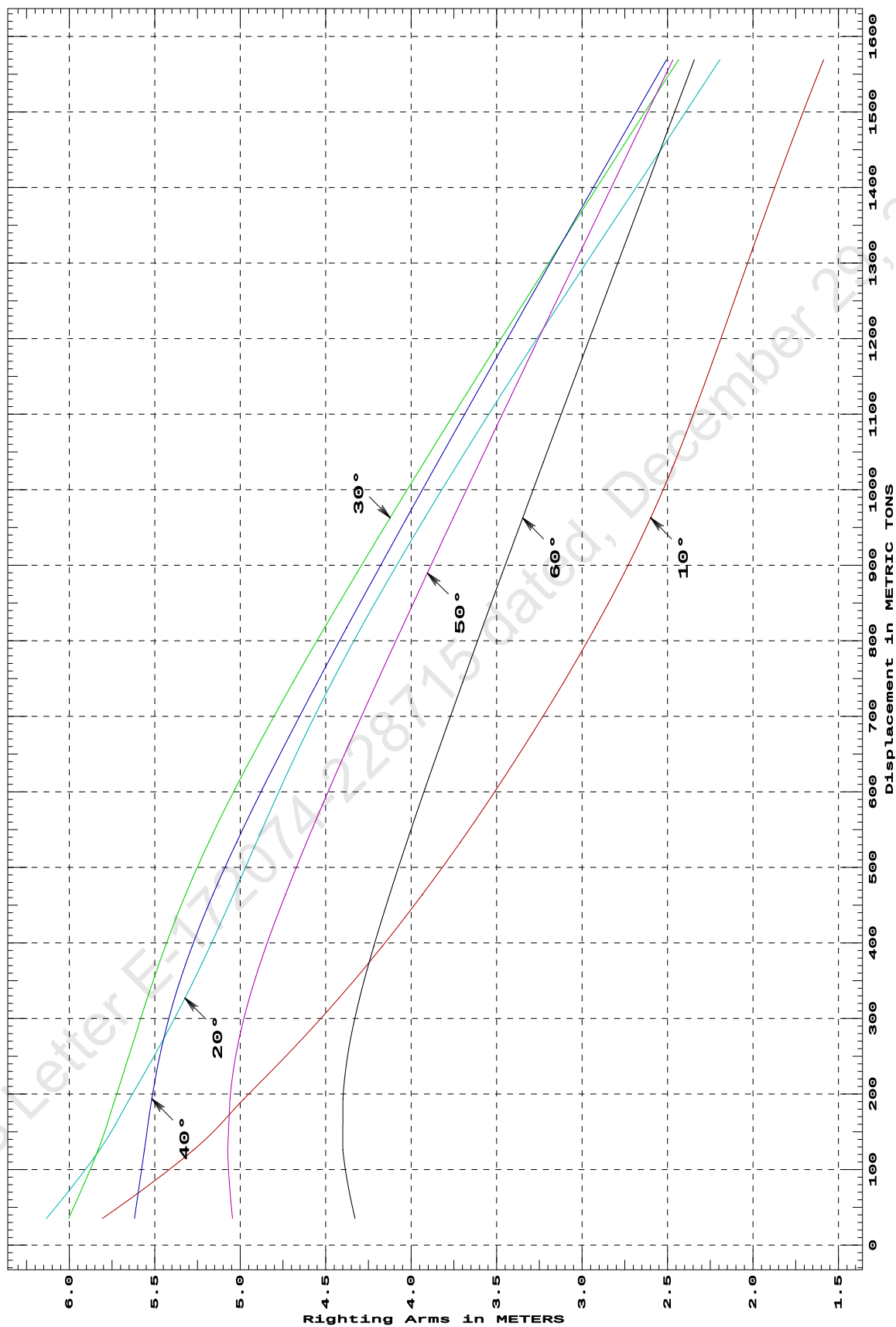
Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
35.31	5.807s	6.134s	6.004s	5.619s	5.045s	4.329s
126.67	5.270s	5.821s	5.830s	5.560s	5.073s	4.400s
193.60	4.978s	5.647s	5.732s	5.517s	5.060s	4.400s
259.63	4.693s	5.480s	5.639s	5.462s	5.019s	4.366s
326.98	4.423s	5.323s	5.543s	5.384s	4.947s	4.301s
395.34	4.170s	5.176s	5.439s	5.282s	4.849s	4.219s
464.24	3.934s	5.039s	5.319s	5.159s	4.735s	4.126s
533.64	3.712s	4.905s	5.183s	5.020s	4.611s	4.025s
603.65	3.501s	4.767s	5.030s	4.870s	4.479s	3.920s
674.25	3.300s	4.621s	4.865s	4.712s	4.342s	3.811s
745.46	3.107s	4.463s	4.690s	4.547s	4.200s	3.698s
817.28	2.924s	4.293s	4.507s	4.377s	4.054s	3.584s
889.68	2.754s	4.111s	4.318s	4.203s	3.905s	3.467s
962.69	2.599s	3.920s	4.123s	4.026s	3.754s	3.348s
1,036.38	2.458s	3.722s	3.924s	3.844s	3.600s	3.227s
1,110.58	2.331s	3.516s	3.721s	3.661s	3.444s	3.105s
1,185.40	2.211s	3.305s	3.514s	3.474s	3.286s	2.981s
1,261.19	2.093s	3.088s	3.303s	3.284s	3.125s	2.855s
1,337.62	1.973s	2.867s	3.089s	3.091s	2.962s	2.728s
1,414.37	1.850s	2.642s	2.873s	2.898s	2.799s	2.601s
1,491.44	1.721s	2.418s	2.656s	2.703s	2.634s	2.473s
1,568.83	1.588s	2.195s	2.436s	2.507s	2.469s	2.344s
Distances in METERS.---Specific Gravity = 1.025.-----						

15-02-19 10:56:39

Cybermarine Technologies PTE. Ltd.

GHS 16.50

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. FWD TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

15-02-19 10:56:39 Cybermarine Technologies PTE. Ltd.
GHS 16.50

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Fwd 0.250/50.000 at zero heel (trim righting arm held at zero)

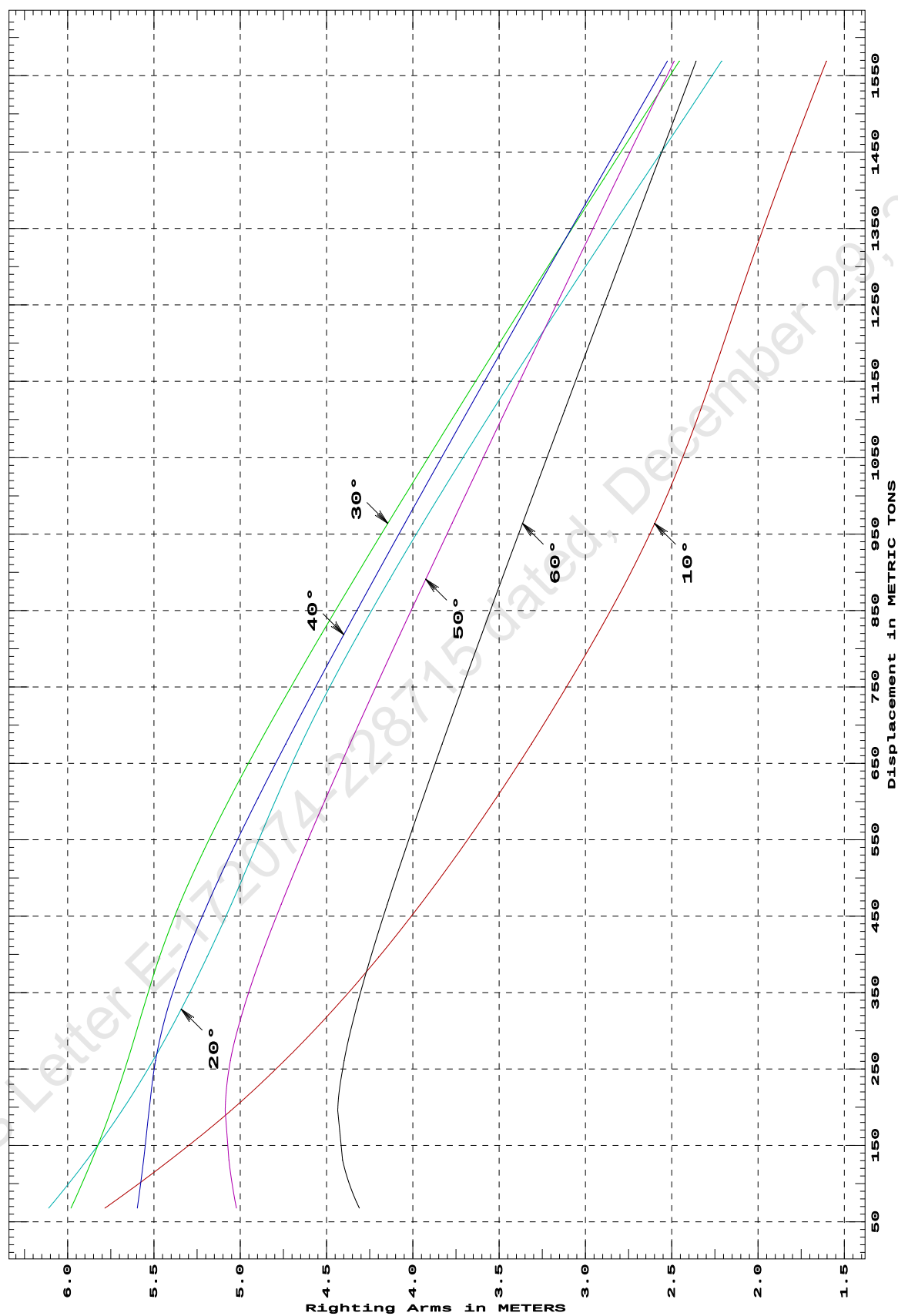
Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
67.99	5.785s	6.110s	5.981s	5.596s	5.022s	4.310s
130.85	5.407s	5.889s	5.859s	5.560s	5.066s	4.407s
195.25	5.056s	5.686s	5.750s	5.529s	5.087s	4.435s
261.01	4.744s	5.506s	5.653s	5.490s	5.057s	4.398s
328.33	4.459s	5.342s	5.563s	5.421s	4.981s	4.329s
396.62	4.197s	5.192s	5.468s	5.317s	4.880s	4.244s
465.44	3.956s	5.053s	5.352s	5.190s	4.763s	4.149s
534.86	3.730s	4.921s	5.214s	5.048s	4.637s	4.047s
604.89	3.516s	4.789s	5.058s	4.896s	4.503s	3.940s
675.51	3.313s	4.647s	4.891s	4.737s	4.364s	3.830s
746.74	3.119s	4.489s	4.714s	4.571s	4.221s	3.717s
818.57	2.932s	4.317s	4.530s	4.400s	4.075s	3.601s
891.00	2.757s	4.135s	4.340s	4.225s	3.925s	3.484s
964.03	2.599s	3.943s	4.145s	4.046s	3.773s	3.365s
1,037.66	2.459s	3.744s	3.945s	3.865s	3.619s	3.244s
1,111.90	2.334s	3.538s	3.742s	3.680s	3.462s	3.121s
1,186.73	2.219s	3.326s	3.534s	3.493s	3.303s	2.997s
1,262.18	2.107s	3.110s	3.324s	3.303s	3.143s	2.871s
1,338.33	1.989s	2.888s	3.110s	3.111s	2.980s	2.744s
1,415.08	1.865s	2.663s	2.893s	2.917s	2.816s	2.616s
1,492.21	1.736s	2.435s	2.675s	2.721s	2.651s	2.487s
1,569.14	1.604s	2.210s	2.456s	2.525s	2.486s	2.358s
Distances in METERS.---Specific Gravity = 1.025.-----						

15-02-19 10:56:39

Cybermarine Technologies PTE. Ltd.

GHS 16.50

CROSS CURVES OF STABILITY - Stbd Heel
at 0.25 M. FWD TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

15-02-19 10:56:39 Cybermarine Technologies PTE. Ltd.
GHS 16.50

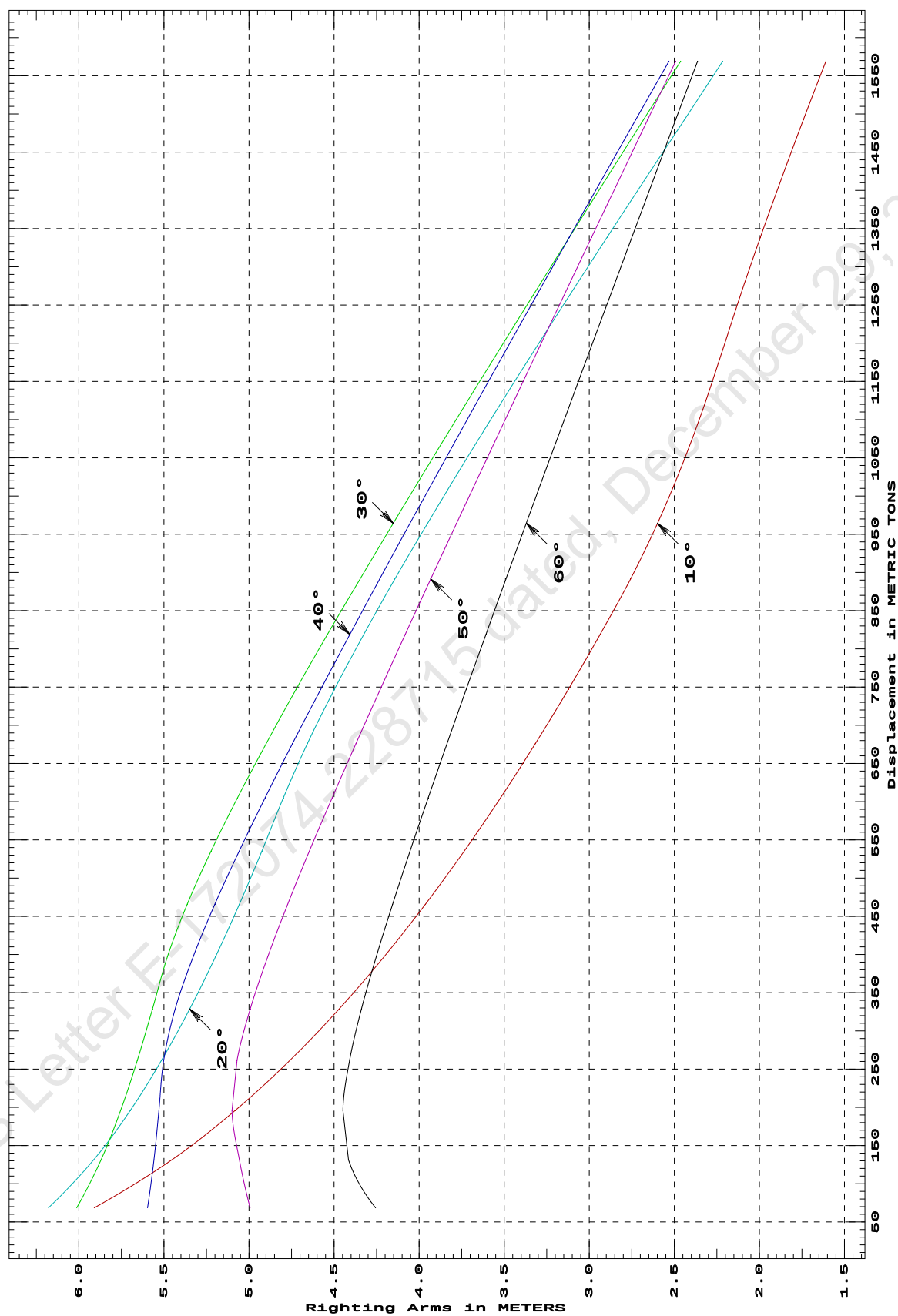
CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
68.53	5.910s	6.179s	6.013s	5.596s	4.993s	4.256s
131.38	5.453s	5.913s	5.870s	5.558s	5.057s	4.416s
195.72	5.082s	5.699s	5.755s	5.529s	5.100s	4.448s
261.52	4.760s	5.514s	5.657s	5.502s	5.071s	4.409s
328.77	4.471s	5.349s	5.568s	5.434s	4.993s	4.339s
396.99	4.206s	5.197s	5.480s	5.328s	4.890s	4.252s
465.83	3.963s	5.057s	5.363s	5.200s	4.772s	4.156s
535.26	3.735s	4.925s	5.223s	5.058s	4.645s	4.054s
605.29	3.521s	4.798s	5.067s	4.905s	4.511s	3.947s
675.92	3.317s	4.657s	4.899s	4.745s	4.372s	3.836s
747.15	3.123s	4.498s	4.723s	4.579s	4.228s	3.723s
818.99	2.936s	4.326s	4.539s	4.407s	4.082s	3.607s
891.43	2.757s	4.143s	4.347s	4.232s	3.932s	3.490s
964.46	2.599s	3.950s	4.152s	4.053s	3.780s	3.370s
1,038.11	2.459s	3.751s	3.952s	3.871s	3.625s	3.249s
1,112.35	2.334s	3.545s	3.748s	3.687s	3.468s	3.126s
1,187.19	2.222s	3.333s	3.541s	3.499s	3.309s	3.002s
1,262.64	2.112s	3.116s	3.330s	3.310s	3.149s	2.877s
1,338.68	1.994s	2.895s	3.117s	3.117s	2.986s	2.750s
1,415.33	1.871s	2.670s	2.900s	2.923s	2.822s	2.621s
1,492.22	1.742s	2.441s	2.682s	2.728s	2.657s	2.493s
1,569.15	1.609s	2.216s	2.463s	2.531s	2.492s	2.363s
Distances in METERS.---Specific Gravity = 1.025.-----						

CROSS CURVES OF STABILITY - Stbd Heel
at LEVEL TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

15-02-19 10:56:39 Cybermarine Technologies PTE. Ltd.

GHS 16.50

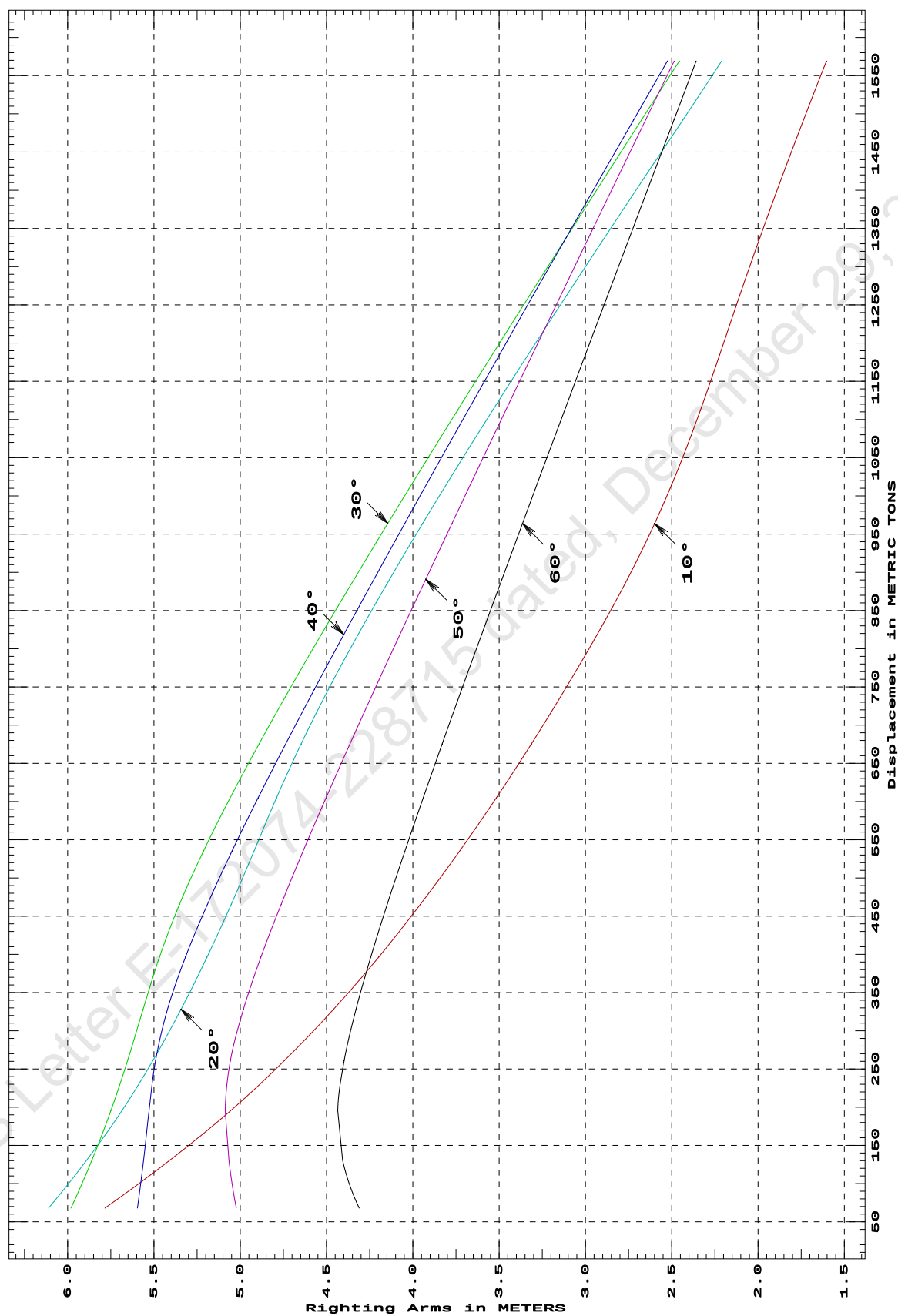
CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Aft 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
67.99	5.784s	6.111s	5.981s	5.596s	5.023s	4.311s
130.84	5.407s	5.889s	5.860s	5.560s	5.066s	4.407s
195.22	5.057s	5.686s	5.750s	5.529s	5.087s	4.435s
260.99	4.744s	5.506s	5.653s	5.490s	5.057s	4.398s
328.32	4.459s	5.343s	5.563s	5.421s	4.981s	4.329s
396.63	4.197s	5.192s	5.468s	5.317s	4.880s	4.244s
465.44	3.956s	5.053s	5.352s	5.190s	4.763s	4.149s
534.88	3.730s	4.921s	5.214s	5.048s	4.637s	4.047s
604.91	3.516s	4.790s	5.058s	4.896s	4.503s	3.940s
675.53	3.313s	4.647s	4.891s	4.737s	4.364s	3.830s
746.76	3.119s	4.489s	4.714s	4.571s	4.221s	3.717s
818.59	2.932s	4.317s	4.530s	4.400s	4.075s	3.601s
891.02	2.757s	4.135s	4.340s	4.225s	3.925s	3.484s
964.05	2.599s	3.943s	4.145s	4.046s	3.773s	3.365s
1,037.68	2.459s	3.743s	3.945s	3.865s	3.619s	3.243s
1,111.92	2.334s	3.538s	3.742s	3.680s	3.462s	3.121s
1,186.75	2.219s	3.326s	3.534s	3.493s	3.303s	2.997s
1,262.19	2.107s	3.110s	3.324s	3.303s	3.143s	2.871s
1,338.33	1.989s	2.888s	3.110s	3.111s	2.980s	2.744s
1,415.08	1.865s	2.663s	2.893s	2.917s	2.816s	2.616s
1,492.15	1.736s	2.435s	2.675s	2.721s	2.651s	2.487s
1,569.13	1.604s	2.210s	2.456s	2.525s	2.486s	2.358s
Distances in METERS.---Specific Gravity = 1.025.-----						

CROSS CURVES OF STABILITY - Stbd Heel
at 0.25 M. AFT TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

15-02-19 10:56:39 Cybermarine Technologies PTE. Ltd.

GHS 16.50

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

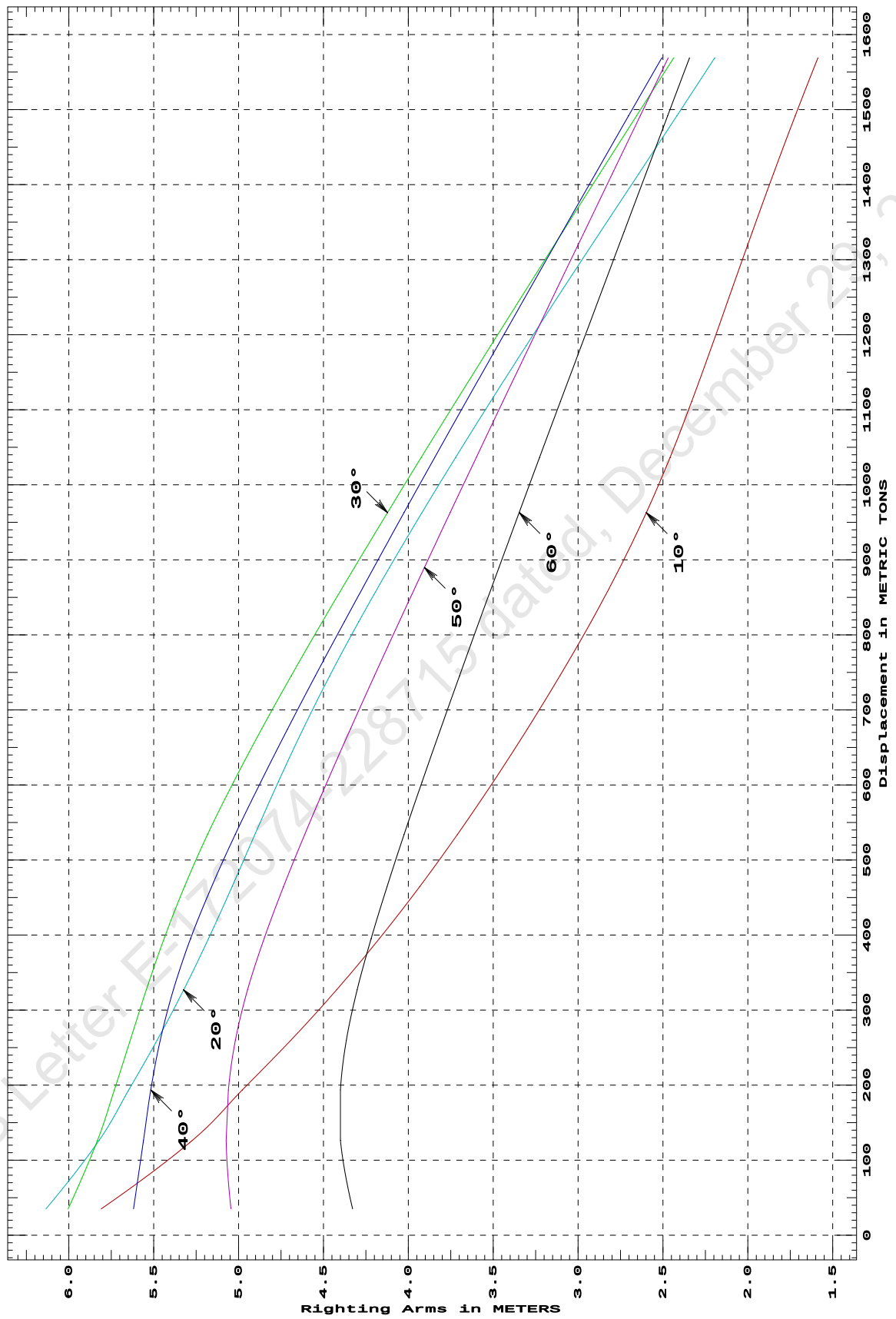
Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
35.14	5.809s	6.135s	6.004s	5.619s	5.045s	4.328s
126.38	5.272s	5.822s	5.831s	5.560s	5.073s	4.400s
193.59	4.978s	5.647s	5.732s	5.517s	5.060s	4.400s
259.63	4.693s	5.481s	5.639s	5.462s	5.019s	4.366s
326.97	4.423s	5.324s	5.543s	5.384s	4.947s	4.301s
395.34	4.170s	5.177s	5.439s	5.282s	4.849s	4.219s
464.31	3.934s	5.039s	5.319s	5.159s	4.735s	4.125s
533.71	3.712s	4.905s	5.183s	5.020s	4.611s	4.025s
603.73	3.501s	4.767s	5.030s	4.870s	4.479s	3.920s
674.39	3.300s	4.621s	4.864s	4.711s	4.341s	3.810s
745.60	3.107s	4.463s	4.689s	4.547s	4.199s	3.698s
817.41	2.924s	4.292s	4.506s	4.377s	4.054s	3.583s
889.82	2.754s	4.111s	4.317s	4.203s	3.905s	3.466s
962.83	2.598s	3.920s	4.123s	4.025s	3.753s	3.347s
1,036.44	2.458s	3.721s	3.924s	3.844s	3.600s	3.227s
1,110.65	2.331s	3.516s	3.721s	3.660s	3.443s	3.105s
1,185.47	2.211s	3.305s	3.514s	3.474s	3.285s	2.981s
1,261.17	2.093s	3.088s	3.303s	3.284s	3.125s	2.855s
1,337.60	1.973s	2.867s	3.089s	3.091s	2.962s	2.728s
1,414.34	1.850s	2.643s	2.873s	2.898s	2.799s	2.601s
1,491.42	1.721s	2.418s	2.656s	2.703s	2.634s	2.473s
1,568.81	1.588s	2.195s	2.437s	2.507s	2.469s	2.344s
Distances in METERS.---Specific Gravity = 1.025.-----						

15-02-19 10:56:39

Cybermarine Technologies PTE. Ltd.

GHS 16.50

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. AFT TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

DEADWEIGHT TABLE

Draft M	Disp T	Dwt T
0.612	396.99	23.62
0.712	465.83	92.46
0.812	535.26	161.89
0.912	605.29	231.92
1.012	675.92	302.55
1.112	747.15	373.78
1.212	818.99	445.62
1.312	891.43	518.06
1.412	964.46	591.09
1.512	1038.11	664.74
1.612	1112.35	738.98
1.712	1187.19	813.82
1.812	1262.64	889.27
1.912	1338.68	965.31
2.012	1415.33	1041.96
2.112	1492.22	1118.85
2.212	1569.15	1195.78

Full Load Displacement

15-02-19 10:57:40 Cybermarine Technologies PTE. Ltd.
GHS 16.50

MAXIMUM VCG vs. DISPLACEMENT

Heeling moment is present from: wind

Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
300.00	15.372	94%	0d	12d	
500.00	14.274	47%	0d	9d	
750.00	12.752	16%	0d	6d	
1,000.00	10.916	0%	0d	5d	
1,250.00	8.412	0%	2d	3d	
1,500.00	5.807	0%	6d	2d	
1,750.00	3.468 3.343	0%	15d	1d	

Heeling moment is present from: wind

Trim = Fwd 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
300.00	15.445	95%	0d	14d	
500.00	14.325	51%	0d	10d	
750.00	12.852	14%	0d	7d	
1,000.00	11.011	0%	0d	5d	
1,250.00	8.477	0%	2d	4d	
1,500.00	5.833	0%	6d	3d	
1,750.00	3.494 3.235	0%	15d	2d	

Heeling moment is present from: wind

Trim = zero at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
300.00	15.471	95%	0d	17d	
500.00	14.340	51%	0d	11d	
750.00	12.878	13%	0d	8d	
1,000.00	11.044	0%	0d	6d	
1,250.00	8.477	0%	2d	4d	
1,500.00	5.832	0%	6d	3d	
1,750.00	3.467 3.161	0%	LARGE	2d	

Heeling moment is present from: wind

Trim = Aft 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
300.00	15.445	95%	0d	14d	
500.00	14.325	51%	0d	10d	
750.00	12.852	14%	0d	7d	
1,000.00	11.011	0%	0d	5d	
1,250.00	8.462	0%	2d	4d	
1,500.00	5.832	0%	6d	3d	
1,750.00	3.493 3.225	0%	LARGE	2d	

MAX VCG DOES NOT COVER THE CRANE OPERATION / LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW WITH THE CRANE IN STOWED POSITION.

MAX VCG IS VERIFIED FOR MAXIMUM DECK CARGO HEIGHT OF 1M ABOVE MAIN DECK.

15-02-19 10:57:40 Cybermarine Technologies PTE. Ltd.
GHS 16.50

Heeling moment is present from: wind
Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)
Displacement --- Margins ---
METRIC TONS Max VCG LIM1 LIM2 LIM3

300.00	15.373	94%	0d	12d
500.00	14.274	47%	0d	9d
750.00	12.752	16%	0d	6d
1,000.00	10.917	0%	0d	5d
1,250.00	8.412	0%	2d	3d
1,500.00	5.805	0%	6d	2d
1,750.00	3.469	0%	15d	1d

Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg

MAX VCG DOES NOT COVER THE CRANE OPERATION / LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW WITH THE CRANE IN STOWED POSITION.

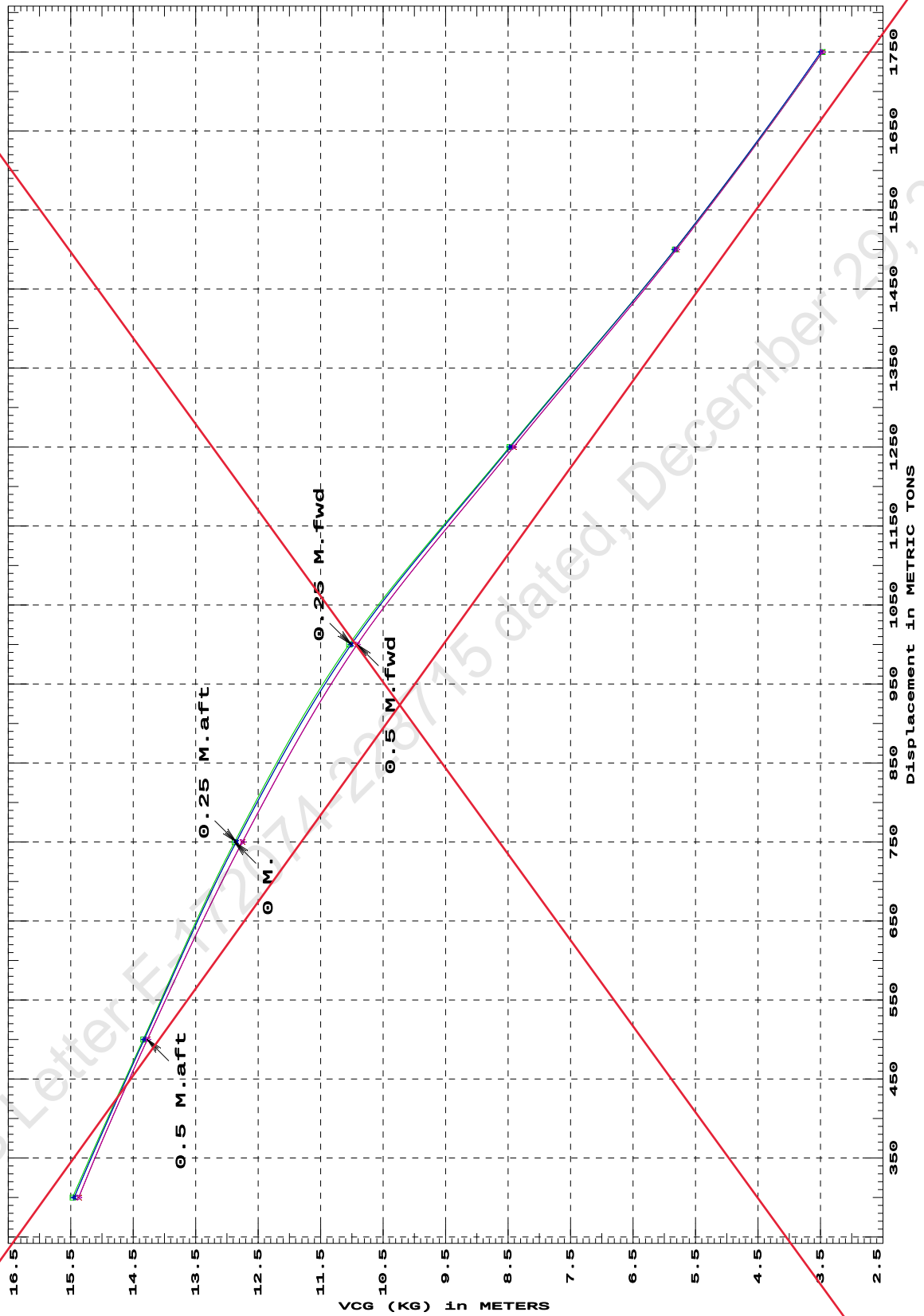
MAX VCG IS VERIFIED FOR MAXIMUM DECK CARGO HEIGHT OF 1M ABOVE MAIN DECK.

15-02-19 10:57:40

Cybermarine Technologies PTE. Ltd.

GHS 16.50

IS CODE 2008 PART B CHAPTER-2 SEC.2 MAXIMUM VCG (KG)
at various trims (initial)



Specific Gravity = 1.025
Trim is per 50 M.
"K" = BPL

LOADING CONDITIONS

SUMMARY OF CONDITIONS

CONDITION NO.	0	1	2	3	4	5
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Per. Ballast	Condition 2 : Lightship + Load at Aft + Ballast Crane Lifting 36.6T @ 18m Radius Aft	Condition 3 : Lightship + Load at Stbd + Ballast Crane Lifting 36.6T @ 18m Radius Stbd	Condition 4 : Lightship + Load at Fwd + Ballast Crane Lifting 36.60T @ 18m Radius Fwd	Condition 5 : Lightship + Crane + Per. Ballast + Deck Cargo (Towing Condition)
TOTAL TANK LOAD (T)	0.000	264.600	264.600	264.600	264.600	264.600
DWT CONST.(T)	0.000	250.000	286.600	286.600	286.600	931.180
DEADWEIGHT (T)	0.000	514.600	551.200	551.200	551.200	1195.780
LIGHT WEIGHT (T)	373.370	373.370	373.370	373.370	373.370	373.370
DISPLACEMENT (T)	373.370	887.970	924.570	924.570	924.570	1569.150
FWD DRAFT (M)	0.575	1.305	1.230	1.350	1.480	2.210
AFT DRAFT (M)	0.580	1.309	1.484	1.355	1.234	2.214
MEAN DRAFT (M)	0.578	1.307	1.357	1.353	1.357	2.212
TRIM (M)	0.005a	0.004a	0.255a	0.004a	0.246f	0.004a
L.C.G. (FROM AP) (M)	24.970f	24.987f	24.275f	24.988f	25.700f	24.993f
L.C.B. (FROM AP) (M)	24.970f	24.987f	24.247f	24.987f	25.728f	24.993f
M.C.T. (T-CM)	22.220	26.530	26.330	26.400	26.330	31.300
KG (M)	3.000	3.431	6.341	6.341	6.341	3.461
GM _r (M)	31.578	12.372	8.956	9.082	8.956	6.766
MAX. GZ (M)	4.197	2.897	1.946	1.284	1.946	1.049

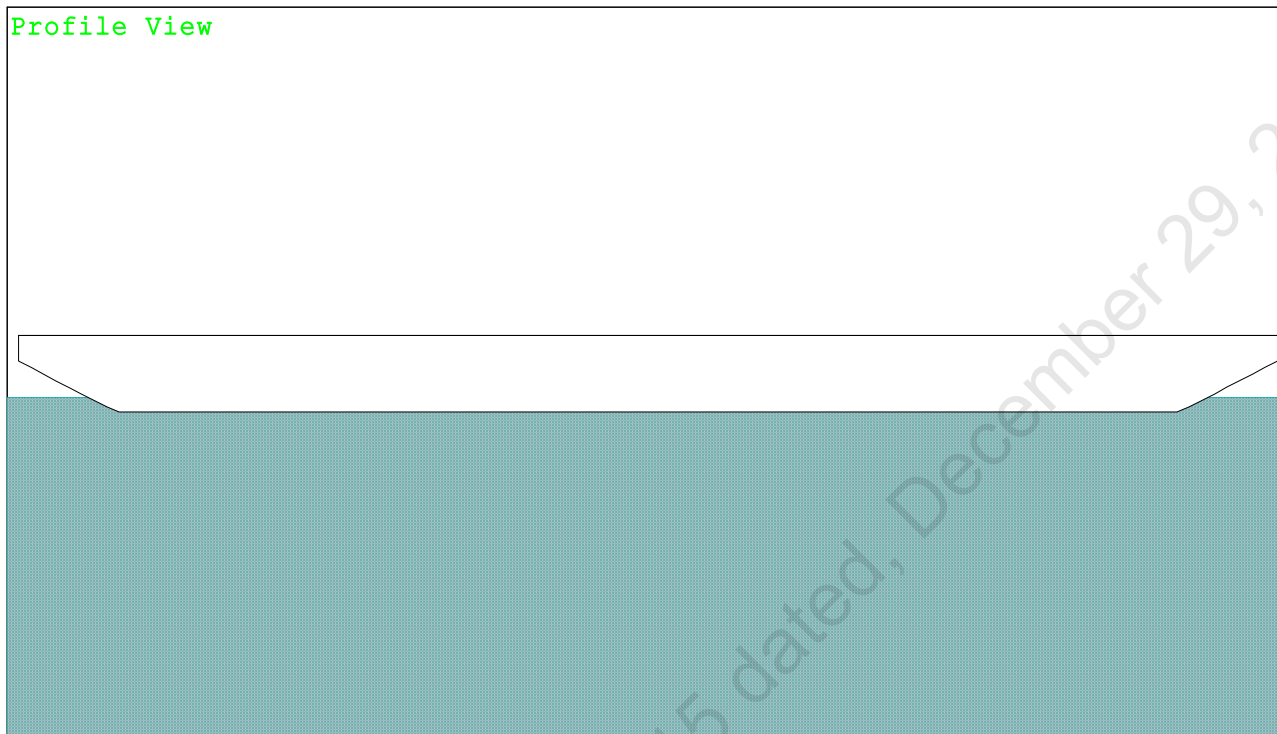
~~USK draft refers to the line:~~

~~0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000~~

Refer IRS Letter E-172074-228715 dated, December 29, 2022

Condition Graphic - Draft: 0.575 @ 50.000f, 0.580 @ 0.000 Heel: zero

Profile View



Plan View



Condition 0 : Lightship Condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 0.575 @ 50.00f, 0.577 @ 25.00f, 0.580 @ 0.00

Trim: Aft 0.005/50.000, Heel: zero

Part-----	Weight (MT)	LCG	TCG	VCG	FSM
WEIGHT	373.37	24.970f	0.000	3.000	
Load-----	SpGr-----	Weight (MT)	LCG	TCG	VCG
Total Tanks----->		0.00			0.00
	Displ (MT)	LCB	TCB	VCB	
HULL	1.025	373.37	24.970f	0.000	0.283

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA	LCF	TCF	BML-----BMT
Total Waterplane---->	1.025	666.5	24.996f	0.000	300.25 34.295
		MT/cm-----m.-	MT/cm-----	GML-----	GMT
		6.83	22.22	297.54	31.578
Distances in METERS.-----		Moments in m.-MT.			

SUMMARY OF LOADING

0.0 Cu.M. (0%) SALT WATER

DRAFT AT FP (M) = 0.575
 DRAFT AT MS (M) = 0.577
 DRAFT AT AP (M) = 0.580
 DRAFT AT FWD MARK (M) = 0.575
 DRAFT AT AFT MARK (M) = 0.580

PONTOON

Condition 0 : Lightship Condition

HYDROSTATIC PROPERTIES

Trim: Aft 0.005/50.000, No Heel, VCG = 3.000

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
 Draft----Weight(MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
 0.577 373.37 24.970f 0.283 6.83 24.996f 22.22 297.54 31.578
 Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 50.00m.

Draft is from USK.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 24.970f TCG = 0.000 VCG = 3.000

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm in Heel	Area	Angle
0.568	0.01a	0.00	373.37	0.000	-0.025	0.0000	14.712
0.568	0.01a	0.04s	373.37	0.000	0.000	-0.0000	14.668
0.547	0.01a	5.04s	373.36	0.000	2.605	0.1137	9.668
0.374	0.01a	10.04s	373.38	0.000	3.755	0.4015	4.668
0.099	0.01a	14.71s	373.37	0.000	4.100	0.7262	0.000
0.077	0.01a	15.04s	373.37	0.000	4.113	0.7500	-0.332
-0.270	0.01a	19.83s	373.18	0.000	4.197	1.0988	-5.117
-0.287	0.01a	20.04s	373.37	0.000	4.197	1.1145	-5.332
-0.692	0.01a	25.04s	373.16	0.000	4.137	1.4792	-10.332
-1.123	0.01a	30.04s	373.17	0.000	3.986	1.8344	-15.332
-1.557	0.01a	35.04s	373.20	0.000	3.739	2.1721	-20.332
-1.978	0.02a	40.04s	373.22	0.000	3.412	2.4847	-25.332
-2.385	0.02a	45.04s	373.23	0.000	3.028	2.7661	-30.332
-2.774	0.02a	50.04s	373.24	0.000	2.601	3.0120	-35.332
-3.141	0.02a	55.04s	373.25	0.000	2.141	3.2191	-40.332
-3.485	0.02a	60.04s	373.26	0.000	1.656	3.3850	-45.332

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

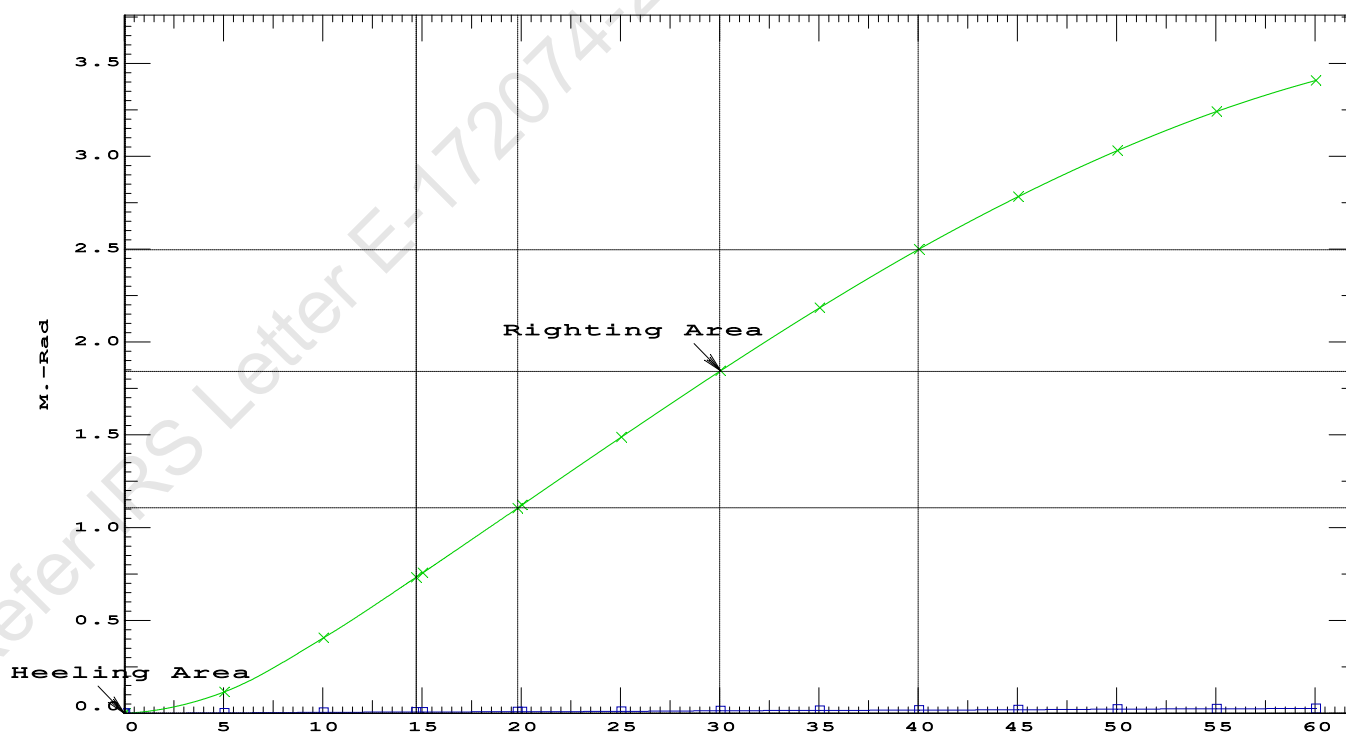
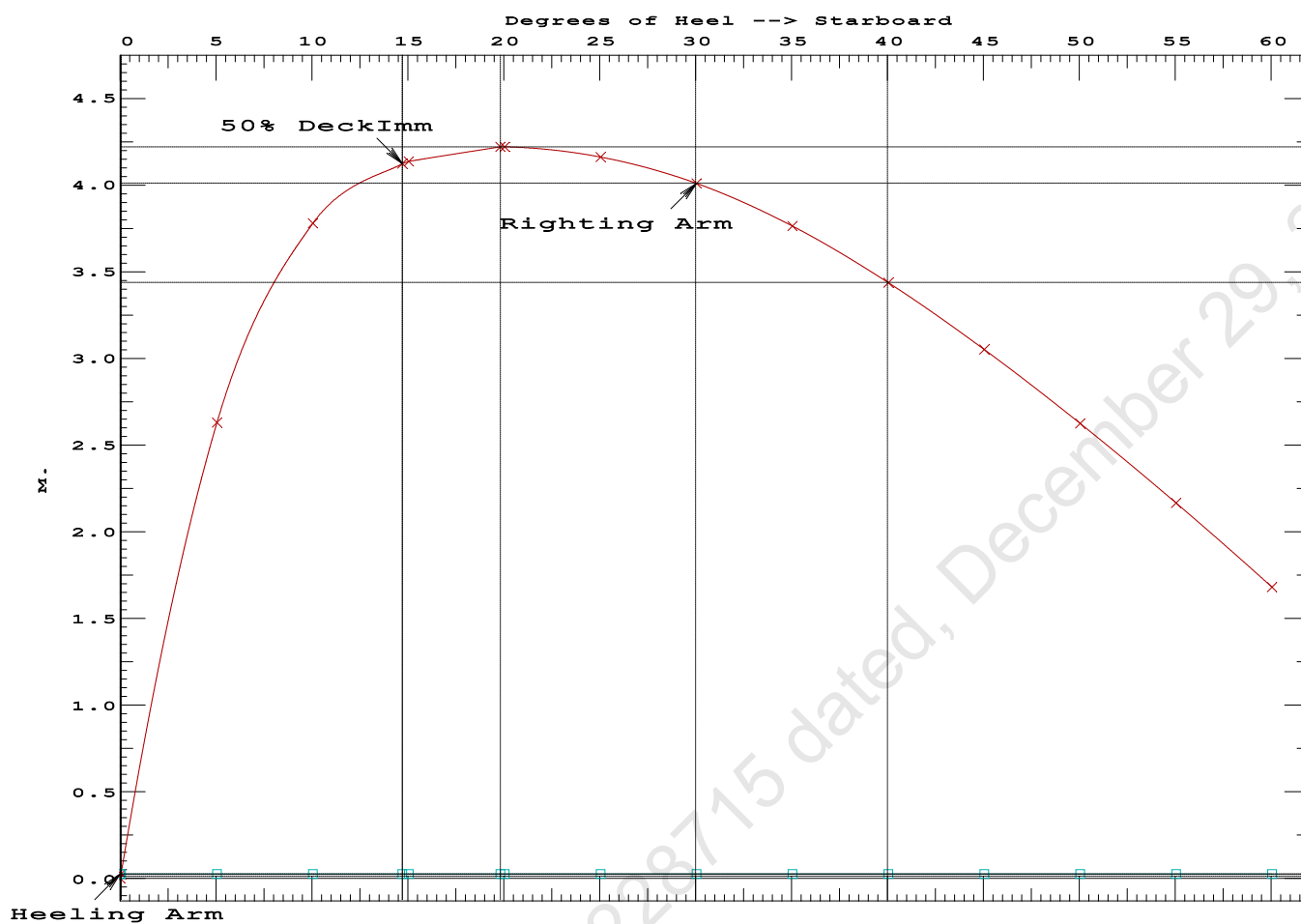
Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the
 wind heeling arms derived from these moments (in m.-MT):

Stbd heeling moment = 9.32 (constant)

LIM-----INTACT CRITERIA-IS CODE 2008PART B ----Min/Max-----Attained
 (1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 1.0988 P
 (2) Angle from Equilibrium to RAzero or Flood > 20.00 deg LARGE
 (3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 14.67 P

PONTOON

Condition 0 : Lightship Condition

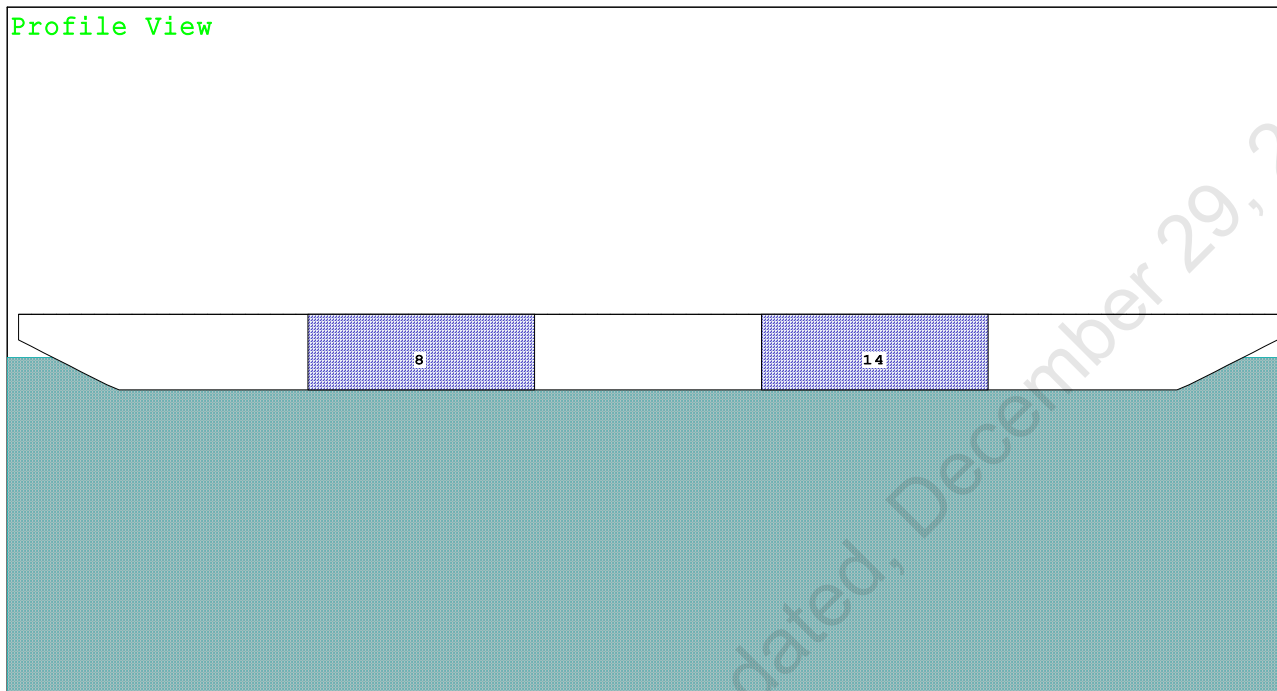
USK draft refers to the line:

0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

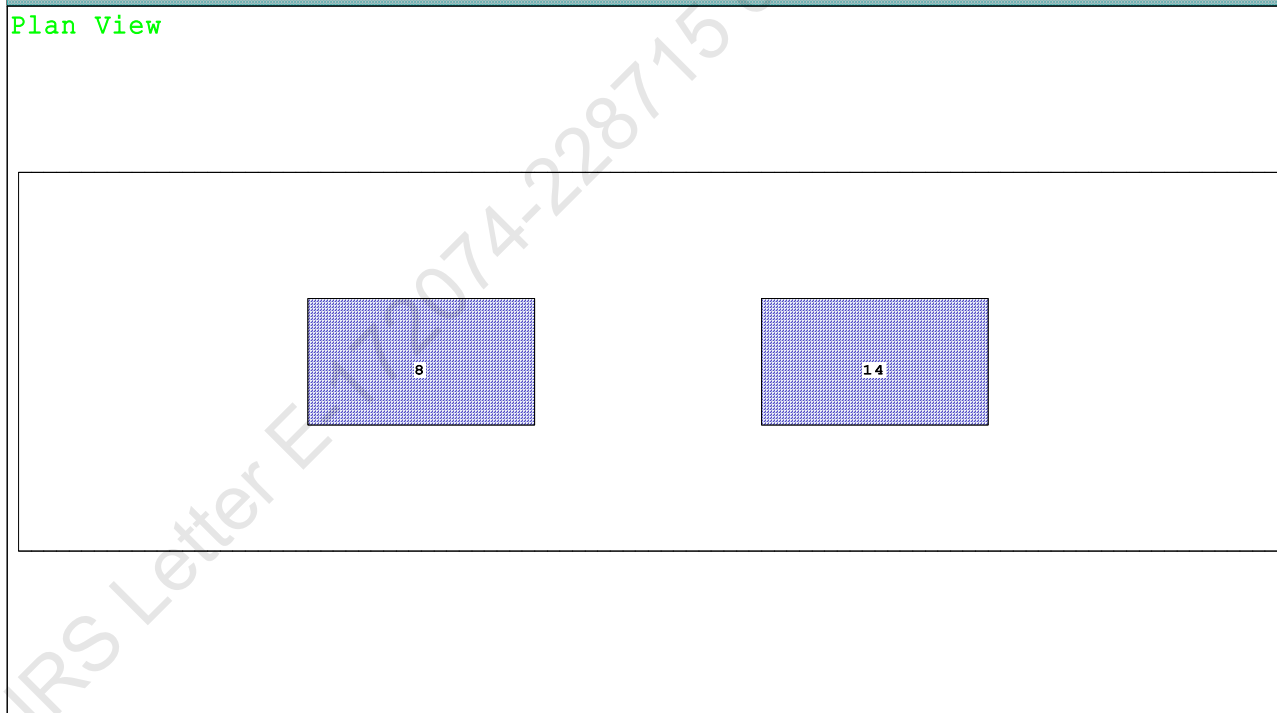
Refer IRS Letter E-172074-228715 dated, December 29, 2022

Condition Graphic - Draft: 1.305 @ 50.000f, 1.309 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

8 03_PFWBTK.C.100% FRESH WATER 14 05_PFWBTK.C.100% FRESH WATER

Condition 1 : Lightship + Crane + Ballast

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.305 @ 50.00f, 1.307 @ 25.00f, 1.309 @ 0.00

Trim: Aft 0.004/50.000, Heel: zero

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			373.37	24.970f	0.000	3.000	
Crane			250.00	25.000f	0.000	6.120	
Total Fixed----->			623.37	24.982f	0.000	4.251	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks----->			264.60	25.000f	0.000	1.500	183.75
Total Weight----->			887.97	24.987f	0.000	3.431	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	887.97	24.987f	0.000	0.660	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		709.3	24.996f	0.000	152.18	15.143*
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			7.27	26.53	149.41	12.372	
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

264.6 Cu.M. (13%) FRESH WATER

250.00 MT Crane

DRAFT AT FP (M) = 1.305
 DRAFT AT MS (M) = 1.307
 DRAFT AT AP (M) = 1.309
 DRAFT AT FWD MARK (M) = 1.305
 DRAFT AT AFT MARK (M) = 1.309

Condition 1 : Lightship + Crane + Ballast

HYDROSTATIC PROPERTIES

Trim: Aft 0.004/50.000, No Heel, VCG = 3.431

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
 Draft----Weight(MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
 1.307 887.97 24.987f 0.660 7.27 24.996f 26.53 149.41 12.372
 Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.987f TCG = 0.000 VCG = 3.431

Free Surface Adjustment: 0.207

Adjusted CG: LCG = 24.987f TCG = 0.000 VCG = 3.638

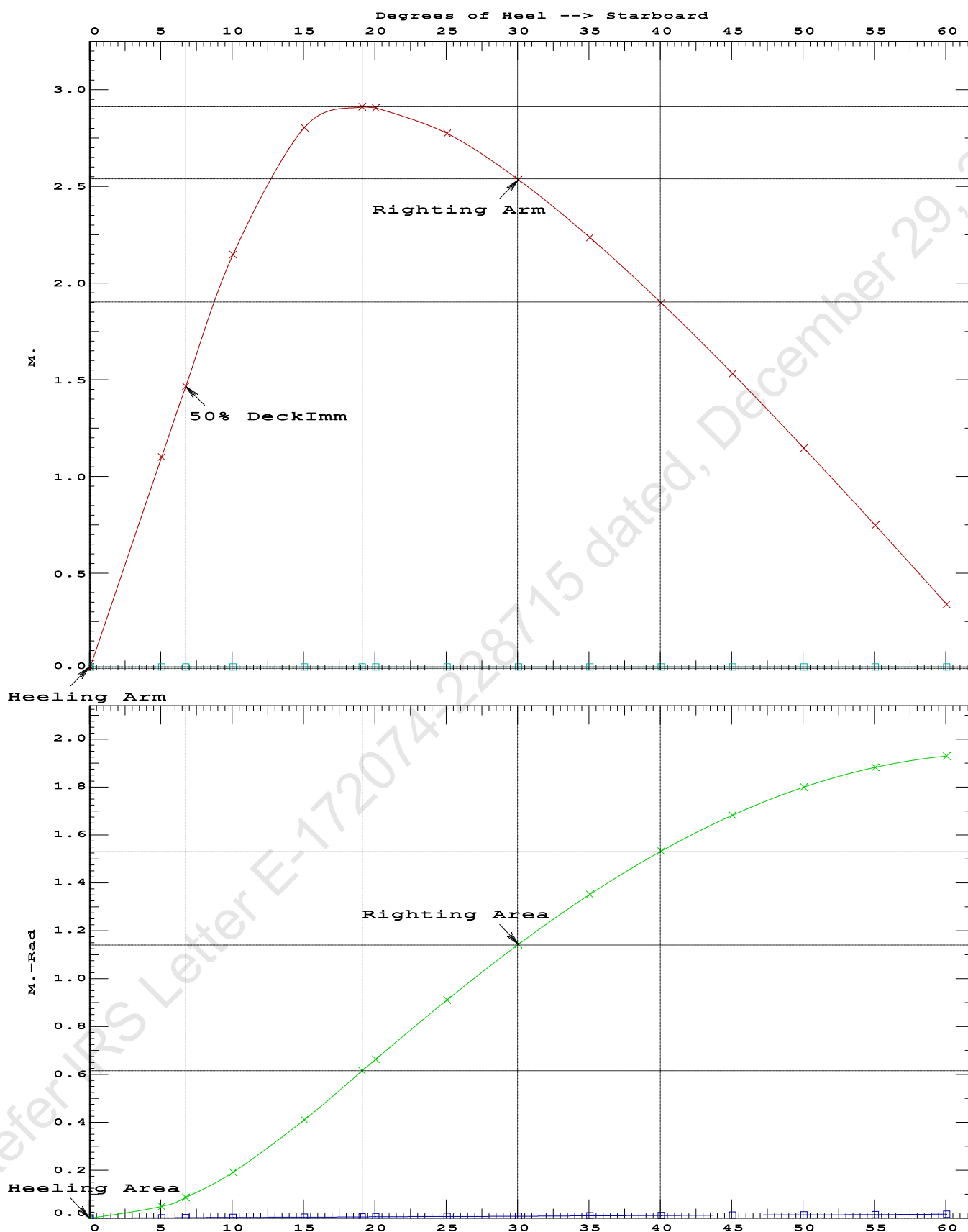
Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight(MT)---	in Trim--in Heel---	Area----	Angle
1.297 0.00a 0.00	887.97	0.000 -0.015	0.0000	6.751
1.297 0.00a 0.07s	887.97	0.000 0.000	-0.0000	6.682
1.286 0.00a 5.07s	887.97	0.000 1.085	0.0473	1.682
1.277 0.00a 6.75s	887.97	0.000 1.451	0.0846	0.000
1.246 0.00a 10.07s	887.96	0.000 2.132	0.1884	-3.318
1.126 0.01a 15.07s	887.97	0.000 2.789	0.4052	-8.318
0.999 0.01a 19.12s	887.97	0.000 2.897	0.6083	-12.367
0.968 0.01a 20.07s	887.97	0.000 2.891	0.6563	-13.318
0.803 0.01a 25.07s	887.97	0.000 2.758	0.9053	-18.318
0.631 0.01a 30.07s	887.77	0.000 2.519	1.1363	-23.318
0.456 0.01a 35.07s	887.96	0.000 2.220	1.3436	-28.318
0.276 0.02a 40.07s	887.97	0.000 1.882	1.5228	-33.318
0.095 0.02a 45.07s	887.97	0.000 1.516	1.6713	-38.318
-0.088 0.02a 50.07s	887.97	0.000 1.131	1.7870	-43.318
-0.269 0.02a 55.07s	887.97	0.000 0.732	1.8684	-48.318
-0.449 0.02a 60.07s	887.97	0.000 0.324	1.9145	-53.318

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
 Stbd heeling moment = 13.65 (constant)

LIM-----INTACT CRITERIA-IS CODE 2008PART B ----Min/Max-----Attained
 (1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.6083 P
 (2) Angle from Equilibrium to RZero or Flood > 20.00 deg LARGE
 (3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 6.68 P

Condition 1 : Lightship + Crane + Ballast

USK draft refers to the line:

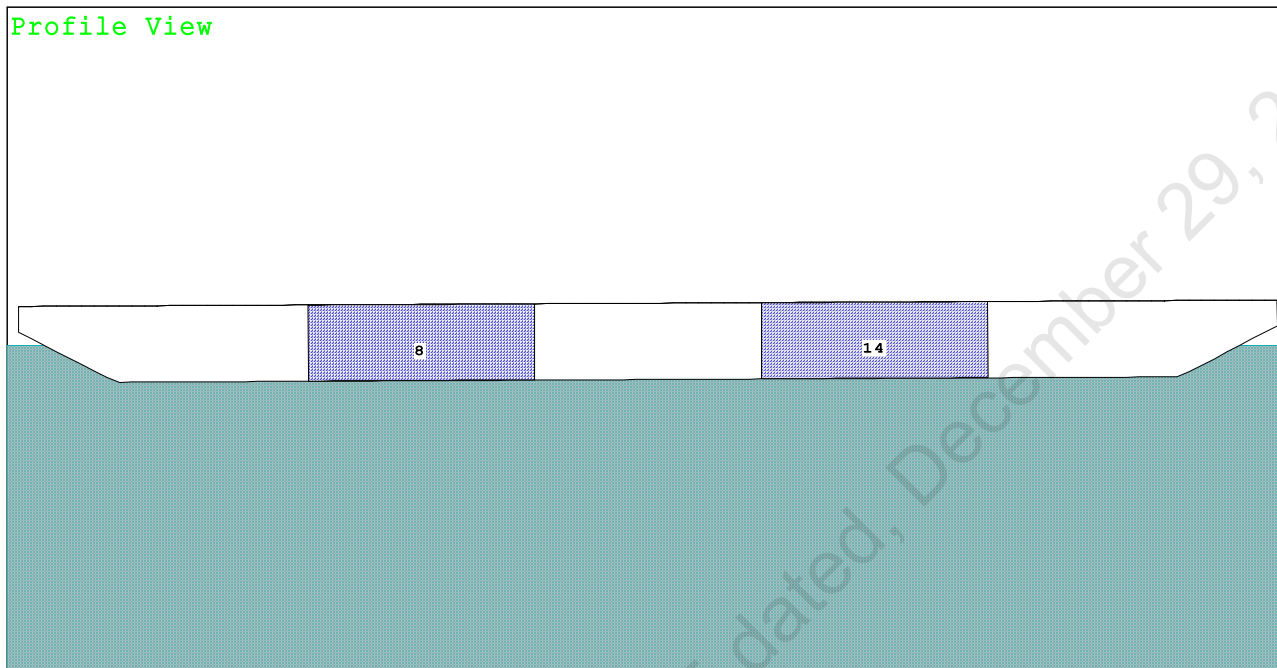
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Refer IRS Letter E-172074-228715 dated, December 29, 2022

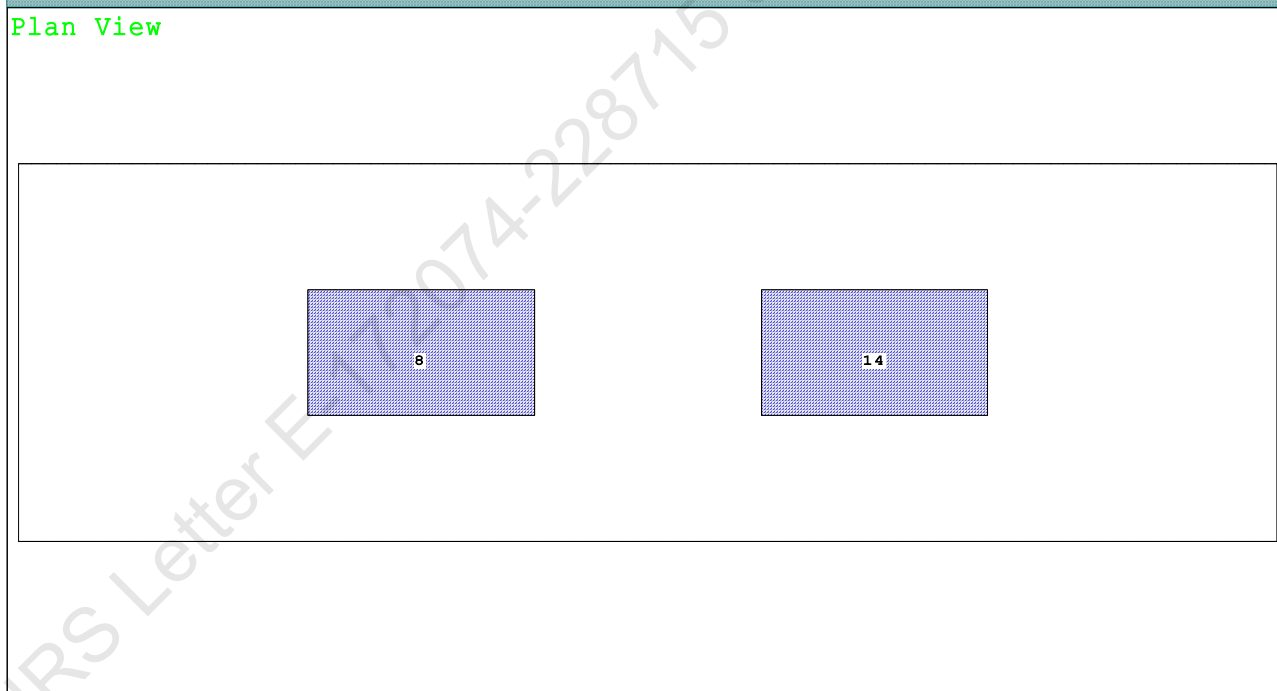
Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

Condition Graphic - Draft: 1.230 @ 50.000f, 1.484 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

8 03_PFWBTK.C.100% FRESH WATER 14 05_PFWBTK.C.100% FRESH WATER

Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.230 @ 50.00f, 1.357 @ 25.00f, 1.484 @ 0.00

Trim: Aft 0.255/50.000, Heel: zero

Part-----	Weight(MT)	LCG-----	TCG-----	VCG	
LIGHT SHIP	373.37	24.970f	0.000	3.000	
Crane	250.00	25.000f	0.000	6.120	
Crane Load	36.60	7.000f	0.000	76.930	
Total Fixed----->	659.97	23.985f	0.000	8.282	
Load-----	SpGr-----	Weight(MT)	LCG-----	TCG-----	VCG-----
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000
Total Tanks----->			264.60	25.000f	0.000
Total Weight----->			924.57	24.275f	0.000
		Displ(MT)	LCB-----	TCB-----	VCB
HULL	1.025	924.57	24.247f	0.000	0.688

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane----->	1.025	712.3	24.764f	0.000	148.04
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		7.30	26.33	142.39	8.956
Distances in METERS.-----					
-----Moments in m.-MT.					

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

264.6 Cu.M. (13%) FRESH WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.230
 DRAFT AT MS (M) = 1.357
 DRAFT AT AP (M) = 1.484
 DRAFT AT FWD MARK (M) = 1.250
 DRAFT AT AFT MARK (M) = 1.464

HYDROSTATIC PROPERTIES

Trim: Aft 0.255/50.000, No Heel, VCG = 6.341

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/	
Draft----	Weight(MT)	LCB-----	VCB-----	cm-----	LCF-----
1.358	924.57	24.247f	0.688	7.30	24.764f
					26.33
					142.39
					8.956

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.275f TCG = 0.000 VCG = 6.341

Free Surface Adjustment: 0.199

Adjusted CG: LCG = 24.276f TCG = 0.000 VCG = 6.540

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---	Trim---	Heel---	Weight (MT)---	in Trim--in Heel--> Area----Angle
1.472	0.29a	0.00	924.57	0.000 -0.014 0.0000 5.923
1.472	0.29a	0.08s	924.57	0.000 0.000 -0.0000 5.841
1.460	0.29a	5.08s	924.57	0.000 0.786 0.0343 0.841
1.456	0.29a	5.92s	924.57	0.000 0.919 0.0468 0.000
1.425	0.30a	10.08s	924.57	0.000 1.544 0.1363 -4.159
1.353	0.38a	15.08s	924.57	0.000 1.943 0.2903 -9.159
1.346	0.39a	15.55s	924.57	0.000 1.946 0.3061 -9.624
1.272	0.51a	20.08s	924.57	0.000 1.791 0.4569 -14.159
1.185	0.64a	25.08s	924.57	0.000 1.420 0.5994 -19.159
1.090	0.78a	30.08s	924.57	0.000 0.958 0.7038 -24.159
0.989	0.91a	35.08s	924.57	0.000 0.449 0.7656 -29.159
0.899	1.02a	39.30s	924.57	0.000 0.000 0.7822 -33.377
0.882	1.04a	40.08s	924.57	0.000 -0.084 0.7816 -34.159
0.768	1.16a	45.08s	924.57	0.000 -0.630 0.7506 -39.159
0.649	1.27a	50.08s	924.57	0.000 -1.178 0.6717 -44.159
0.524	1.37a	55.08s	924.57	0.000 -1.722 0.5452 -49.159
0.394	1.46a	60.08s	924.57	0.000 -2.257 0.3715 -54.159

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

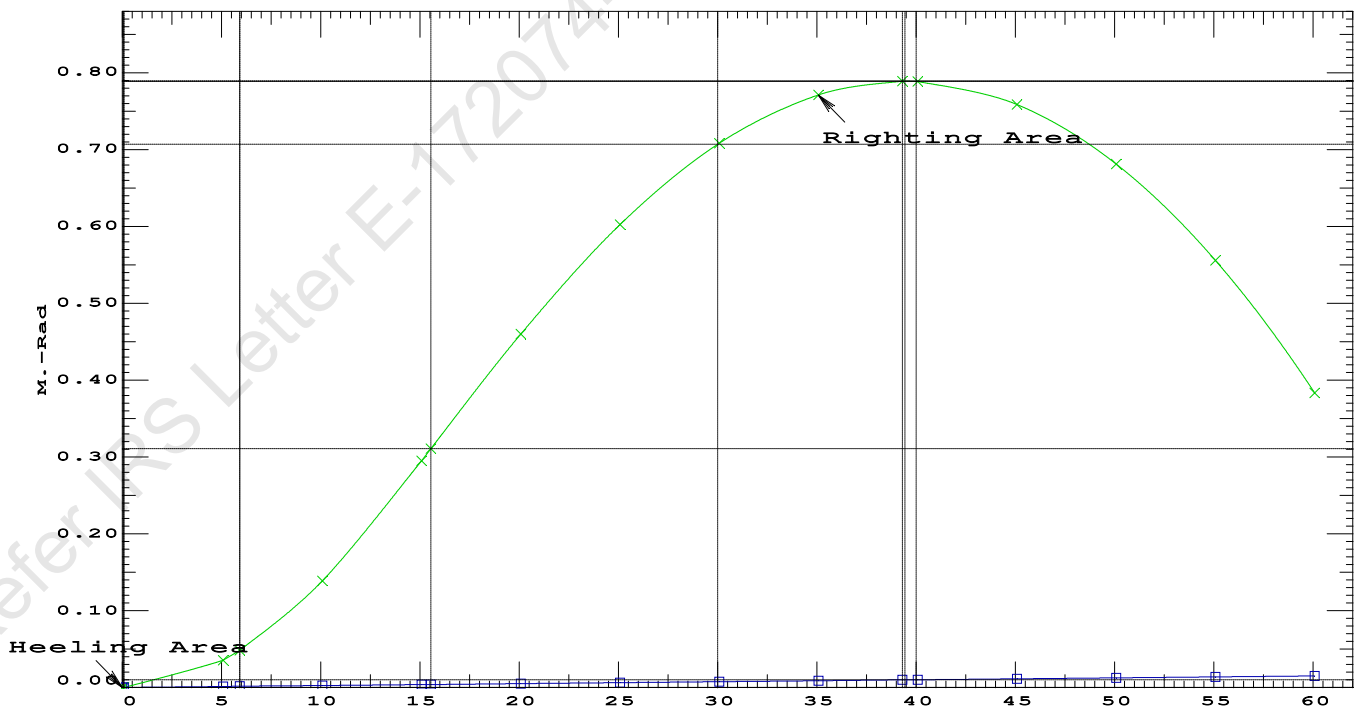
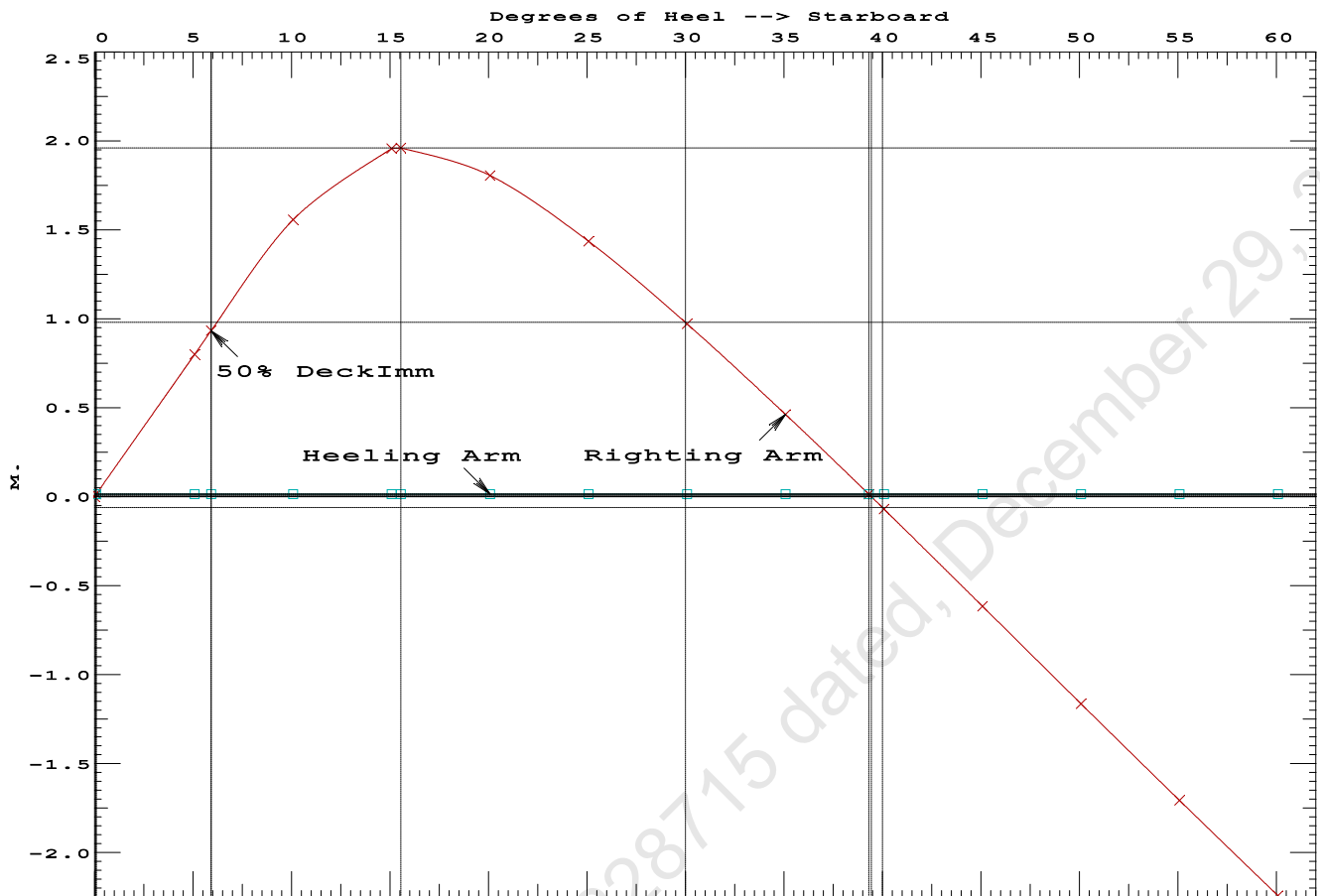
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
 Stbd heeling moment = 13.29 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.3061 P
(2) Angle from Equilibrium to RZero or Flood	>	20.00 deg	39.22 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	5.84 P

PONTOON

Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft



PONTOON

Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.275f TCG = 0.000 VCG = 6.341
Free Surface Adjustment: 0.199
Adjusted CG: LCG = 24.276f TCG = 0.000 VCG = 6.540

Origin	Degrees of		Displacement	Residual Arms		Res.
Depth	Trim	Heel	Weight (MT)	in Trim	in Heel	Area
1.472	0.29a	0.00	924.57	0.000	0.000	0.0000
1.460	0.29a	5.00p	924.57	0.000	0.786	0.0343
1.425	0.30a	10.00p	924.57	0.000	1.545	0.1362
1.354	0.38a	15.00p	924.57	0.000	1.956	0.2915
1.346	0.39a	15.56p	924.57	0.000	1.959	0.3107
1.274	0.51a	20.00p	924.57	0.000	1.809	0.4594
1.186	0.64a	25.00p	924.57	0.000	1.440	0.6036
1.091	0.78a	30.00p	924.57	0.000	0.979	0.7098
0.990	0.91a	35.00p	924.57	0.000	0.470	0.7734
0.896	1.02a	39.42p	924.57	0.000	0.000	0.7917
0.883	1.04a	40.00p	924.57	0.000	-0.063	0.7913
0.770	1.16a	45.00p	924.57	0.000	-0.608	0.7622
0.651	1.27a	50.00p	924.57	0.000	-1.156	0.6853
0.526	1.37a	55.00p	924.57	0.000	-1.700	0.5606
0.396	1.46a	60.00p	924.57	0.000	-2.235	0.3889

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

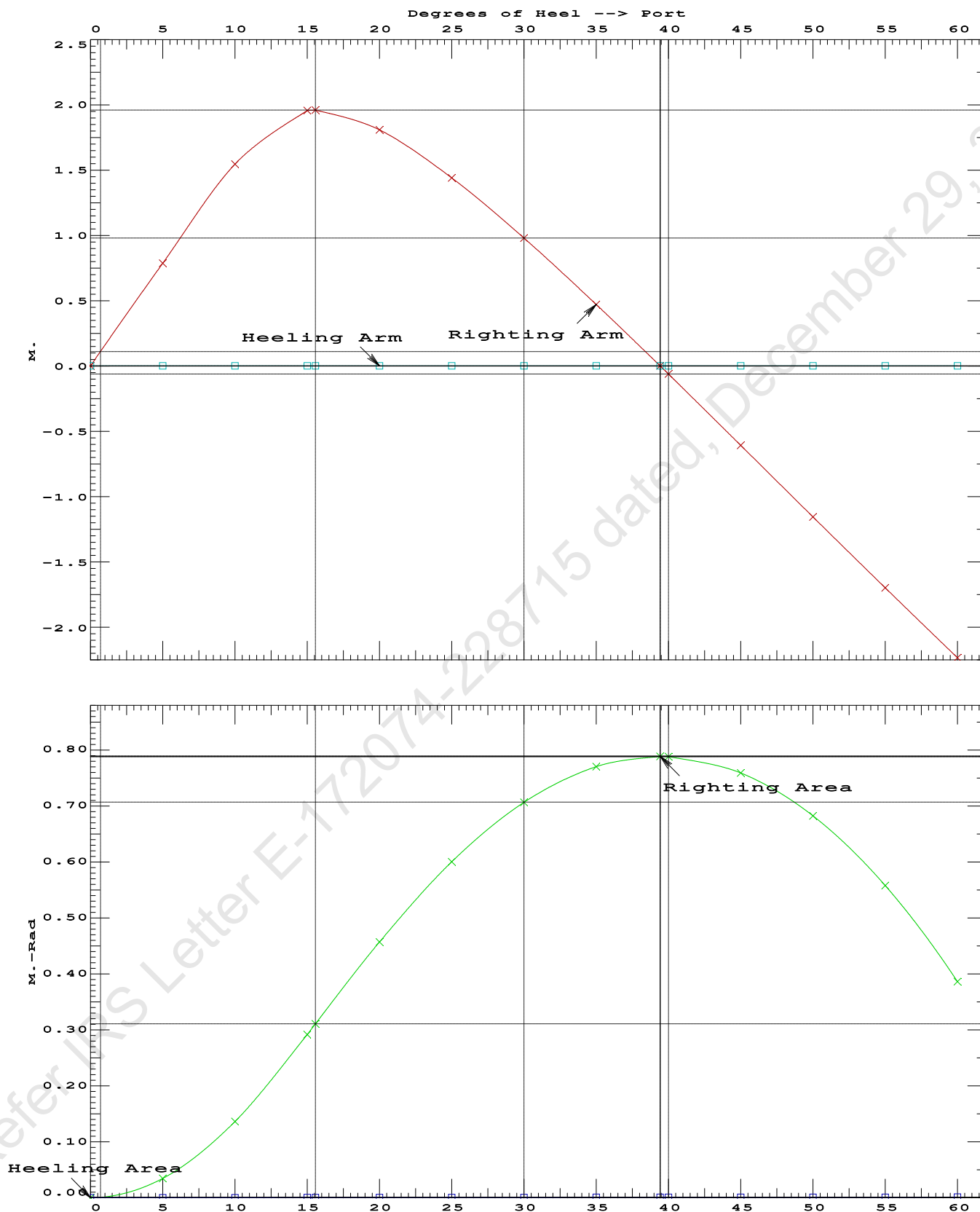
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	181117 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1064.88 P

Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft



USK draft refers to the line:

0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Refer IRS Letter E-172074-228715 dated, December 29, 2022

Condition 2 : Lightship + Load at Aft + BallastCrane Lifting 36.60T @ 18m Radius Aft

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.229 @ 50.00f, 1.357 @ 25.00f, 1.484 @ 0.00

Trim: Aft 0.254/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			373.37	24.970f	0.000	3.000	
Crane			250.00	25.000f	0.000	6.120	
Crane Load			36.60	7.000f	0.000	76.930	
Total Fixed----->			659.97	23.985f	0.000	8.282	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks----->			264.60	25.000f	0.000	1.500	183.75
Total Weight----->			924.57	24.275f	0.000	6.341	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	924.46	24.247f	0.001p	0.688	

	Righting Arms:			0.000	-0.001s		
	External Arms:			0.000	-0.001p		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		709.1	24.873f	0.017p	146.04	14.563*
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			7.27		25.96	140.38	8.910
Distances in METERS.-----							
-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.305 @ 50.00f, 1.307 @ 25.00f, 1.309 @ 0.00

Trim: Aft 0.004/50.000, Heel: 0.00 deg.

Part	Weight (MT)	LCG	TCG	VCG	FSM		
LIGHT SHIP	373.37	24.970f	0.000	3.000			
Crane	250.00	25.000f	0.000	6.120			
Total Fixed	623.37	24.982f	0.000	4.251			
Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks			264.60	25.000f	0.000	1.500	183.75
Total Weight			887.97	24.987f	0.000	3.431	
HULL	1.025		887.97	24.987f	0.001	0.660	
Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.000			
Part	SpGr	WPA	LCF	TCF	BML	BMT	
Total Waterplane	1.025	708.7	24.997f	0.005s	151.80	15.119*	
MT/cm		7.26		26.47	149.03	12.348	
Distances in METERS.							Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

```

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

```

Theta3 is 0.07 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Area
1.297	0.00a	0.00	887.97	0.000	0.000	0.0000
1.297	0.00a	0.07s	887.97	0.000	0.015	0.0000
1.286	0.00a	5.00s	887.97	0.000	1.103	0.0481
1.247	0.00a	10.00s	887.96	0.000	2.169	0.1912
1.128	0.01a	15.00s	887.97	0.000	2.853	0.4131
0.991	0.01a	19.36s	887.97	0.000	2.979	0.6377
0.970	0.01a	20.00s	887.97	0.000	2.977	0.6711
0.805	0.01a	25.00s	887.97	0.000	2.863	0.9284
0.633	0.01a	30.00s	887.78	0.000	2.641	1.1693

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

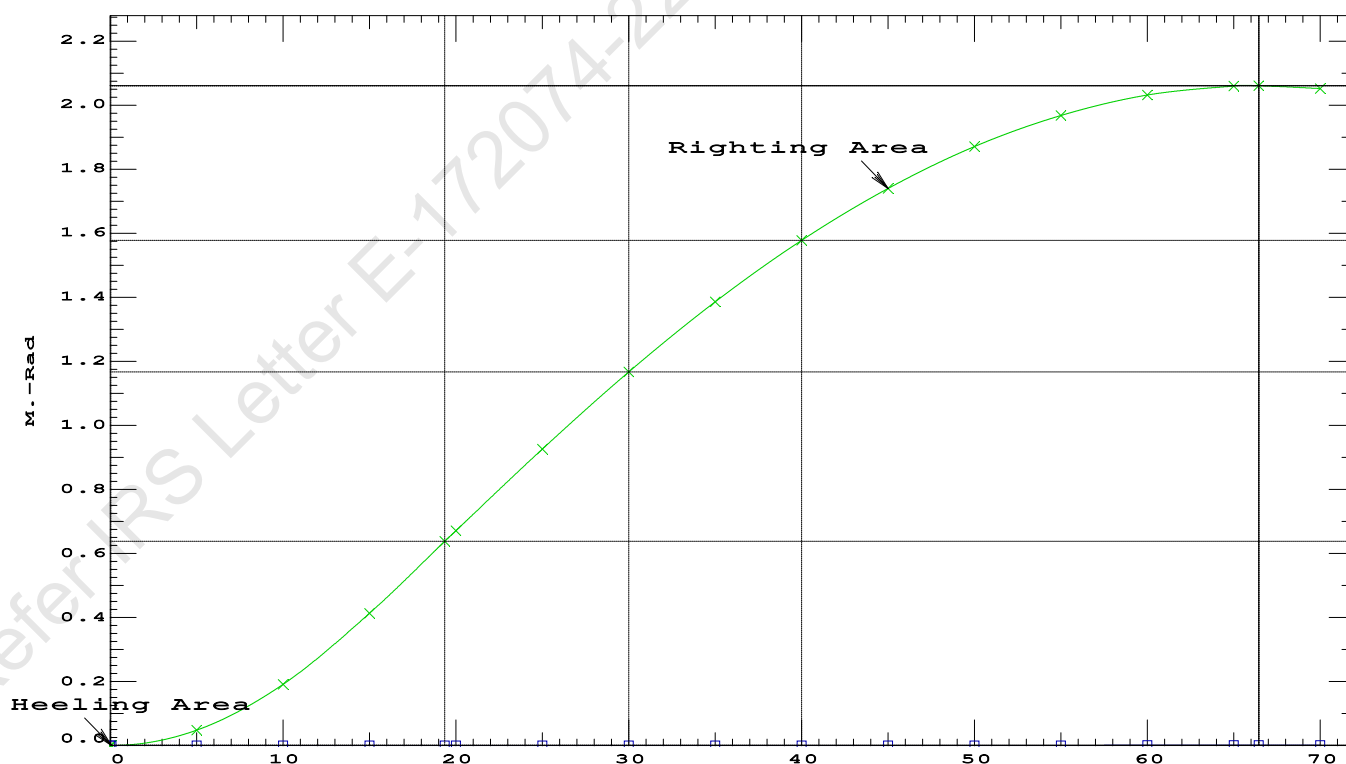
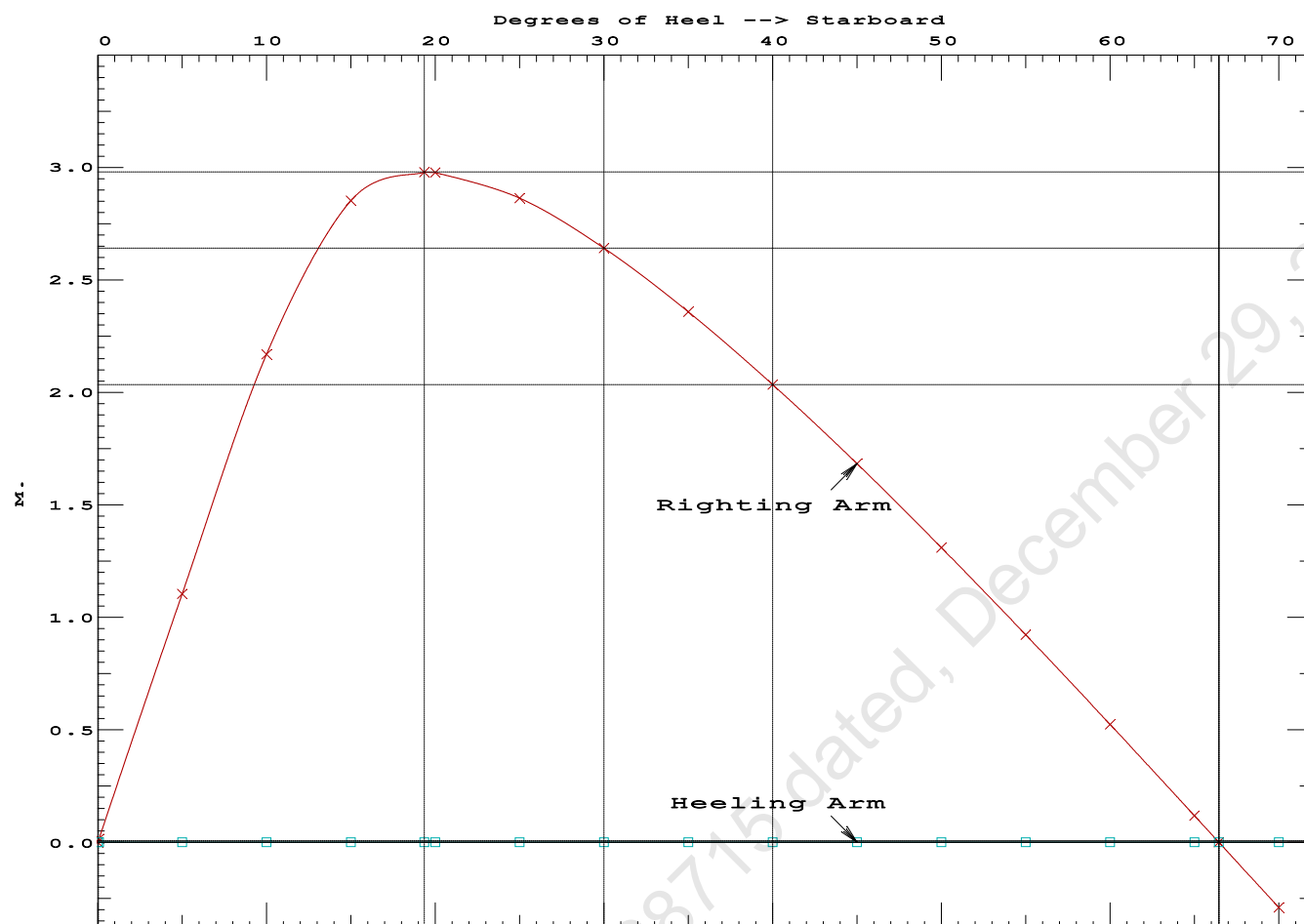
Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---Weight (MT)---in Trim--in Heel---> Area				
1.297	0.00a	0.00	887.97	0.0000
1.297	0.00a	0.07s	887.97	0.0000
1.286	0.00a	5.00s	887.97	0.0481
1.247	0.00a	10.00s	887.96	0.1912
1.128	0.01a	15.00s	887.97	0.4131
0.991	0.01a	19.36s	887.97	0.6377
0.970	0.01a	20.00s	887.97	0.6711
0.805	0.01a	25.00s	887.97	0.9284
0.633	0.01a	30.00s	887.78	1.1693
0.458	0.01a	35.00s	887.97	1.3879
0.279	0.02a	40.00s	887.97	1.5798
0.097	0.02a	45.00s	887.97	1.7422
-0.085	0.02a	50.00s	887.96	1.8729
-0.267	0.02a	55.00s	887.97	1.9704
-0.446	0.02a	60.00s	887.97	2.0335
-0.622	0.02a	65.00s	887.97	2.0615
-0.672	0.02a	66.45s	887.97	2.0629
-0.794	0.03a	70.00s	887.97	2.0540

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAzero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAzero > 20.00 deg 66.38 P



USK draft refers to the line:

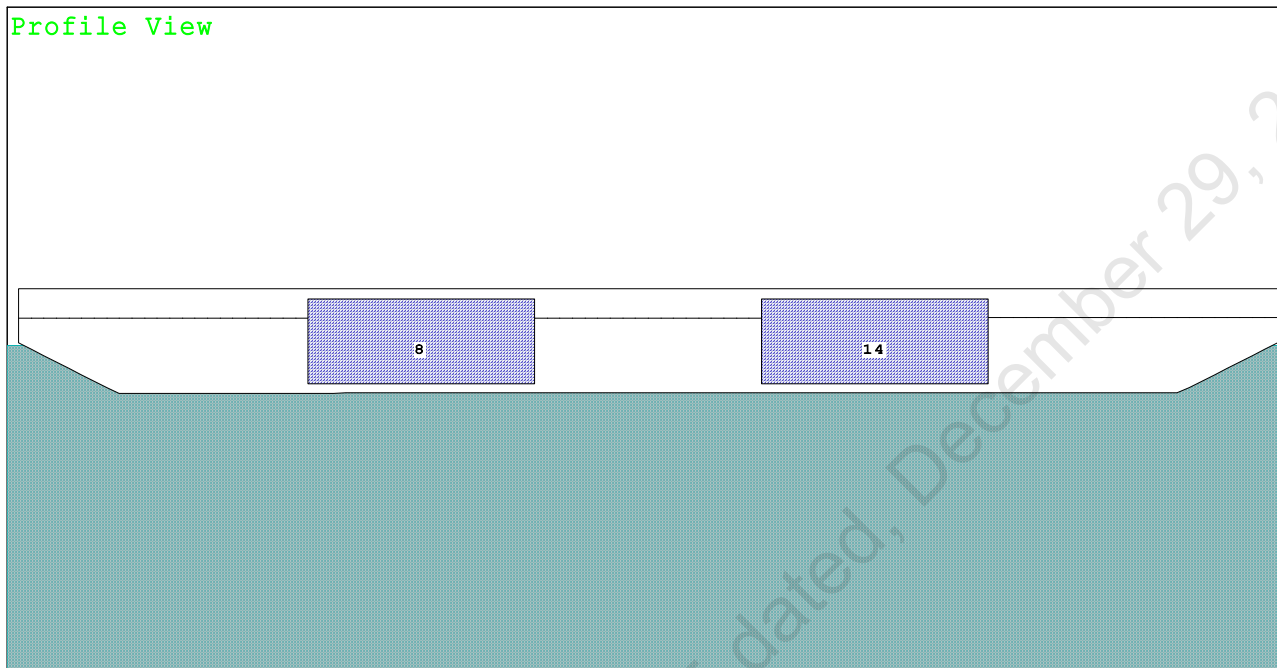
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Refer IRS Letter E-172074-228715 dated, December 29, 2022

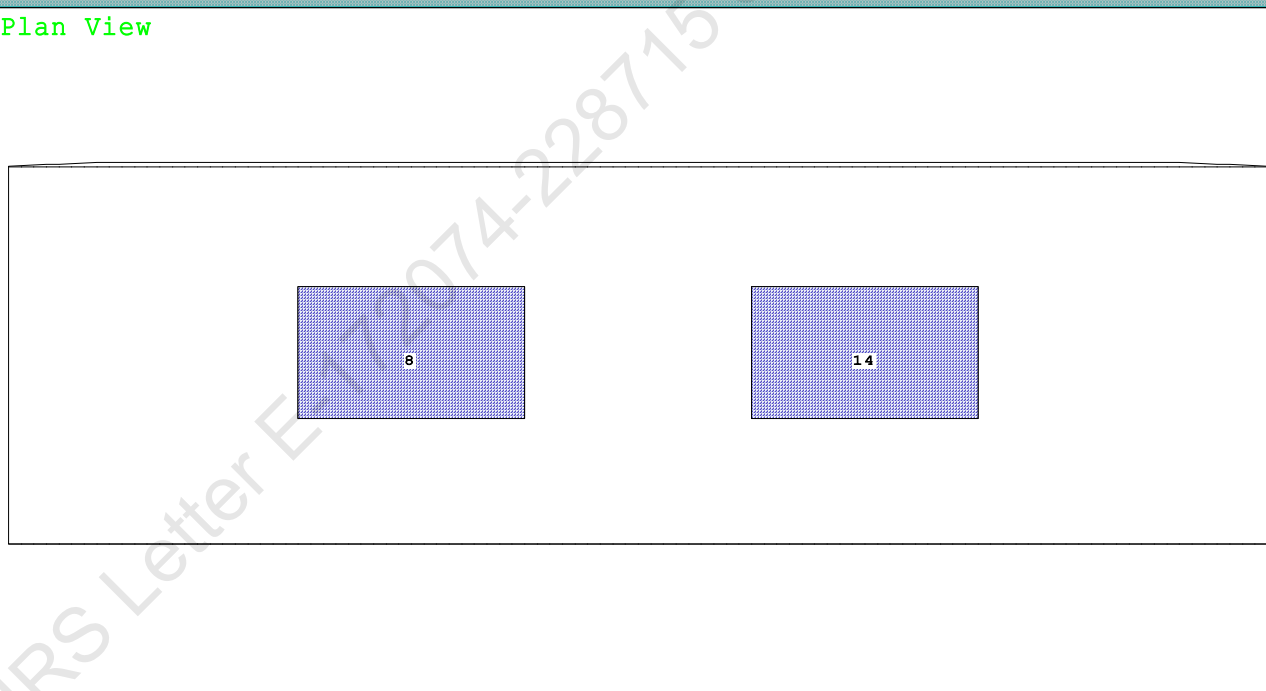
Condition 3 : Lightship + Load at Stbd + BallastCrane Lifting 36.6T @ 18m Radius Stbd

CG - Draft: 1.350 @ 50.000f, 1.355 @ 0.000 Heel: stbd 4.42 deg.

Profile View



Plan View



Tanks

8 03_PFWBTK.C.100% FRESH WATER 14 05_PFWBTK.C.100% FRESH WATER

Condition 3 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.350 @ 50.00f, 1.352 @ 25.00f, 1.355 @ 0.00

Trim: Aft 0.004/50.000, Heel: Stbd 4.42 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP	373.37	24.970f	0.000	3.000	
Crane	250.00	25.000f	0.000	6.120	
Crane Load	36.60	25.000f	18.000s	76.930	
Total Fixed----->	659.97	24.983f	0.998s	8.282	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000
Total Tanks----->			264.60	25.000f	0.000
Total Weight----->			924.57	24.988f	0.713s
			Displ(MT)----	LCB-----	TCB-----
HULL	1.025		924.57	24.987f	1.146s

Righting Arms:	0.000	0.000s		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----
Total Waterplane---->	1.025	713.7	24.996f	0.125s
		MT/cm-----	m.-MT/cm-----	GML-----
		7.32	26.40	142.76

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

264.6 Cu.M. (13%) FRESH WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.350
 DRAFT AT MS (M) = 1.352
 DRAFT AT AP (M) = 1.355
 DRAFT AT FWD MARK (M) = 1.351
 DRAFT AT AFT MARK (M) = 1.354

HYDROSTATIC PROPERTIES

Trim: Aft 0.004/50.000, Heel: Stbd 4.42 deg., VCG = 6.341

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCB-----	cm trim-----
1.352	924.57	24.987f	0.731	7.32
				24.996f
				26.40
				142.76
				9.082

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

Condition 3 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.988f TCG = 0.713s VCG = 6.341

Free Surface Adjustment: 0.199

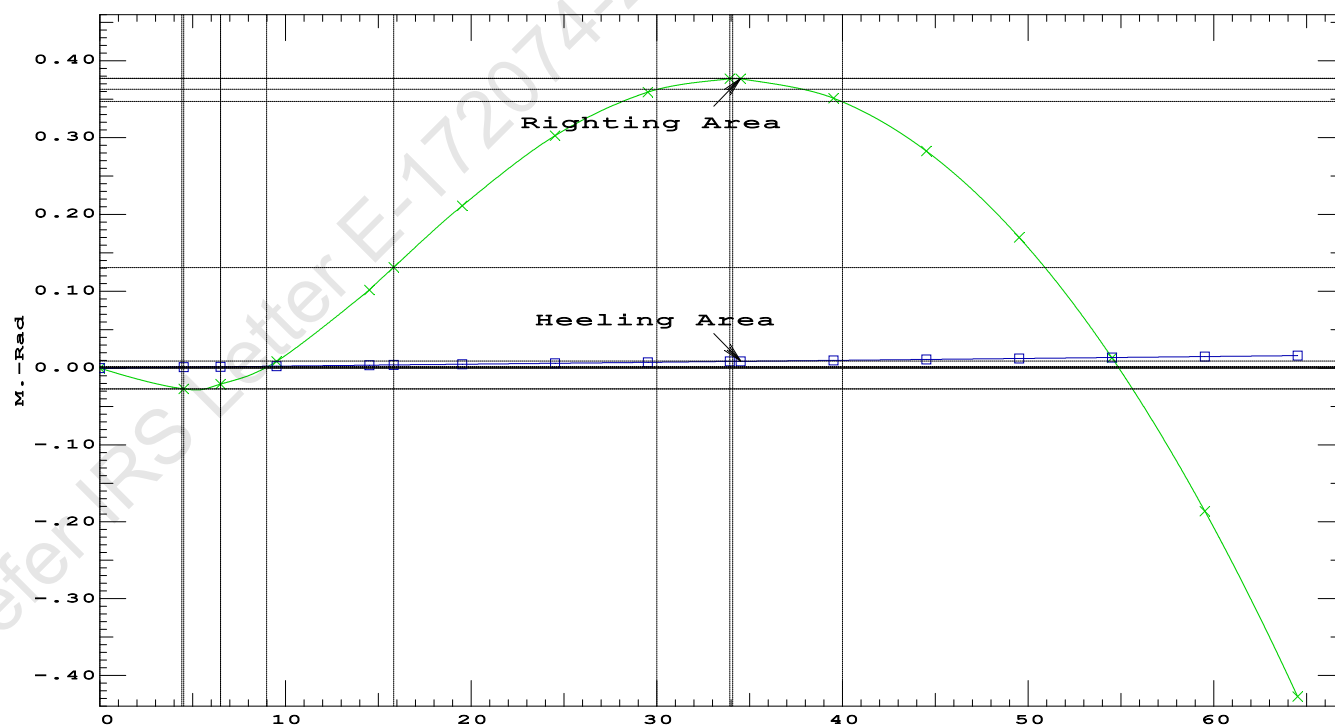
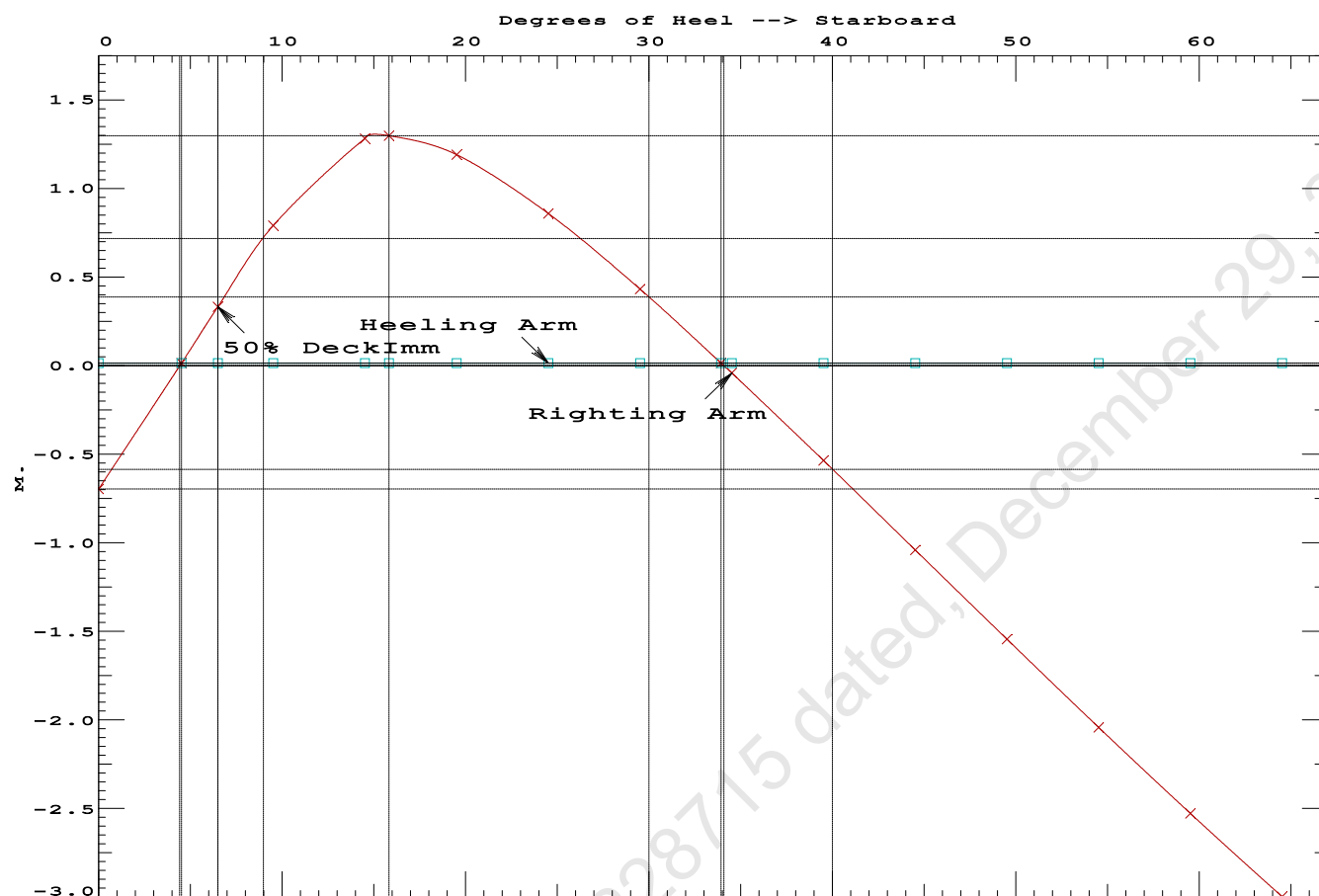
Adjusted CG: LCG = 24.988f TCG = 0.697s VCG = 6.539

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area----	Angle
1.348 0.00a 0.00	924.57	0.000 -0.711	0.0000	6.496
1.338 0.00a 4.51s	924.57	0.000 0.000	-0.0280	1.987
1.329 0.00a 6.50s	924.57	0.000 0.317	-0.0225	0.000
1.304 0.00a 9.51s	924.56	0.000 0.776	0.0063	-3.013
1.205 0.01a 14.51s	924.57	0.000 1.268	0.0969	-8.013
1.170 0.01a 15.82s	924.57	0.000 1.284	0.1262	-9.327
1.069 0.01a 19.51s	924.57	0.000 1.176	0.2069	-13.013
0.925 0.01a 24.51s	924.57	0.000 0.844	0.2961	-18.013
0.774 0.01a 29.51s	924.57	0.000 0.417	0.3518	-23.013
0.635 0.01a 33.93s	924.57	0.000 0.000	0.3681	-27.436
0.617 0.01a 34.51s	924.57	0.000 -0.056	0.3678	-28.013
0.455 0.02a 39.51s	924.57	0.000 -0.551	0.3415	-33.013
0.290 0.02a 44.51s	924.57	0.000 -1.055	0.2715	-38.013
0.123 0.02a 49.51s	924.57	0.000 -1.559	0.1574	-43.013
-0.045 0.02a 54.51s	924.57	0.000 -2.057	-0.0004	-48.013
-0.213 0.02a 59.51s	924.57	0.000 -2.543	-0.2013	-53.013
-0.380 0.02a 64.51s	924.57	0.000 -3.013	-0.4438	-58.013
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
 Stbd heeling moment = 13.41 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1262 P
(2) Angle from Equilibrium to RZero or Flood	>	20.00 deg	29.42 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	1.99 P

Condition 3 : Lightship + Load at Stbd + BallastCrane Lifting 36.6T @ 18m Radius Stbd

PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.988f TCG = 0.713s VCG = 6.341

Free Surface Adjustment: 0.199

Adjusted CG: LCG = 24.988f TCG = 0.697s VCG = 6.539

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area	
1.339 0.00a 4.42s	924.57	0.000 0.000	0.0000	
1.347 0.00a 0.58p	924.57	0.000 0.789	0.0344	
1.333 0.00a 5.58p	924.57	0.000 1.573	0.1375	
1.291 0.01a 10.58p	924.56	0.000 2.313	0.3074	
1.181 0.01a 15.42p	924.57	0.000 2.641	0.5191	
1.177 0.01a 15.58p	924.57	0.000 2.641	0.5267	
1.039 0.01a 20.58p	924.57	0.000 2.437	0.7522	
0.893 0.01a 25.58p	924.57	0.000 2.030	0.9486	
0.741 0.01a 30.58p	924.57	0.000 1.533	1.1047	
0.583 0.02a 35.58p	924.57	0.000 0.987	1.2150	
0.420 0.02a 40.58p	924.57	0.000 0.414	1.2764	
0.304 0.02a 44.11p	924.57	0.000 0.000	1.2892	
0.255 0.02a 45.58p	924.57	0.000 -0.173	1.2870	
0.087 0.02a 50.58p	924.57	0.000 -0.767	1.2460	
-0.081 0.02a 55.58p	924.57	0.000 -1.360	1.1531	

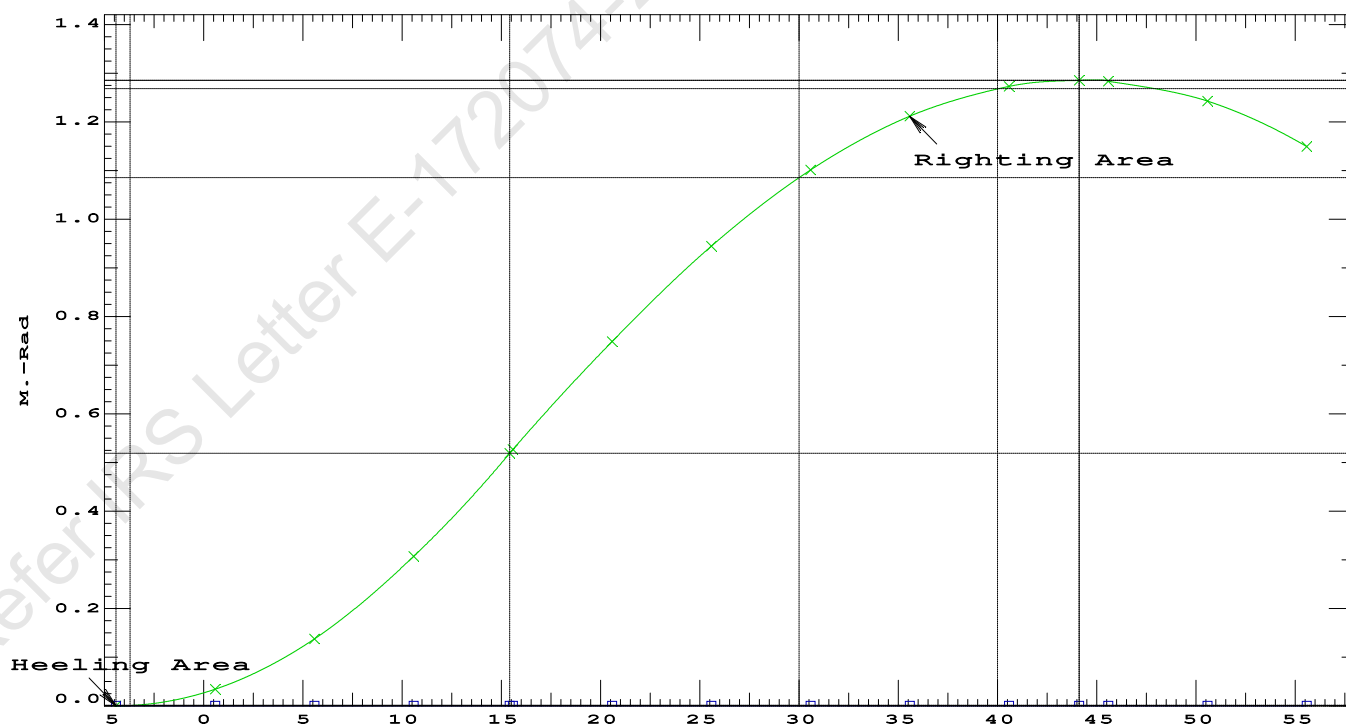
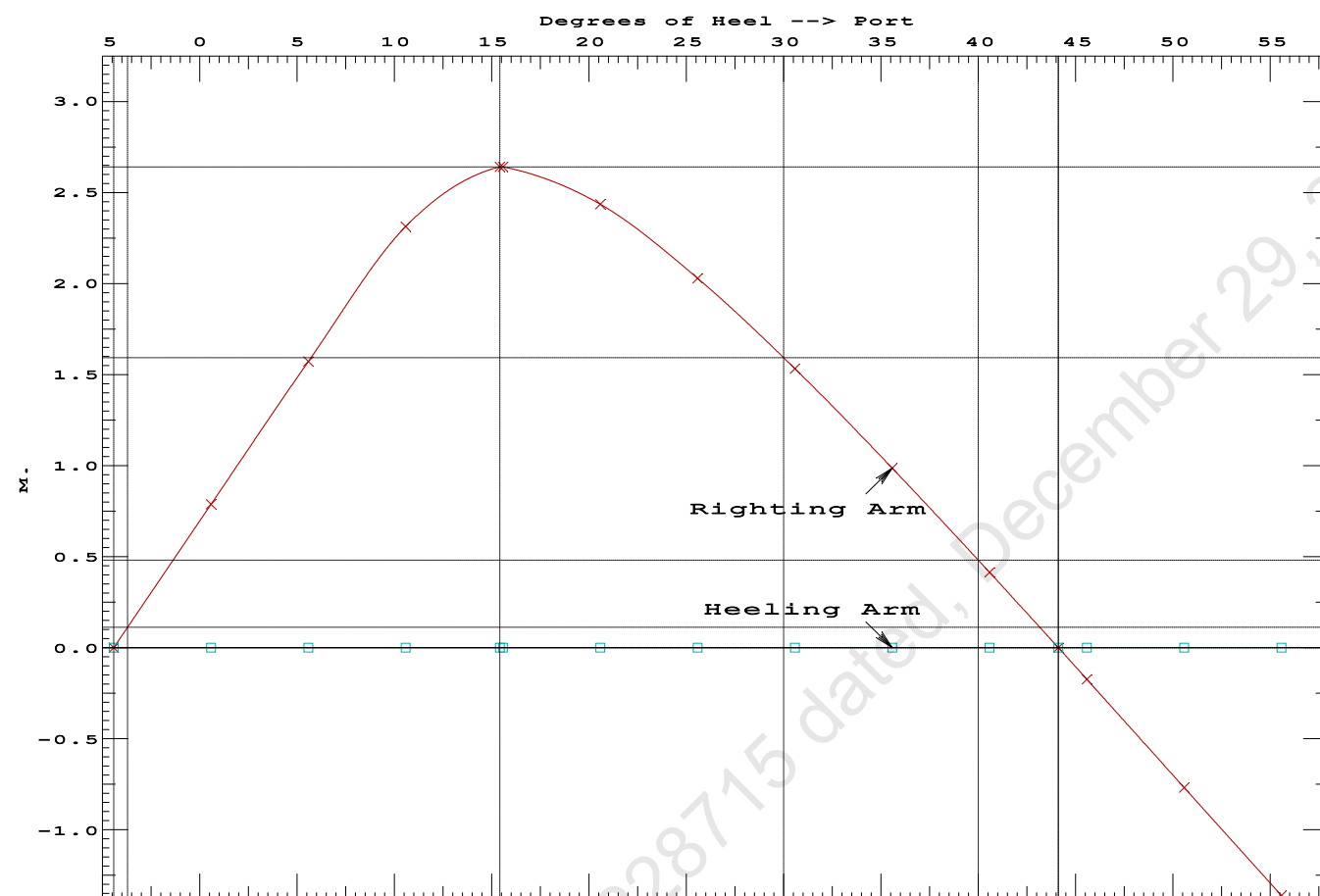
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.10

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	4.42 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	2442210 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	14072.5 P

Condition 3 : Lightship + Load at Stbd + BallastCrane Lifting 36.6T @ 18m Radius Stbd

USK draft refers to the line:

0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Refer IRS Letter E-172074-228715 dated, December 29, 2022

Condition 3 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.350 @ 50.00f, 1.352 @ 25.00f, 1.355 @ 0.00

Trim: Aft 0.004/50.000, Heel: Stbd 4.41 deg.

Part-----			Weight (MT)	---LCG---	TCG---	VCG	
LIGHT SHIP			373.37	24.970f	0.000	3.000	
Crane			250.00	25.000f	0.000	6.120	
Crane Load			36.60	25.000f	18.000s	76.630	
Total Fixed----->			659.97	24.983f	0.998s	8.265	
	Load----	SpGr----	Weight (MT)	---LCG---	TCG---	VCG-----	FSM
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks----->			264.60	25.000f	0.000	1.500	183.75
Total Weight----->			924.57	24.988f	0.713s	6.329	
			Displ (MT)	---LCB---	TCB---	VCB	
HULL		1.025	924.57	24.987f	1.145s	0.731	

	Righting Arms:			0.000	0.000s		
Part-----		SpGr----	WPA----	LCF---	TCF---	BML----	BMT
Total Waterplane---->	1.025		713.7	24.996f	0.125s	148.38	14.708*
			MT/cm-----	m.-MT/cm-----	GML----	GMT	
			7.32		26.40	142.77	9.093
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 4.41

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.305 @ 50.00f, 1.307 @ 25.00f, 1.309 @ 0.00

Trim: Aft 0.004/50.000, Heel: 0.00 deg.

Part	Weight (MT)	LCG	TCG	VCG	FSM		
LIGHT SHIP	373.37	24.970f	0.000	3.000			
Crane	250.00	25.000f	0.000	6.120			
Total Fixed	623.37	24.982f	0.000	4.251			
Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks			264.60	25.000f	0.000	1.500	183.75
Total Weight			887.97	24.987f	0.000	3.431	
HULL	1.025		887.86	24.987f	0.001	0.660	
Righting Arms:			0.000	-0.001s			
External Arms:			0.000	-0.001			
Residual Righting Arms:			0.000	0.000			
Part	SpGr	WPA	LCF	TCF	BML	BMT	
Total Waterplane	1.025	708.7	24.997f	0.005p	151.81	15.121*	
MT/cm		7.26		26.47	149.04	12.350	
Distances in METERS.							Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 3014.26; t3: 30.00; t: 15.00
limmarg: 1000.30; t3: 15.00; t: 7.50
limmarg: 188.14; t3: 7.50; t: 3.75
limmarg: -27.99; t3: 3.75; t: 1.88
limmarg: 65.33; t3: 5.68; t: 0.94
limmarg: 15.09; t3: 4.74; t: 0.47
limmarg: -6.63; t3: 4.27; t: 0.23
limmarg: 5.10; t3: 4.53; t: 0.12
limmarg: -0.40; t3: 4.41; t: 0.06
limmarg: 1.87; t3: 4.46; t: 0.03
limmarg: 0.51; t3: 4.43; t: 0.01

Theta3 is 4.43 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Area
1.288	0.00a	4.41s	887.97	0.000	-0.974	0.0000
1.297	0.00a	0.00	887.97	0.000	0.000	-0.0375
1.288	0.00a	4.43p	887.97	0.000	0.977	0.0002
1.286	0.00a	5.00p	887.97	0.000	1.103	0.0106
1.247	0.00a	10.00p	887.96	0.000	2.169	0.1536
1.128	0.01a	15.00p	887.97	0.000	2.853	0.3755
0.991	0.01a	19.36p	887.97	0.000	2.979	0.6000
0.970	0.01a	20.00p	887.97	0.000	2.977	0.6334
0.805	0.01a	25.00p	887.97	0.000	2.863	0.8908
0.633	0.01a	30.00p	887.78	0.000	2.641	1.1317

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -4.4 deg to abs 4.4 > 1.000 1.005 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

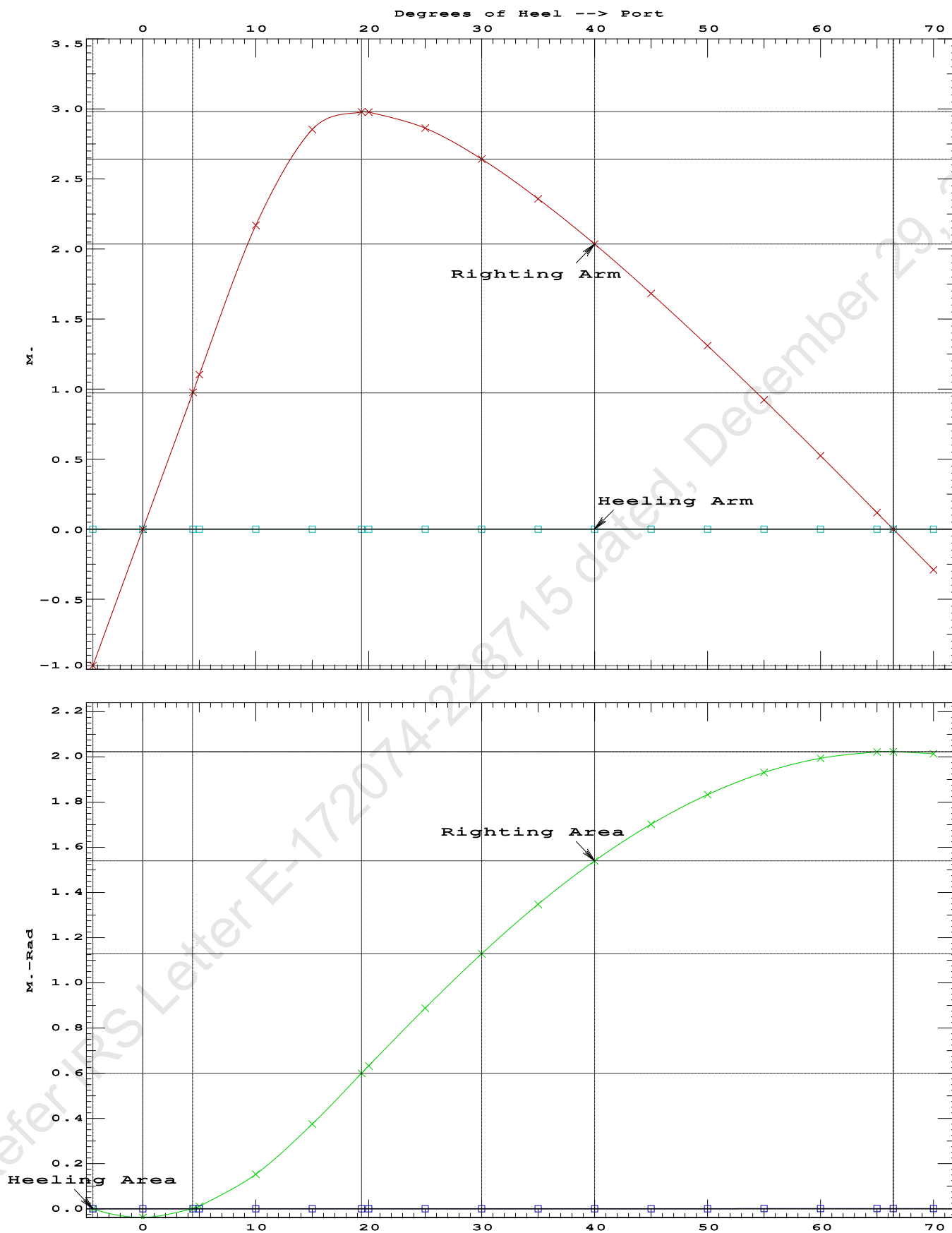
Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---Weight (MT)---in Trim--in Heel---> Area				
1.288	0.00a	4.41s	887.97	0.0000
1.297	0.00a	0.00	887.97	-0.0375
1.288	0.00a	4.43p	887.97	0.0002
1.286	0.00a	5.00p	887.97	0.0106
1.247	0.00a	10.00p	887.96	0.1536
1.128	0.01a	15.00p	887.97	0.3755
0.991	0.01a	19.36p	887.97	0.6000
0.970	0.01a	20.00p	887.97	0.6334
0.805	0.01a	25.00p	887.97	0.8908
0.633	0.01a	30.00p	887.78	1.1317
0.458	0.01a	35.00p	887.97	1.3502
0.279	0.02a	40.00p	887.97	1.5422
0.097	0.02a	45.00p	887.97	1.7045
-0.085	0.02a	50.00p	887.97	1.8352
-0.267	0.02a	55.00p	887.97	1.9327
-0.446	0.02a	60.00p	887.97	1.9958
-0.622	0.02a	65.00p	887.97	2.0238
-0.672	0.02a	66.45p	887.97	2.0253
-0.794	0.03a	70.00p	887.97	2.0163

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -4.4 deg to RAzero > 1.000 54.939 P
(2) Angle from abs 4.4 deg to RAzero > 20.00 deg 62.02 P



USK draft refers to the line:

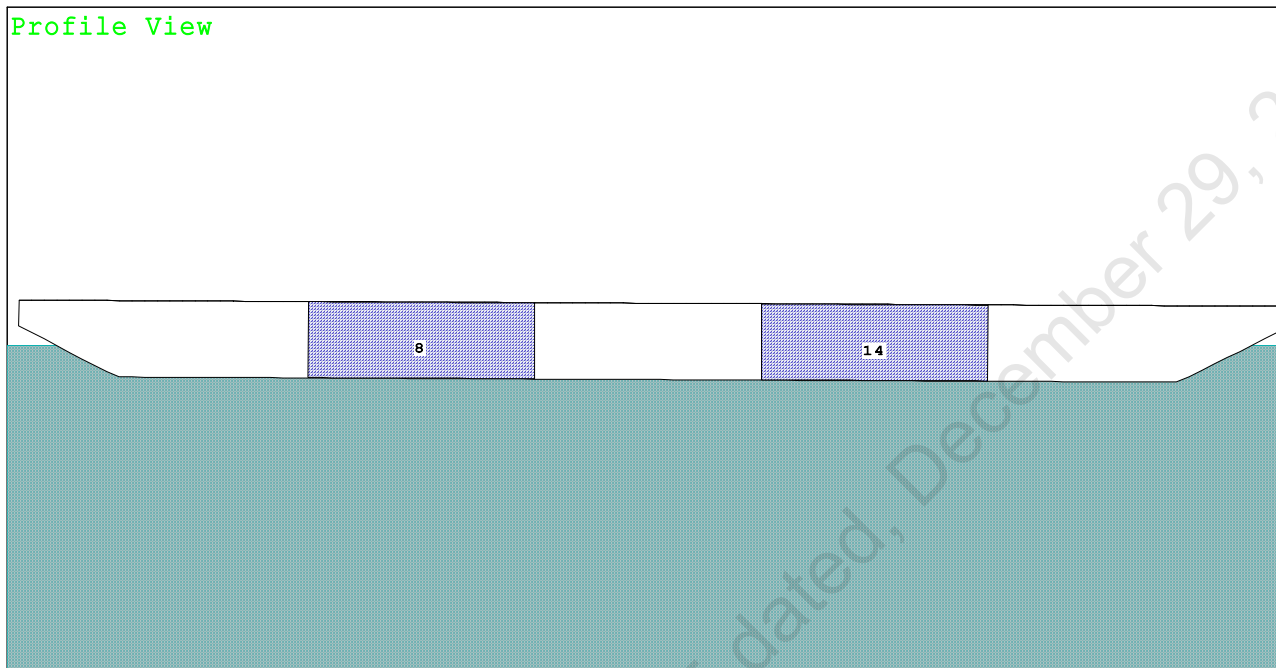
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Refer IRS Letter E-172074-228715 dated, December 29, 2022

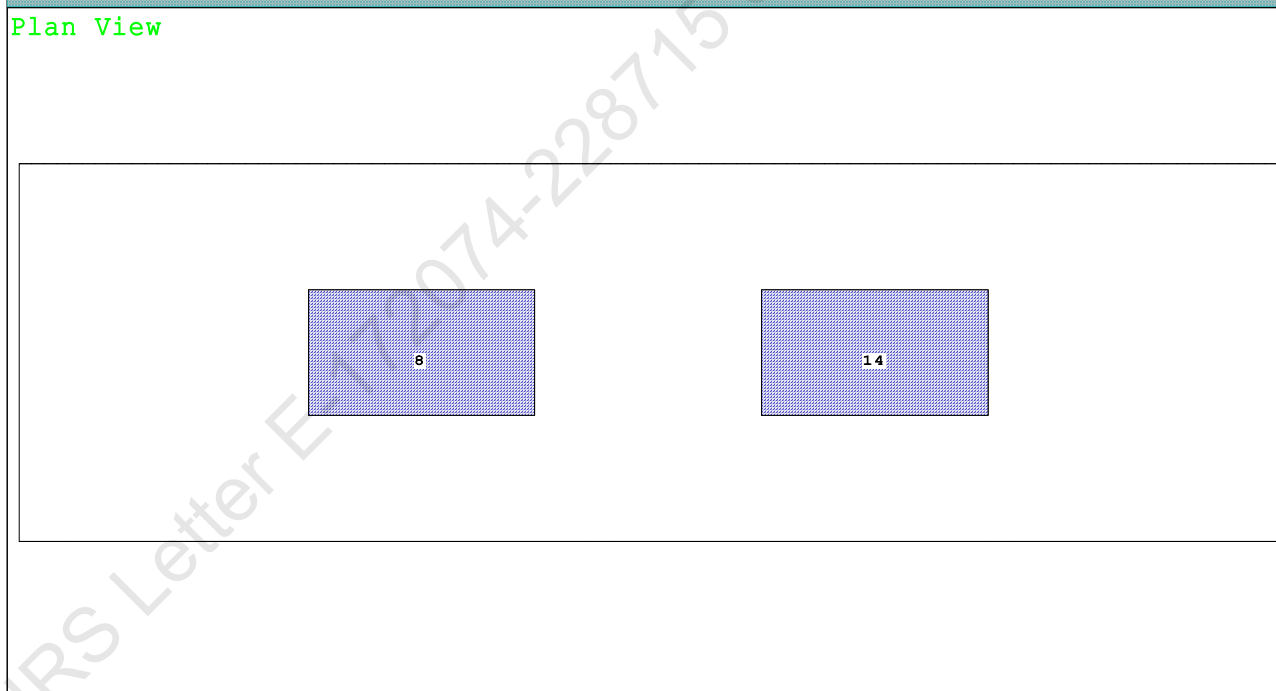
Condition 4 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

Condition Graphic - Draft: 1.480 @ 50.000f, 1.234 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

8 03_PFWBTK.C.100% FRESH WATER 14 05_PFWBTK.C.100% FRESH WATER

Condition 4 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.480 @ 50.00f, 1.357 @ 25.00f, 1.234 @ 0.00

Trim: Fwd 0.246/50.000, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP	373.37	24.970f	0.000	3.000	
Crane	250.00	25.000f	0.000	6.120	
Crane Load	36.60	43.000f	0.000	76.930	
Total Fixed----->	659.97	25.981f	0.000	8.282	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000
Total Tanks----->			264.60	25.000f	0.000
Total Weight----->			924.57	25.700f	0.000
		Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	924.53	25.728f	0.000	0.688

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane---->	1.025	712.3	25.228f	0.000	148.04
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----
		7.30	26.33	142.39	8.956
Distances in METERS.-----					
-----Moments in m.-MT.					

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

264.6 Cu.M. (13%) FRESH WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.480
 DRAFT AT MS (M) = 1.357
 DRAFT AT AP (M) = 1.234
 DRAFT AT FWD MARK (M) = 1.460
 DRAFT AT AFT MARK (M) = 1.254

HYDROSTATIC PROPERTIES

Trim: Fwd 0.246/50.000, No Heel, VCG = 6.341

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/	
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----
1.358	924.53	25.728f	0.688	7.30	25.228f
					26.33
					142.39
					8.956
Distances in METERS.-----					
Specific Gravity = 1.025.-----					
Moment in m.-MT.					
Trim is per 50.00m.					

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

Condition 4 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.700f TCG = 0.000 VCG = 6.341

Free Surface Adjustment: 0.199

Adjusted CG: LCG = 25.699f TCG = 0.000 VCG = 6.540

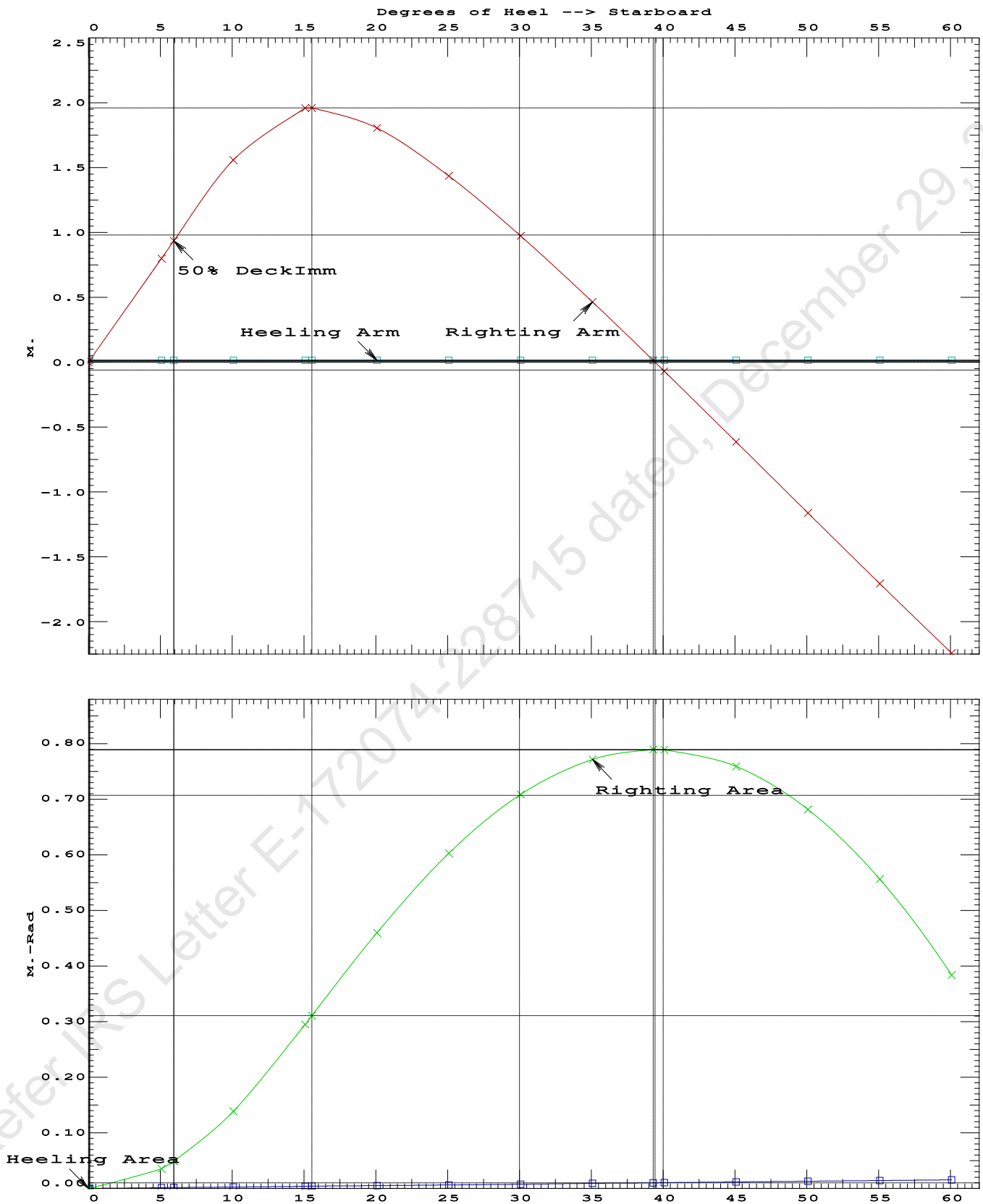
Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area----	Angle
1.222 0.28f 0.00	924.53	0.000 -0.014	0.0000	5.942
1.222 0.28f 0.08s	924.57	0.000 0.000	-0.0000	5.857
1.211 0.28f 5.08s	924.57	0.000 0.786	0.0343	0.857
1.207 0.28f 5.94s	924.57	0.000 0.922	0.0471	0.000
1.169 0.29f 10.08s	924.57	0.000 1.544	0.1363	-4.143
1.027 0.37f 15.08s	924.57	0.000 1.944	0.2903	-9.143
1.010 0.38f 15.55s	924.57	0.000 1.946	0.3060	-9.604
0.834 0.49f 20.08s	924.57	0.000 1.791	0.4569	-14.143
0.632 0.62f 25.08s	924.57	0.000 1.421	0.5995	-19.143
0.423 0.75f 30.08s	924.57	0.000 0.958	0.7039	-24.143
0.208 0.88f 35.08s	924.57	0.000 0.449	0.7656	-29.143
0.026 0.98f 39.30s	924.64	0.000 -0.000	0.7823	-33.361
-0.009 1.00f 40.08s	924.57	0.000 -0.084	0.7817	-34.143
-0.226 1.12f 45.08s	924.57	0.000 -0.630	0.7507	-39.143
-0.442 1.23f 50.08s	924.57	0.000 -1.178	0.6718	-44.143
-0.654 1.33f 55.08s	924.57	0.000 -1.722	0.5453	-49.143
-0.859 1.41f 60.08s	924.57	0.000 -2.257	0.3716	-54.143
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
 Stbd heeling moment = 13.58 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.3060 P
(2) Angle from Equilibrium to RZero or Flood	>	20.00 deg	39.22 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	5.86 P

PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.700f TCG = 0.000 VCG = 6.341

Free Surface Adjustment: 0.199

Adjusted CG: LCG = 25.699f TCG = 0.000 VCG = 6.540

Origin	Degrees of		Displacement	Residual Arms		Res.
Depth	Trim	Heel	Weight (MT)	in Trim	in Heel	Area
1.222	0.28f	0.00	924.45	0.000	0.000	0.0000
1.211	0.28f	5.00p	924.50	0.000	0.786	0.0343
1.170	0.29f	10.00p	924.57	0.000	1.545	0.1363
1.030	0.36f	15.00p	924.57	0.000	1.956	0.2916
1.009	0.38f	15.56p	924.57	0.000	1.960	0.3107
0.837	0.49f	20.00p	924.57	0.000	1.809	0.4595
0.635	0.62f	25.00p	924.57	0.000	1.441	0.6037
0.426	0.75f	30.00p	924.57	0.000	0.979	0.7100
0.212	0.88f	35.00p	924.57	0.000	0.471	0.7737
0.020	0.99f	39.43p	924.62	0.000	0.000	0.7920
-0.005	1.00f	40.00p	924.57	0.000	-0.062	0.7917
-0.223	1.12f	45.00p	924.57	0.000	-0.607	0.7626
-0.438	1.23f	50.00p	924.57	0.000	-1.155	0.6857
-0.650	1.33f	55.00p	924.57	0.000	-1.700	0.5611
-0.856	1.41f	60.00p	924.57	0.000	-2.235	0.3893

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

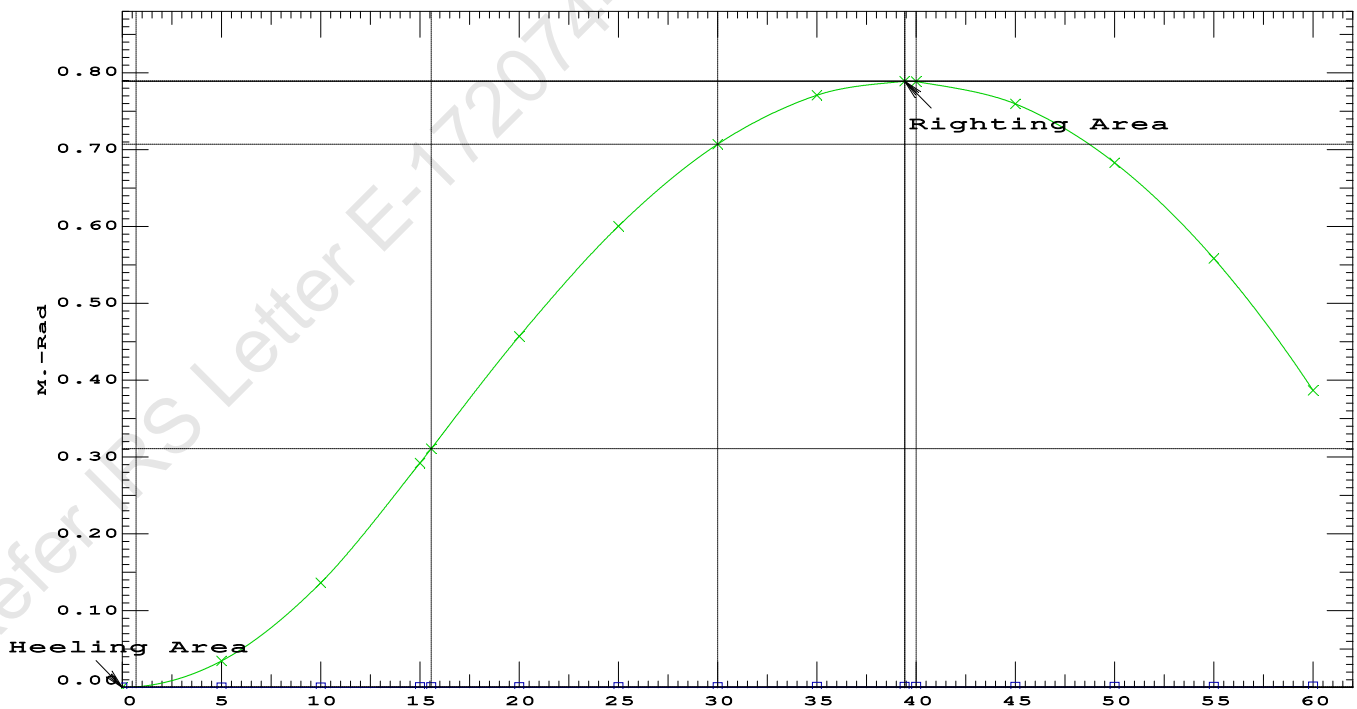
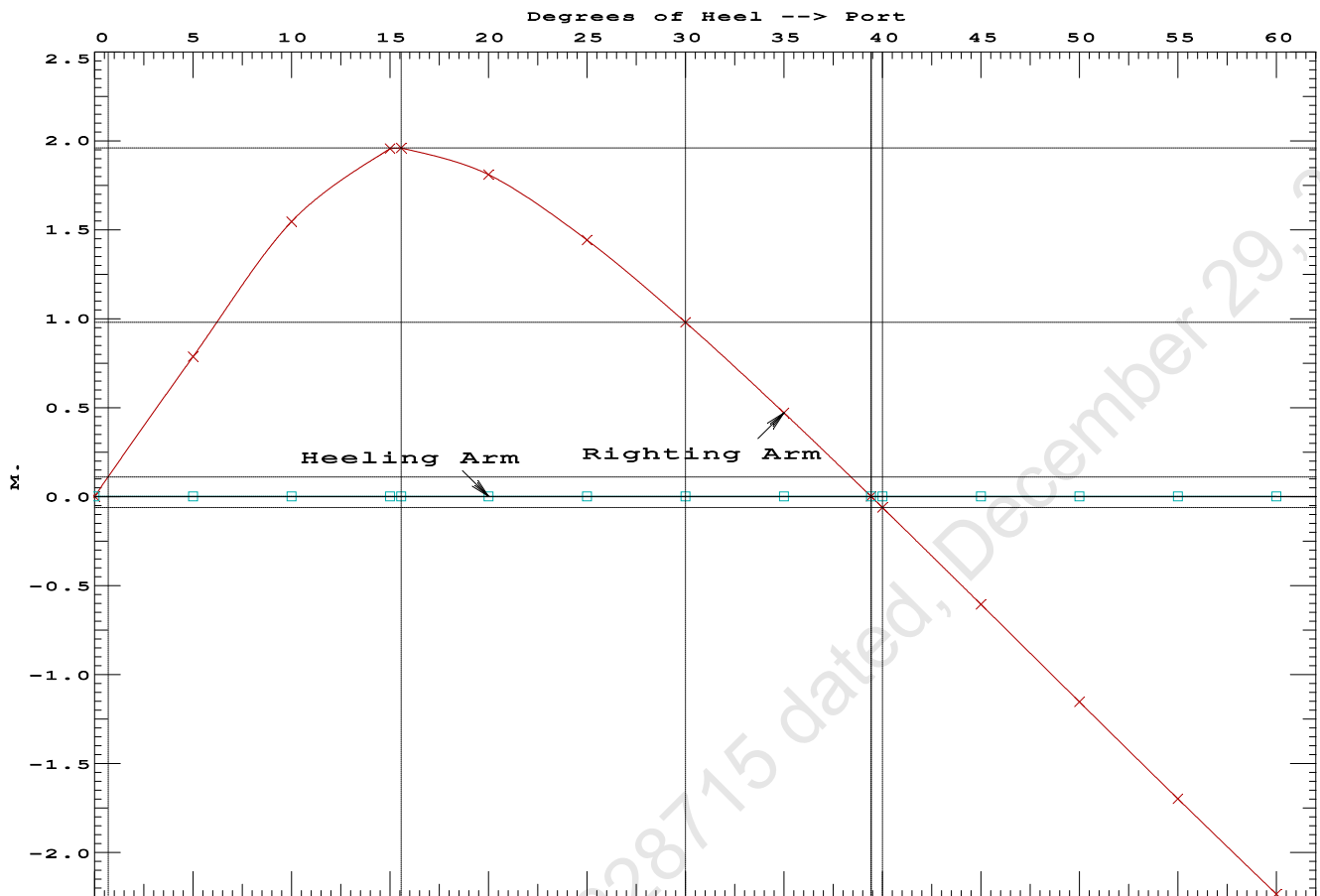
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	181152 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1065.13 P

PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd



USK draft refers to the line:

0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Refer IRS Letter E-172074-228715 dated, December 29, 2022

Condition 4 : Lightship + Load at Fwd + BallastCrane Lifting 36.60T @ 18m Radius Fwd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.480 @ 50.00f, 1.357 @ 25.00f, 1.234 @ 0.00

Trim: Fwd 0.246/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			373.37	24.970f	0.000	3.000	
Crane			250.00	25.000f	0.000	6.120	
Crane Load			36.60	43.000f	0.000	76.930	
Total Fixed----->			659.97	25.981f	0.000	8.282	
	Load----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks----->			264.60	25.000f	0.000	1.500	183.75
Total Weight----->			924.57	25.700f	0.000	6.341	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	924.45	25.728f	0.001p	0.688	

	Righting Arms:			0.000	-0.001s		
	External Arms:			0.000	-0.001p		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		709.2	25.123f	0.017p	146.10	14.562*
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			7.27	25.97	140.45	8.909	
Distances in METERS.-----Moments in m.-MT							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.305 @ 50.00f, 1.307 @ 25.00f, 1.309 @ 0.00

Trim: Aft 0.004/50.000, Heel: 0.00 deg.

Part	Weight (MT)	LCG	TCG	VCG	FSM		
LIGHT SHIP	373.37	24.970f	0.000	3.000			
Crane	250.00	25.000f	0.000	6.120			
Total Fixed	623.37	24.982f	0.000	4.251			
Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks			264.60	25.000f	0.000	1.500	183.75
Total Weight			887.97	24.987f	0.000	3.431	
HULL	1.025		887.97	24.987f	0.001	0.660	
Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.000			
Part	SpGr	WPA	LCF	TCF	BML	BMT	
Total Waterplane	1.025	708.7	24.997f	0.005s	151.80	15.119*	
MT/cm		7.26		26.47	149.03	12.348	
Distances in METERS.							Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

Origin Depth	Degrees of Trim	Heel	Displacement Weight (MT)	Residual Arms in Trim	in Heel	Res. Area
1.297	0.00a	0.00	887.97	0.000	0.000	0.0000
1.297	0.00a	0.07s	887.97	0.000	0.015	0.0000
1.286	0.00a	5.00s	887.97	0.000	1.103	0.0481
1.247	0.00a	10.00s	887.96	0.000	2.169	0.1912
1.128	0.01a	15.00s	887.97	0.000	2.853	0.4131
0.991	0.01a	19.36s	887.97	0.000	2.979	0.6377
0.970	0.01a	20.00s	887.97	0.000	2.977	0.6711
0.805	0.01a	25.00s	887.97	0.000	2.863	0.9284
0.633	0.01a	30.00s	887.78	0.000	2.641	1.1693

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

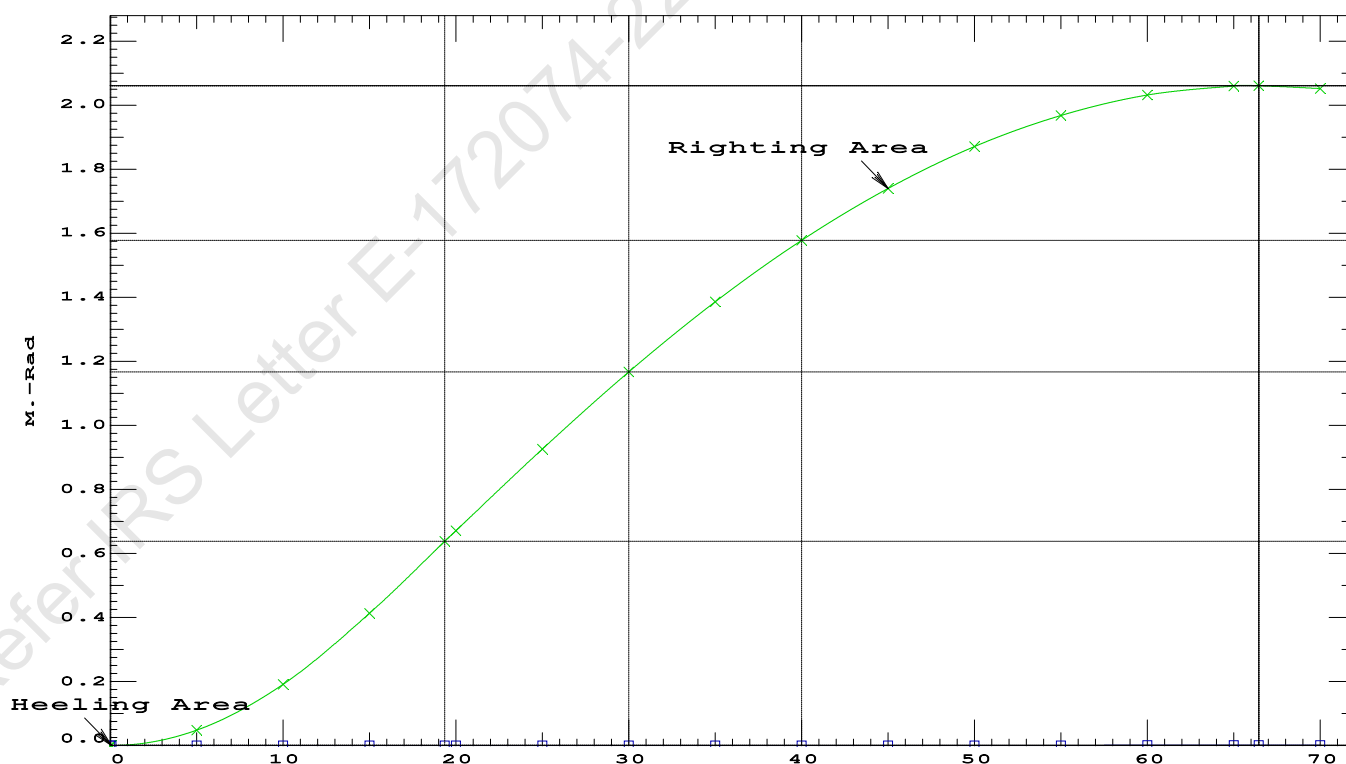
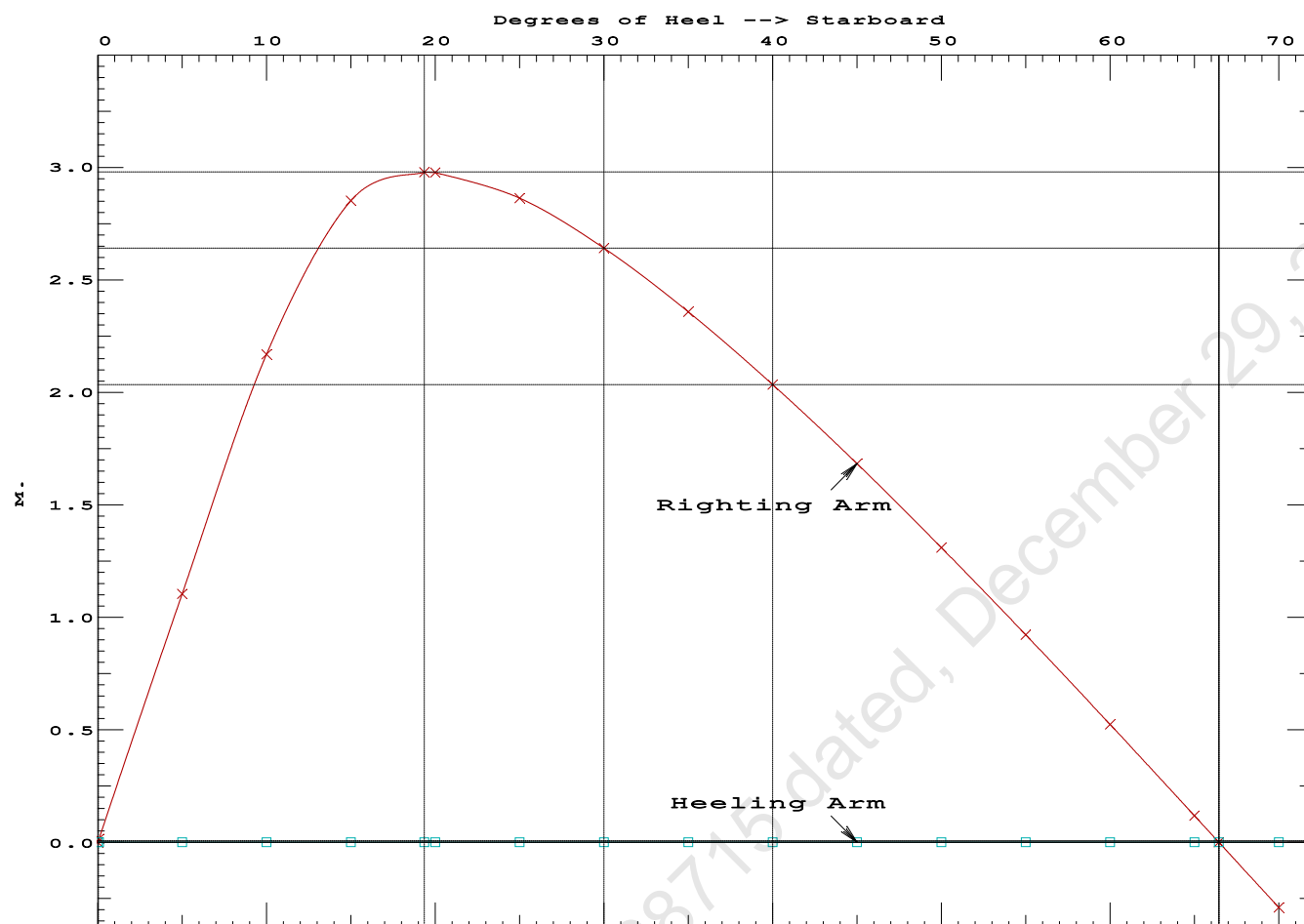
Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---Weight (MT)---in Trim--in Heel---> Area				
1.297	0.00a	0.00	887.97	0.0000
1.297	0.00a	0.07s	887.97	0.0000
1.286	0.00a	5.00s	887.97	0.0481
1.247	0.00a	10.00s	887.96	0.1912
1.128	0.01a	15.00s	887.97	0.4131
0.991	0.01a	19.36s	887.97	0.6377
0.970	0.01a	20.00s	887.97	0.6711
0.805	0.01a	25.00s	887.97	0.9284
0.633	0.01a	30.00s	887.78	1.1693
0.458	0.01a	35.00s	887.97	1.3879
0.279	0.02a	40.00s	887.97	1.5798
0.097	0.02a	45.00s	887.97	1.7422
-0.085	0.02a	50.00s	887.96	1.8729
-0.267	0.02a	55.00s	887.97	1.9704
-0.446	0.02a	60.00s	887.97	2.0335
-0.622	0.02a	65.00s	887.97	2.0615
-0.672	0.02a	66.45s	887.97	2.0629
-0.794	0.03a	70.00s	887.97	2.0540

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 66.38 P



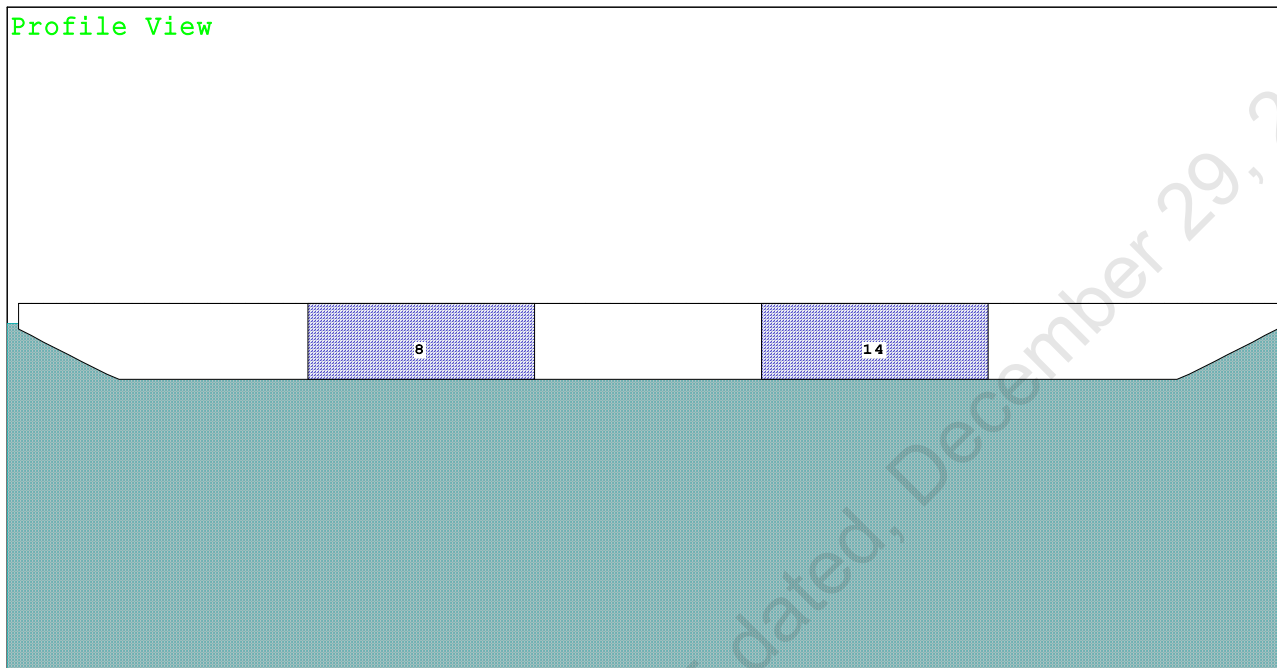
USK draft refers to the line:

0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

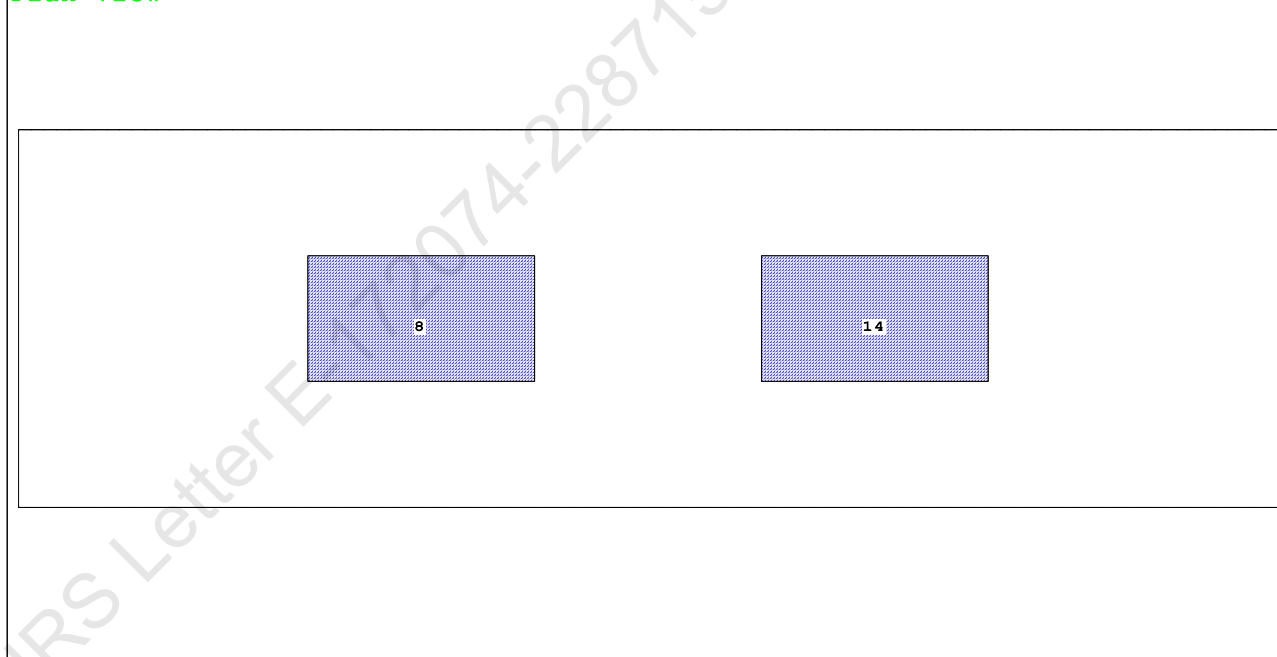
Refer IRS Letter E-172074-228715 dated, December 29, 2022

Condition Graphic - Draft: 2.210 @ 50.000f, 2.214 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

8 03_PFWBTK.C.100% FRESH WATER 14 05_PFWBTK.C.100% FRESH WATER

Condition 5 : Lightship + Crane + Ballast +Deck cargoTowing Condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.210 @ 50.00f, 2.212 @ 25.00f, 2.214 @ 0.00

Trim: Aft 0.004/50.000, Heel: zero

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			373.37	24.970f	0.000	3.000	
Crane			250.00	25.000f	0.000	6.120	
Deck Cargo			681.18	25.000f	0.000	3.500	
Total Fixed----->			1,304.55	24.991f	0.000	3.859	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks----->			264.60	25.000f	0.000	1.500	183.75
Total Weight----->			1,569.15	24.993f	0.000	3.461	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,569.15	24.993f	0.000	1.134	

	Righting Arms:			0.000	0.000		
Part-----			SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->		1.025		750.6	25.000f	0.000	102.05 9.092*
				MT/cm-----	m.-MT/cm-----	GML-----	GMT-----
				7.69	31.30	99.73	6.766
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

264.6 Cu.M. (13%) FRESH WATER

250.00 MT Crane

681.18 MT Deck Cargo

DRAFT AT FP (M) = 2.210
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.214
 DRAFT AT FWD MARK (M) = 2.210
 DRAFT AT AFT MARK (M) = 2.214

Condition 5 : Lightship + Crane + Ballast +Deck cargo**Towing Condition**

HYDROSTATIC PROPERTIES

Trim: Aft 0.004/50.000, No Heel, VCG = 3.461

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
 Draft---Weight(MT)---LCB---VCB---cm---LCF---cm trim---GML---GMT
 2.212 1,569.15 24.993f 1.134 7.69 25.000f 31.30 99.73 6.766
 Distances in METERS.---Specific Gravity = 1.025.---Moment in m.-MT.
 Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.993f TCG = 0.000 VCG = 3.461

Free Surface Adjustment: 0.117

Adjusted CG: LCG = 24.993f TCG = 0.000 VCG = 3.578

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---Weight(MT)---in Trim--in Heel---> Area---Angle				
2.202	0.00a	0.00	1,569.1	0.000 -0.006 0.0000 3.079
2.202	0.00a	0.05s	1,569.1	0.000 0.000 -0.0000 3.030
2.198	0.00a	3.08s	1,569.1	0.000 0.359 0.0095 0.000
2.192	0.00a	5.05s	1,569.1	0.000 0.591 0.0258 -1.970
2.223	0.01a	10.05s	1,569.1	0.000 0.984 0.0959 -6.970
2.353	0.01a	14.58s	1,569.1	0.000 1.049 0.1781 -11.500
2.369	0.01a	15.05s	1,569.1	0.000 1.049 0.1867 -11.970
2.562	0.01a	20.05s	1,569.1	0.000 0.985 0.2768 -16.970
2.768	0.01a	25.05s	1,569.1	0.000 0.851 0.3574 -21.970
2.955	0.01a	30.05s	1,569.2	0.000 0.665 0.4239 -26.970
3.120	0.01a	35.05s	1,569.1	0.000 0.452 0.4729 -31.970
3.262	0.02a	40.05s	1,569.1	0.000 0.223 0.5025 -36.970
3.372	0.02a	44.74s	1,569.1	0.000 0.000 0.5117 -41.657
3.378	0.02a	45.05s	1,569.1	0.000 -0.015 0.5116 -41.970
3.469	0.02a	50.05s	1,569.1	0.000 -0.258 0.4998 -46.970
3.534	0.02a	55.05s	1,569.1	0.000 -0.502 0.4666 -51.970
3.571	0.02a	60.05s	1,569.1	0.000 -0.744 0.4123 -56.970

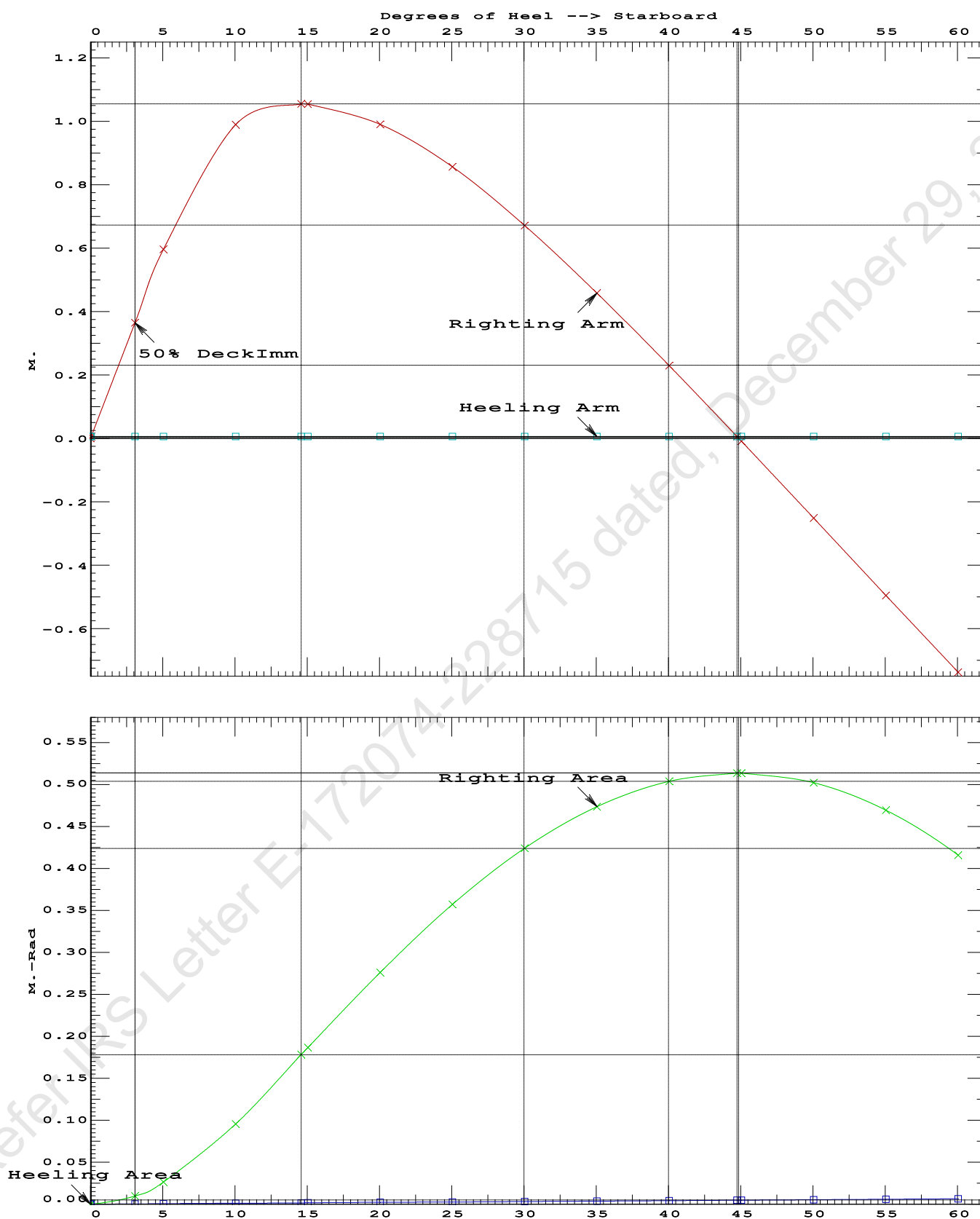
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
 Stbd heeling moment = 9.39 (constant)

LIM-----INTACT CRITERIA-IS CODE 2008PART B ----Min/Max-----Attained
 (1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.1781 P
 (2) Angle from Equilibrium to RZero or Flood > 20.00 deg 44.69 P
 (3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 3.03 P

PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck cargo
Towing Condition

LIGHTSHIP PARTICULARS

Lightship Weight	=	373.37	T
VCG	=	3.000	M above Baseline
LCG	=	24.970	M fwd of AP
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are taken from approved draft survey report.)

18-08-22 10:33:06 Cybermarine Technologies PTE. Ltd.
GHS 18.24A PONT00N

LATERAL PLANE STATUS

USK draft: 0.500 @ 50.00f, 0.500 @ 25.00f, 0.500 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	21.6	25.000f	-0.248	121.7	25.000f	1.289
CRANE.C				35.2	7.500f	4.712
DECKCARGO.C				40.0	25.000f	3.512
Total Lateral Plane->	21.6	25.000f	-0.248	196.9	21.871f	2.352
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.600 @ 50.00f, 0.600 @ 25.00f, 0.600 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	26.0	25.000f	-0.297	117.2	25.000f	1.236
CRANE.C				35.2	7.500f	4.612
DECKCARGO.C				40.0	25.000f	3.412
Total Lateral Plane->	26.0	25.000f	-0.297	192.4	21.799f	2.306
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.700 @ 50.00f, 0.700 @ 25.00f, 0.700 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	30.5	25.000f	-0.346	112.8	25.000f	1.183
CRANE.C				35.2	7.500f	4.512
DECKCARGO.C				40.0	25.000f	3.312
Total Lateral Plane->	30.5	25.000f	-0.346	188.0	21.723f	2.259
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.800 @ 50.00f, 0.800 @ 25.00f, 0.800 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	35.0	25.000f	-0.395	108.3	25.000f	1.130
CRANE.C				35.2	7.500f	4.412
DECKCARGO.C				40.0	25.000f	3.212
Total Lateral Plane->	35.0	25.000f	-0.395	183.5	21.642f	2.214
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.900 @ 50.00f, 0.900 @ 25.00f, 0.900 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	39.6	25.000f	-0.443	103.7	25.000f	1.077
CRANE.C				35.2	7.500f	4.312
DECKCARGO.C				40.0	25.000f	3.112
Total Lateral Plane->	39.6	25.000f	-0.443	178.9	21.557f	2.169
Distances in METERS.-----						

18-08-22 10:33:06 Cybermarine Technologies PTE. Ltd.
GHS 18.24A PONTON

LATERAL PLANE STATUS

USK draft: 1.000 @ 50.00f, 1.000 @ 25.00f, 1.000 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	44.1	25.000f	-0.492	99.1	25.000f	1.025
CRANE.C				35.2	7.500f	4.212
DECKCARGO.C				40.0	25.000f	3.012
Total Lateral Plane->	44.1	25.000f	-0.492	174.3	21.467f	2.124

Distances in METERS.

LATERAL PLANE STATUS

USK draft: 1.100 @ 50.00f, 1.100 @ 25.00f, 1.100 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	48.8	25.000f	-0.540	94.5	25.000f	0.972
CRANE.C				35.2	7.500f	4.112
DECKCARGO.C				40.0	25.000f	2.912
Total Lateral Plane->	48.8	25.000f	-0.540	169.7	21.370f	2.081

Distances in METERS.

LATERAL PLANE STATUS

USK draft: 1.200 @ 50.00f, 1.200 @ 25.00f, 1.200 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	53.4	25.000f	-0.589	89.8	25.000f	0.920
CRANE.C				35.2	7.500f	4.012
DECKCARGO.C				40.0	25.000f	2.812
Total Lateral Plane->	53.4	25.000f	-0.589	165.0	21.267f	2.038

Distances in METERS.

LATERAL PLANE STATUS

USK draft: 1.300 @ 50.00f, 1.300 @ 25.00f, 1.300 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	58.2	25.000f	-0.637	85.1	25.000f	0.868
CRANE.C				35.2	7.500f	3.912
DECKCARGO.C				40.0	25.000f	2.712
Total Lateral Plane->	58.2	25.000f	-0.637	160.3	21.158f	1.997

Distances in METERS.

LATERAL PLANE STATUS

USK draft: 1.400 @ 50.00f, 1.400 @ 25.00f, 1.400 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	62.9	25.000f	-0.685	80.4	25.000f	0.817
CRANE.C				35.2	7.500f	3.812
DECKCARGO.C				40.0	25.000f	2.612
Total Lateral Plane->	62.9	25.000f	-0.685	155.6	21.040f	1.956

Distances in METERS.

18-08-22 10:33:06 Cybermarine Technologies PTE. Ltd.
GHS 18.24A PONT00N

LATERAL PLANE STATUS

USK draft: 1.500 @ 50.00f, 1.500 @ 25.00f, 1.500 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	67.7	25.000f	-0.733	75.6	25.000f	0.765
CRANE.C				35.2	7.500f	3.712
DECKCARGO.C				40.0	25.000f	2.512
Total Lateral Plane->	67.7	25.000f	-0.733	150.8	20.915f	1.916
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.600 @ 50.00f, 1.600 @ 25.00f, 1.600 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	72.5	25.000f	-0.781	70.8	25.000f	0.714
CRANE.C				35.2	7.500f	3.612
DECKCARGO.C				40.0	25.000f	2.412
Total Lateral Plane->	72.5	25.000f	-0.781	146.0	20.780f	1.878
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.700 @ 50.00f, 1.700 @ 25.00f, 1.700 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	77.4	25.000f	-0.829	65.9	25.000f	0.663
CRANE.C				35.2	7.500f	3.512
DECKCARGO.C				40.0	25.000f	2.312
Total Lateral Plane->	77.4	25.000f	-0.829	141.1	20.634f	1.841
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.800 @ 50.00f, 1.800 @ 25.00f, 1.800 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	82.3	25.000f	-0.876	61.0	25.000f	0.612
CRANE.C				35.2	7.500f	3.412
DECKCARGO.C				40.0	25.000f	2.212
Total Lateral Plane->	82.3	25.000f	-0.876	136.2	20.477f	1.806
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.900 @ 50.00f, 1.900 @ 25.00f, 1.900 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	87.2	25.000f	-0.924	56.0	25.000f	0.561
CRANE.C				35.2	7.500f	3.312
DECKCARGO.C				40.0	25.000f	2.112
Total Lateral Plane->	87.2	25.000f	-0.924	131.2	20.307f	1.772
Distances in METERS.-----						

18-08-22 10:33:06 Cybermarine Technologies PTE. Ltd.
GHS 18.24A PONTON

LATERAL PLANE STATUS

USK draft: 2.000 @ 50.00f, 2.000 @ 25.00f, 2.000 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	92.2	25.000f	-0.971	51.1	25.000f	0.511
CRANE.C				35.2	7.500f	3.212
DECKCARGO.C				40.0	25.000f	2.012
Total Lateral Plane->	92.2	25.000f	-0.971	126.3	20.123f	1.739
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 2.100 @ 50.00f, 2.100 @ 25.00f, 2.100 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	97.2	25.000f	-1.019	46.1	25.000f	0.461
CRANE.C				35.2	7.500f	3.112
DECKCARGO.C				40.0	25.000f	1.912
Total Lateral Plane->	97.2	25.000f	-1.019	121.3	19.922f	1.709
Distances in METERS.-----						



IRCLASS
Indian Register of Shipping
CIN: U61100MH1975NPL018244

52 A, Adi Shankaracharya Marg, Opp. Powai Lake,
Powai, Mumbai - 400 072, India
Phone : +91 22 30519400 / +91 22 7119 9400
Fax : +91 22 2570 3611
E-mail : ho@irclass.org ★ Website : www.irclass.org

Our ref: E-171937-228542

21-November-2022

Company : Cybermarine Knowledge System Pvt. Ltd.

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project Details	
Project number	NC002816
Project name	Chisti Marine Engineering Works - CHISTI/023 and CHISTI/025
Specific Yard No(s):	CHISTI/025
Project Manager	M. Saravanan
Project In-Charge	Srinivas Kundrapu

Document Details	
Plan No.	CM22-VI-112-0208-B
Plan Title	Draft Survey Report_025
Rev.No.	0
Upload.No.	1
Submitted Date	19-November-2022
Review Status	Approved
Parent Plan Title	

Notes

1. The review status Approved indicates:

The following lightship particulars are approved based on the Draft survey for Yard No. CHISTI/025 conducted on 19th November 2022:

Lightship Weight : 373.37 T
Lightship L.C.G : 24.970 m forward of AP(Frame no. 0)
Lightship V.C.G : 3.000 m above Baseline (VCG taken at the level of main deck)
Lightship T.C.G : 0.000 m off Centreline.

Final stability booklet prepared based on the approved lightship particulars should be submitted for approval.
Copy of the approved Draft survey report should form part of the Final Stability booklet

Thanking you,
Yours Sincerely,

For INDIAN REGISTER OF SHIPPING

Vimal R
Surveyor

This is an electronically published document and does not require signature.



MEMORANDUM OF FREEBOARDS

14 NOVEMBER 2022

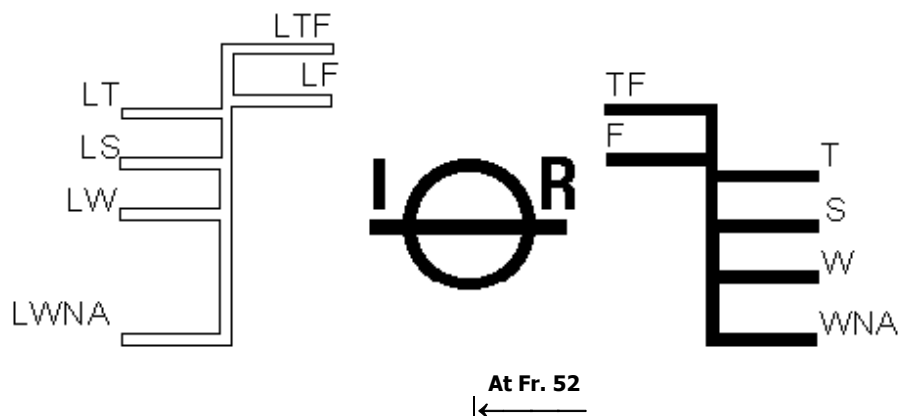
Chisti Marine Engineering Works
Yard No. 025

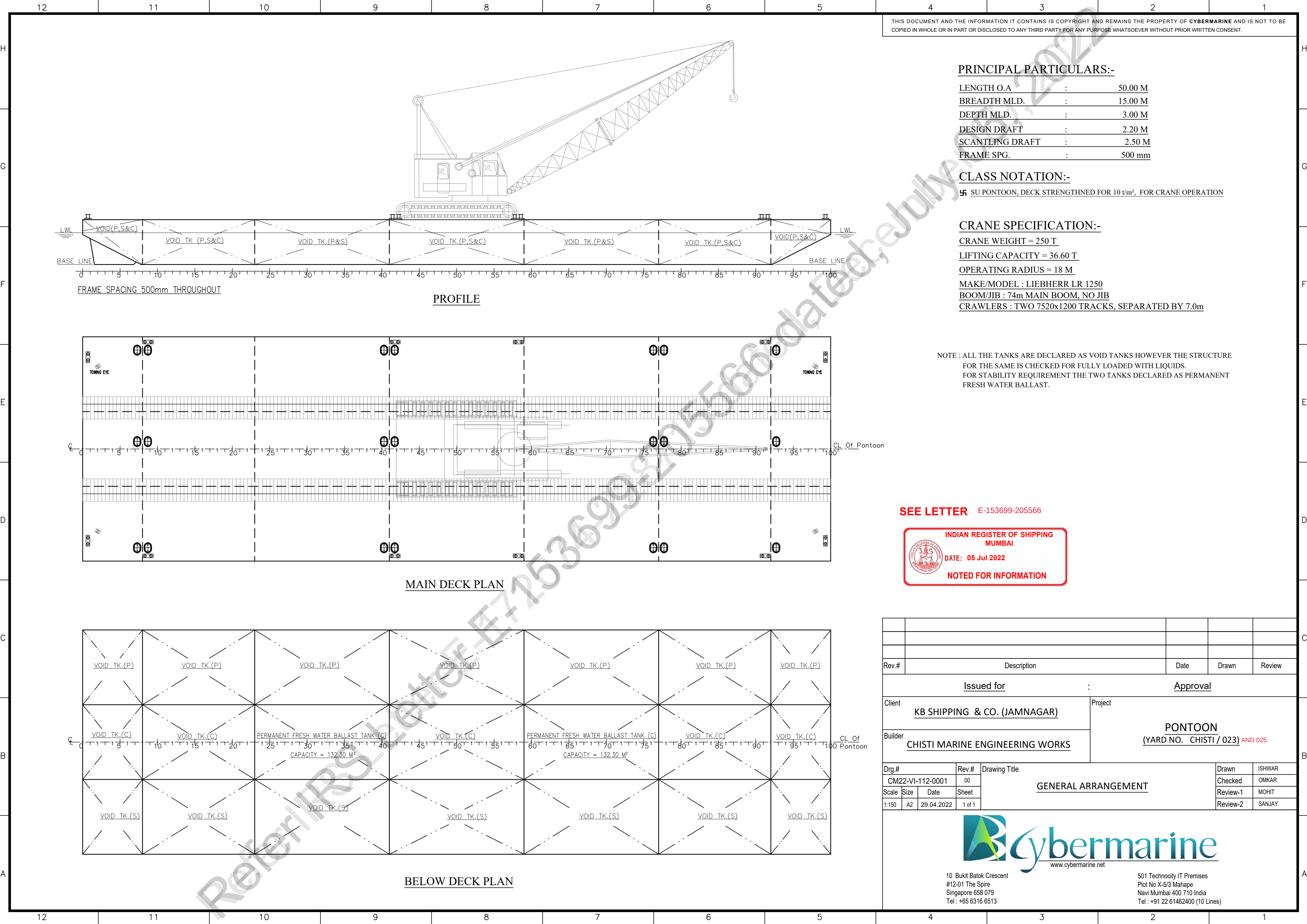
Type B freeboard
Freeboard Length : 48.000 m
Summer WL Extreme Draught : 2.212 m
- loadline amidships

The upper edge of the deckline from which these freeboards are measured is 0 mm below top of *
~~Wood~~ / *steel Deck *at side / *continued to side.

Freeboards from Deck Line		Seasonal, Fresh Water, or Timber allowances		
Tropical Fresh Water	713	mm.	97	mm. above summer
Fresh Water	759	mm.	51	mm. above summer
Tropical	764	mm.	46	mm. above summer
Summer	810	mm.	Upper edge of line through centre	
Winter	856	mm.	46	mm. below summer
Winter North Atlantic	906	mm.	96	mm. below summer
Timber Tropical fresh water	—	mm.	—	mm. above timber summer
Timber fresh water	—	mm.	—	mm. above timber summer
Timber Tropical	—	m m.	—	mm. above timber summer
Timber Summer	—	mm.	—	mm. above centre of ring
Timber Winter	—	mm.	—	mm. below timber summer
Timber Winter North Atlantic	—	mm.	—	mm. below timber summer

Valid for unmanned tow only.





THIS DOCUMENT AND THE INFORMATION IT CONTAINS IS COPYRIGHT AND REMAINS THE PROPERTY OF CYBERMARINE AND IS NOT TO BE COPIED IN WHOLE OR IN PART OR DISCLOSED TO ANY THIRD PARTY FOR ANY PURPOSE WHATSOEVER WITHOUT PRIOR WRITTEN CONSENT.

PRINCIPAL PARTICULARS:-

LENGTH O.A	:	50.00 M
BREADTH MLD.	:	15.00 M
DEPTH MLD.	:	3.00 M
DESIGN DRAFT	:	2.20 M
SCANTLING DRAFT	:	2.50 M
FRAME SPG.	:	500 mm

CLASS NOTATION:-

SU PONTOON, DECK STRENGTHNED FOR 10 t/m², FOR CRANE OPERATION

CRANE SPECIFICATION:-

CRANE WEIGHT = 250 T
LIFTING CAPACITY = 36.60 T
OPERATING RADIUS = 18 M
MAKE/MODEL : LIEBHERR LR 1250
BOOM/JIB : 74m MAIN BOOM, NO JIB
CRAWLERS : TWO 7520x1200 TRACKS, SEPARATED BY 7.0m

NOTE : ALL THE TANKS ARE DECLARED AS VOID TANKS HOWEVER THE STRUCTURE FOR THE SAME IS CHECKED FOR FULLY LOADED WITH LIQUIDS.
FOR STABILITY REQUIREMENT THE TWO TANKS DECLARED AS PERMANENT FRESH WATER BALLAST.

SEE LETTER E-153699-205566



Rev.#	Description		Date	Drawn	Review		
Issued for			Approval				
Client KB SHIPPING & CO. (JAMNAGAR)			Project PONTOON (YARD NO. CHISTI / 023) AND 025				
Builder CHISTI MARINE ENGINEERING WORKS							
Drg.#		Rev.#	Drawing Title GENERAL ARRANGEMENT		Drawn	ISHWAR	
CM22-VI-112-0001		00			Checked	OMKAR	
Scale	Size	Date			Sheet	Review-1	MOHIT
1:150	A2	29.04.2022			1 of 1	Review-2	SANJAY



10 Bukit Batok Crescent
#12-01 The Spire
Singapore 658 079
Tel : +65 6316 6513

501 Technocity IT Premises
Plot No X-5/3 Mahape
Navi Mumbai 400 710 India
Tel : +91 22 61462400 (10 Lines)

01	GENERAL ARRANGEMENT	CM22-VI-112-0001	00
#	Drawing / document Name	Drawing #	Rev.

List of Reference drgs / documents used for drg.



Rev.#	Description				Date	Drawn	Review
<u>Issued for</u>					:	<u>Approval</u>	
Client <u>KB SHIPPING & CO. (JAMNAGAR)</u>					Project <u>PONTOON</u> <u>(YARD NO. CHISTI / 023)</u>		
Builder <u>CHISTI MARINE ENGINEERING WORKS</u>							
Drg.#		Rev.#		<u>DISPOSITION OF DRAFT MARK</u>		Drawn	ISHWAR
CM22-VI-112-4001		00				Checked	OMKAR
Scale	Size	Date	Sheet			Review-1	SHAILESH
1:50	A4/A3	09.05.2021	1 of 2			Review-2	

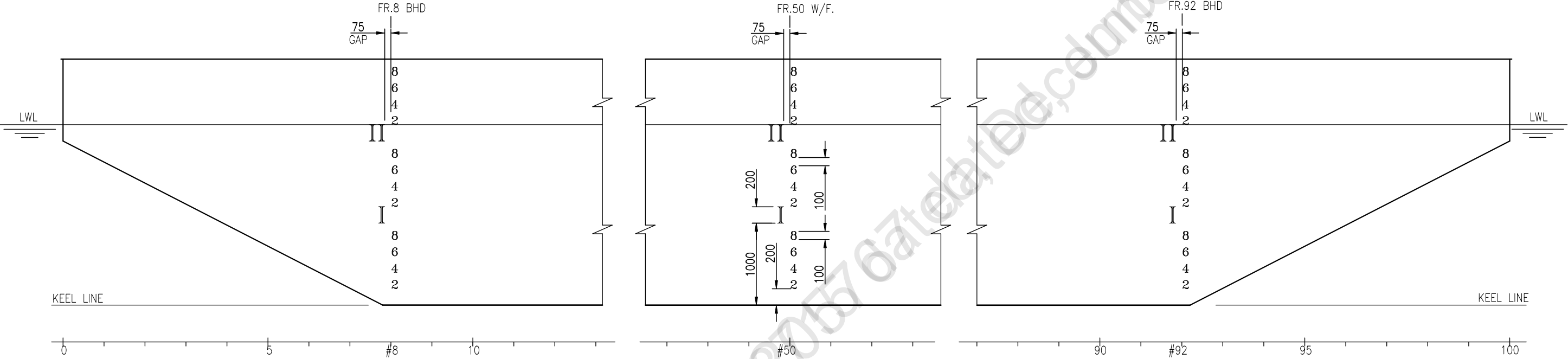


Cybermarine
www.cybermarine.net info@cybermarine.net

10, Bukit Batok Crescent,
#12-01, The Spire Building,
Singapore-658079.

501, Technocity IT Premises Co-op. Soc. Ltd,
Plot No. X-5/3, Mahape,
Maharashtra- 400710, India.

22503, Katy Freeway #7,
Katy, Texas,
USA 77450.



PROFILE
(PORT SHOWN AND STBD SIMILAR)

PRINCIPAL PARTICULARS:-

LENGTH O.A	:	50.00 M
BREADTH MLD.	:	15.00 M
DEPTH MLD.	:	3.00 M
DESIGN DRAFT	:	2.20 M

NOTE:-

- 1) DRAFT MARKS ARE TO BE CUT OF 6 MM PLT. AND TO BE WELDED TO THE HULL.
- 2) MARKS ARE TO BE PAINTED WITH COATS OF PRIMER TWO COATS OF ANTICORROSIVE PAINTS AND TWO COATS OF WHITE FINISH